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Who Runs Short in a Drought? Water Rights and Canal Delivery in Arizona*

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Abstract

In Arizona, water rights and irrigation district rules determine how much water each farm may use, while the water reaches farms through canals shared with other farms. Along a canal, water reaches upstream farms first, so most of what they use is no longer available to farms farther down. I use recent droughts to ask whether farms with the same rights share water shortages equally. I trace Arizona's canals from their sources to 28,427 fields. Each month, I measure the water used by the fields upstream and downstream of each field. In a drought month, a field uses less water when the fields upstream of it use more, and not when the fields downstream do. In junior-rights districts, which lose surface water first, position decides who goes short. Fields that usually sit low on the canal are left fallow more often in drought. Planted fields grow less when their upstream neighbours plant more than usual. In senior-rights districts, fields lose water when the whole canal draws more. Over the drought years of 2019–2023, position costs junior fields $3.4 million at net return, mostly through fallowing. The drier years of 2008–2015, when deliveries were not cut, show no such effect. For large surface-water districts with monthly water records, I predict each field's surface-water shortfall each month from the records, crop demand and the irrigation network. In Maricopa–Stanfield, fields without a well use about 18 percent less water in a month when canal water is predicted to meet none of their crop demand than when it meets all of it. Where district wells pump into the canals marks which fields keep water. In the Upper Gila, which rotates water, equal sharing within a canal fits best. Farms with the same rights do not share the costs of delivery shortfalls equally.

1 Introduction

Laws and rules often decide how much of a shared resource each user may take. In many cases, the resource also reaches those users one after another. When the resource is plentiful, the order in which users are served makes little difference. When it is scarce, the order can matter. If each

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user's use could be measured and controlled exactly, the order would not matter even then, because the rules alone would decide who gets what. In practice, use is rarely measured that precisely, so the users the resource reaches first could affect how much is left for everyone after them.

Irrigation water is a clear example of this problem. Water rights and the rules of irrigation districts, the local organizations that run the canals, determine how much water each farm is entitled to. But when there is not enough water for everyone, the amount a farm actually receives can be different from what it is entitled to. Water reaches farms through branching canals shared with other farms, and along a canal, delivery is sequential: water passes the upstream farms before it reaches the farms below. Such a system is hard to monitor and to operate exactly, so farms upstream can use more water than the rules intend. A district may also serve the upstream farms first when water is short. Any extra water they use is water the farms downstream do not receive. In ordinary years these shortfalls are small, and reservoirs, wells, and adjustments to delivery timing cover them. When a drought lasts several years, these buffers run low. Every field in crop upstream then draws on the same shrinking flow, and the water a field gets can start to depend on how much water the fields upstream of it use, even though its own land and rights have not changed.[1]

Suppose ten farms on one canal are each entitled to a tenth of its water, and a long drought cuts the canal's flow by 30 percent. On paper, each farm loses 30 percent. In practice, the water reaches the upstream farms first. If each of them consumes even a little more than its reduced share, that water is no longer in the canal for the farms below. These small amounts add up along the canal, so the farms at the end lose much more than 30 percent. If some of the upstream farms leave their fields fallow that year, more water stays in the canal for the farms at the end. Even where water rights are well developed, the order in which water reaches farms, and not only their rights, can decide who faces a shortfall.

In the United States, this problem has received little attention. The only direct test, in small ditches in Colorado, found no difference between farms at the head and at the end of a canal (Cody 2018a). I ask whether, when farms hold formal water rights, a long drought makes the water a field gets depend on what upstream fields use. Using data from Arizona, I find that it does in junior-rights districts. Farms with the same rights therefore do not share a shortage equally. The effect follows the direction of water flow along the canals, not distance on the map. Its timing follows water deliveries, not the weather.

Arizona is a good place to study this question, for two reasons. First, its water rules are well developed. Water rights and contracts determine which users lose water first when the Colorado River runs short, and irrigation districts deliver the water to farms through their own canals. Second, Arizona has faced a long and severe drought. The years 2000–2021 were the driest 22-year period in southwestern North America since at least 800 CE (Williams, Cook, and Smerdon 2022). In 2022 the shortage reached Arizona's farms directly. That year brought the first shortage ever declared on the lower Colorado River, known as Tier 1 (Reclamation 2021). The Central Arizona Project (CAP) sets aside a low-priority block of its water, the Agricultural Settlement Pool, for central Arizona's irrigation districts. In 2022 these districts lost about two-thirds of their CAP

[1] Appendix A9 documents delivery constraints outside the Central Arizona Project: historical lower-end losses and contemporary gate losses in the Gila Valley, capacity pressure in senior Yuma districts, and downstream storage operations in Roosevelt Irrigation District. Tables A11–A13 give source citations, dates, sample coverage, and interpretive limits.

water, even after replacement water (ADWR and CAP 2021). In 2023 the pool received no water at all (CAP 2023). These cuts were made at the source and followed the rules, so the users with the weakest claims lost water first.

The simplest approach would compare farms at the head of a canal with farms at its end. Farms at the end often get less water every year and adjust to it. They may grow different crops or rely more on wells, and some stop farming. In years with enough water, they may not look worse at all. So I follow the same fields over time. I compare drought months when the fields upstream of a field use more with months when they use less. Measuring that needs a map of the whole irrigation network and each field's water use. Distance alone is not enough. A main canal splits into smaller canals. The water reaching a field has passed only the fields on the canals that feed its own branch. Two fields at the same distance from the head can have very different demand above them. That demand changes from month to month while the distance stays the same.

I build both for Arizona. I trace the canals from their sources to every field, so each field has one route (Section 5). The irrigation network shows which fields take water before each field (its upstream fields) and which take water after it (its downstream fields). Monthly water use comes from satellite estimates of evapotranspiration, for 2016–2023. A field's upstream demand is the water its upstream fields use in the month, and downstream demand is the same for its downstream fields. Only fields more than 3 km away count. I call upstream minus downstream demand the field's *position*, and their sum the *canal total*. I study water use each month, a satellite yield proxy (GCVI) each crop season, and whether a field is left fallow.

I set drought for each irrigation district and year. The measure comes from a standard index that compares rainfall with how much water the air could evaporate over two years (SPEI-24). Arizona's irrigation districts receive their water from the state and divide it among their own fields. In winter, a district sends each farm a letter with its limit. The farmer then decides what to plant and what to leave fallow. The water model compares months within a field's year, so the field's crop, rights and wells that year are held fixed.

The main concern is that the water upstream fields use responds to the same heat and the same cuts as the field's own. Ordinary least squares would then mix "the fields upstream took the water" with "everyone got less." I answer in three ways. First, I instrument upstream and downstream demand with predicted demand, a field's acres times the typical water use of its crop. It moves only when fields change crops or go fallow, which farmers decide before the season. Second, I compare upstream demand with downstream demand in the same regression. Downstream fields sit on the same canals and face the same weather and cuts, but they cannot take water before the field. If water follows the canal, upstream demand lowers a field's water use in drought and downstream demand does not. If the whole canal shares a shortage, both lower it alike. Third, nearby fields share local shocks and go fallow together, so I leave out fields within 3 km.

I report three main findings. First, in drought, fields lose water to the fields upstream of them. In a drought month, upstream demand lowers a field's water use and downstream demand does not (Section 7.1). If upstream fields use 10 percent more and downstream fields 10 percent less, a field's water use falls by about 0.3 percent. The effect is in junior-rights districts, whose surface water is cut first, and there position decides who goes short. In senior-rights districts, demand on either side lowers a field's water use alike, as on a canal that cannot carry its peak (Section 7.2.1).

The loss shows on two margins. Before the season, fields that usually sit low on junior canals are left fallow more often in drought. In 2021, the share left fallow rose 3.9 percentage points in the tail third of junior fields. In the head third, it rose 0.8 points (Section 7.2.2). During the season,

a planted field grows less when its upstream neighbours plant more than usual.

The timing follows deliveries. Junior fields lose most in 2021, the year their irrigation districts were in drought. The drier years of 2008–2015, when CAP deliveries were not cut, show no position effect (Section 7.2.4).

Over the drought years of 2019–2023, position costs junior fields $3.4 million at net return (95 percent range $1.9–4.9 million). It reaches up to $26.0 million at crop revenue. Most of the loss comes from leaving fields fallow. Position accounts for 13,364 extra fallowed acres, 12,399 of them in 2021. Over all fields, the total is $10.2 million at net return, mostly yield loss (Section 7.2.3).

Second, monthly deliveries show how the gap arises. For large surface-water districts with monthly water records, the records, crop demand, and the irrigation network predict each field's surface-water shortfall. In Maricopa–Stanfield, fields without a well lose water when canal water is predicted to meet none of their crop demand. In such a month, they use about 18 percent less water than when canal water meets all of it. New Magma, an irrigation district the model had not seen, repeats the pattern. District wells that pump into the canals mark which fields keep water. I cannot show that a field's own well does. Shuffling delivery dates or positions does not reproduce so large an effect. The comparison is within a field, across months (Section 8).

Third, district rules can decide which position matters. Nine of eleven irrigation districts show what their rules and records predict, including four senior-rights districts where water does not fall short. In the Upper Gila, which rotates water, fields served by canals with larger supply shortfalls use less water. Within a canal, equal sharing fits best. In the Central Arizona Irrigation and Drainage District, district wells come first and the records do not say which wells ran. There I cannot tell, and in Hohokam the model goes wrong. When CAP water was cut, farms in the four CAP districts left much more land fallow. Between 2019 and 2022, their share of fields left fallow rose 13 to 19 percentage points more than in other irrigation districts.[2] The 2022 cut was announced in August 2021. Within it, the same shortfall predicts which fields lost water and went fallow, and where canal pumps stand predicts which fields kept water (Section 8.5).

The rules also point to a remedy. The shortfalls and the estimated slopes put a price on delivery. In ordinary years shortfalls cost Maricopa–Stanfield's fields without a well about $213,000 a year, about 0.5 percent of their crop revenue. Serving those fields first, as the district did in 2023, recovers all of it before pumping costs. It pays if the fields with a well can pump the water they give up for less than about $63 an acre-foot, and my estimates do not show their wells replacing it. New wells and lining the Upper Gila canals are not worth their cost in ordinary years (Section 8.8).

The findings have a simple interpretation. When there is enough water, any extra water used upstream is small, and reservoirs and wells cover it. When water is scarce, the same extra use leaves less water for the fields below, and those fields go short. Water rights and contracts determine how much water each district receives at the source. But inside the district, how much water the fields upstream use still affects which fields face the shortage. The findings do not show who causes the gap. A district may deliver water in order down the canal, so fields at the end are cut first. Or farms upstream may use more water than the rules intend. Both would produce the same pattern, and either way the order in which water reaches farms affects who faces the shortage.

The estimates also have limits. I observe water use from satellites, not the water delivered to each field. The delivery model predicts that water, and the predictions pass the shuffle tests, but

[2]The fallowing range comes from two ways of assigning fields to districts. Differences between the districts before the cut, and the choice of comparison districts, make the changes descriptive.

they remain predictions. The design measures demand, so it cannot say how much water a field lacked.

The paper contributes to the literature on common-pool resources with sequential access, in which irrigation is the main example (Ostrom 1990; Ostrom and Gardner 1993). Theory and field studies show that farms at the head of a canal have an advantage over farms at the end, and that enforcement and local institutions affect how large this advantage is (Ray and Williams 2002; Akram and Mendelsohn 2017; Jacoby and Mansuri 2020; Jacoby, Mansuri, and Fatima 2021; Das and Dutta 2026). Most of this evidence comes from developing countries, where water rules are often weakly enforced. In the western United States, Cody (2018a) finds no clear difference in crop growth between the head and the end of 60 small ditches in Colorado's San Luis Valley. These ditches do not branch, and the farms on them take turns using the water. It has therefore been unclear whether the head-end advantage matters where water rights are well developed. I find that it does when a long drought leaves the canals short. To my knowledge, this is the first study in the United States to test for this advantage field by field across the canal networks of a whole state.

The paper also contributes a way to measure who is upstream and who is downstream of each farm field. Studies of irrigation rarely observe which fields take water before which others. They usually compare farms near the head of a canal with farms near its end, which misses how canals branch. I trace each canal from its source to every field it serves. Each field has one route and a set of fields upstream and downstream of it. I combine the irrigation network with satellite estimates of each field's monthly water use. The network shows how much water the fields upstream of each field use each month. It also shows how that water use changes when fields change crops or go fallow. The method can be applied to other irrigated regions.

Finally, the paper adds to recent work on how water is allocated among farms. Rafey (2023) measures the value of reallocating water through markets in Australia's Murray–Darling Basin, and Rafey (2026) measures water misallocation in California. Harris (2024) studies how Arizona farmers changed their crops and planted acreage when CAP deliveries were cut. These studies look at how water is divided among rights holders, or at how farms respond to a cut at the source. This paper looks inside a district's canals. It asks which fields face a shortage once the water has reached the district.

The problem studied here is not limited to water. Wherever rules decide each user's share of a resource that reaches its users one after another, reading the rules is not enough to know who faces a shortfall. Rules that seem to work while the resource is plentiful may stop deciding the outcome once it becomes scarce, so they need to be judged by how they work in a shortage. Natural gas pipelines and electricity grids that cut supply during shortages are possible examples. The method used here could be applied to them as well, because it maps which users are connected to each location and tracks how their use changes over time.

2 Literature Review

2.1 Sequential Delivery and Long-Run Spatial Equilibrium

Because canal water diverted upstream is unavailable downstream, and monitoring extraction along kilometers of open canal is costly and incomplete, canal systems combine rivalry with costly exclusion, the defining features of a common-pool resource. The literature anchored by Ostrom (1990) studies how communities govern such resources. Ostrom and Gardner (1993) and Ray and

Williams (2002) make sequential head-end/tail-end asymmetry, unequal access for users near the source and users at the end of the canal, central to irrigation governance. Their core insight is directly relevant here: allocation rules operate within a physical system in which the order of access can shape realized incidence, that is, who actually absorbs a shortfall. The distinction is the one between legal and economic property rights (Barzel 1989): an entitlement is what the rule assigns, realized access is what enforcement delivers, and in a sequentially delivered resource the gap between them has a spatial order. In western water that order is layered on prior appropriation, whose emergence and enforcement Leonard and Libecap (2019) and Libecap (2011) study; this paper asks what the priority layer leaves ungoverned below the headgate.

Field evidence establishes both the spatial gradient and the importance of implementation. Wade (1988) shows that tail-end scarcity induces stronger collective organization in South India. Manero et al. (2019) finds that position along secondary canals predicts yields and income in Tanzania. Akram and Mendelsohn (2017) pairs GPS position with farm-level flow measurements in Pakistan and finds that head farms receive more water than tail farms. Jacoby and Mansuri (2020) provides the most direct institutional comparison, contrasting the proportional allocation on paper with measured water availability across hundreds of channels in Pakistan's Indus Basin. Political influence tilts both water and land values toward head-end users, demonstrating that formal formulas and realized delivery can diverge inside an administered system.

Institutions do not move the gradient in a uniform direction. Decentralization increased upstream water appropriation and shifted wealth toward head-end users in Pakistan (Jacoby, Mansuri, and Fatima 2021). Longer exposure to locally elected canal bodies reduced distance-based misallocation and raised revenue for distant farmers in Odisha (Das and Dutta 2026). Water-user associations disproportionately raised productivity at tail-end farms in Pakistani Punjab (Mekonnen, Channa, and Ringler 2015). These studies establish that monitoring, political power, and governance can weaken or deepen head-to-tail differences along a canal; the district characteristics examined in Section 7.2.1 are observational counterparts rather than identified institutional treatments.

For this paper, the key implication is that canal position is itself an equilibrium object, an outcome of long-run choices and not an assigned treatment. Most irrigation studies compare users at different fixed locations. Land development, infrastructure, rights, groundwater investment, crops, and exit can all adjust around a persistent hierarchy. A static gradient in crop outcomes therefore combines physical delivery order with sorting and long-run adaptation. Direct flow measurements such as those in Akram and Mendelsohn (2017) and Jacoby and Mansuri (2020) observe the allocation margin more closely than satellite outcomes do. They do not follow, for the same farm, how the water used by a branching set of farms upstream of it changes over time. The design here follows the same field over time and uses changes in the water its upstream and downstream fields use.

2.2 Western Water Allocation and Delivery Under Scarcity

Burness and Quirk (1979) formalize the efficiency properties of prior appropriation (the Western rule that serves older, senior water rights before newer, junior rights during shortage) under stochastic flows, clearly defined priorities, and costless enforcement. Rafey (2023) shows that market reallocation on river networks can generate substantial gains: water trading in Australia's Murray–Darling Basin increased output by 4–6%, with the largest gains during drought. This

literature concerns who holds and reallocates claims to water. A complementary empirical literature shows how shortage risk and adaptation vary with those claims: drought reduces agricultural yields (Kuwayama et al. 2019), groundwater buffers the damage (Mieno et al. 2024), and junior rights holders absorb more drought risk through fallowing and crop adjustment (Li, Xu, and Zhu 2019; Cobourn et al. 2022).

Two papers by Cody come closest in the Western United States, at two scales. Cody (2018b) compares ditches in Colorado's snowmelt-fed San Luis Valley: seniority and catchment size, not a ditch's upstream rank on the stream, predict diverted volume, months of diversion, and irrigated share.[3] Cody (2018a) moves inside 60 of these systems and relates field-level irrigation and peak greenness to distance from the diversion in four annual cross-sections.[4] Rotational delivery, with or without flexible allocation, leaves no significant head-to-tail difference in crop growth. Both studies stop at small, unbranched ditches in which a field's position is a fixed distance to a single headgate; neither observes a branching network, an upstream set, or within-field variation over time. And in Cody's extreme-drought year position stops mattering and seniority bites harder, the opposite of the sustained-drought pattern examined here.

I change the comparison and do not reopen that result. Cody asks which system-level rules make outcomes at fixed positions more equal. I hold a field and its irrigation network fixed and ask whether its water use depends more on the water used upstream of it in drought months.

Arizona-specific evidence further separates source-level allocation from last-mile delivery to individual fields. Bautista et al. (2000) documents operational service inside one district, studying arranged service with volumetric pricing and turnout meters, meters at the gates through which water leaves the canal for a field, in the Maricopa-Stanfield Irrigation and Drainage District. Farmer interviews associate long travel times along laterals, the branch canals that carry water from a main canal toward fields, and a common service window with delivery-rate and timing problems reported more often downstream, although 85% of respondents describe service as equitable. Harris (2024) identifies average adjustment to a source-level policy shock, studying Central Arizona Project policies that reduced Colorado River deliveries during 2015–2021: farmers reduce cropped acreage, change crop mix, and respond in ways consistent with substituting groundwater for surface water.

Recent network-based water studies ask related questions. Rafey (2026) combines field microdata, historical rights, remotely sensed evapotranspiration, and California's hydrological network to value rights and evaluate reallocation. Garcia and Belmar (2025) combines administrative, hydrological, geographic, and satellite data to study whether irrigator organizations limit upstream extraction capacity in Chilean water markets, finding downstream gains. I ask a different question of similar data: how the water used upstream of a field relates to the same field's water use, in drought and in other months.

Peer-effects designs on undirected networks have no comparison in the reverse direction (Manski 1993). A canal has one, the fields downstream of a field. Direction-based empirical designs are established outside managed irrigation. Garg et al. (2018) map upstream and downstream river relationships and relate sanitation upstream to health downstream. Keiser and Shapiro (2019) attach monitoring sites to national hydrography and compare outcomes along water flow. Both

[3]The sample is 1984–2015 averages for 402 ditches.

[4]The sample is 6,711 fields in four annual cross-sections spanning the 2011–2014 drought; the shortage-sharing and rotational-delivery rules come from a 2013 survey of system managers.

studies show how a directed graph separates effects that travel with the flow from ordinary spatial association. Satellite measurement of canal performance is established as well. Peña-Arancibia and Ahmad (2021) studies head-to-tail equity and groundwater buffering across 46 Indus canal commands. BenYishay et al. (2025) measures crop growth around rehabilitated Afghan canals. Remote sensing supplies the outcomes here.

3 A Model of Positional Rationing

The model takes one node of Arizona's layered allocation, a canal serving users in sequence, and abstracts from the rest. It shows how physical position, water scarcity, and institutional design interact to disadvantage downstream users. Arizona agriculture operates within a formal allocation regime whose layers govern different nodes between source and field. The layers include state-law rights and priority, federal and Central Arizona Project contracts, district entitlements, grandfathered groundwater rights, and local delivery rules.[5] The model assumes no farmer holds an observed annual volumetric entitlement, and the empirical work never observes one. Some districts deliver on rotational schedules and apply pro-rata curtailment protocols, equal proportional cuts, during shortage. Appendix S4 reports how many have adopted them.

The model formalizes one accounting identity, that the water available to a field depends on diversions upstream of it, never downstream. The water a downstream field loses because users above it divert more than their entitlements is the *wedge*. The identity signs five patterns. The first four are a head-to-tail level gradient, drought amplification of the wedge, a directional upstream-only structure, and canal-size scaling. Through comparative statics on its parameters, it also signs institutional weakening of the wedge. Written for the months of a season, it signs three more patterns (Section 3.3). Table 1 collects them.[6] The identity does the work. The behavioral story attached to it below, undetected over-diversion, is one illustrative polar mechanism. Rotation whose turns shrink or end early when the canal runs short, pro-rata rationing with undetected over-diversion on top of it, branch scheduling, conveyance loss, and directional district management generate the same residual-supply arithmetic and the same reduced-form signs. A shortage shared exactly equally generates no wedge. The analysis observes none of the leak, the deliveries, the shortfall, or the entitlement at any field. Appendix A1 works the pro-rata case through explicitly.

3.1 Position and Priority

Consider a sub-canal serving two farmers, one senior and one junior in rights. When the junior sits upstream, the downstream senior can invoke a "call on the river," so legal priority restrains upstream extraction.[7] Reverse the positions and the downstream junior has no physical control

[5] Prior appropriation ranks surface-water rights by historical seniority ("first in time, first in right") and sets the curtailment order among them during shortage (Arizona Department of Water Resources and Central Arizona Project 2023). It governs one layer. District distribution, canal capacity, and turnout scheduling are distinct operational objects. Roughly 72% of fields here match no recorded surface-water right (Section 4).

[6] The head-to-tail mechanism is well documented in the irrigation literature reviewed in Section 2 (Ostrom and Gardner 1993; Ray and Williams 2002; Akram and Mendelsohn 2017; Jacoby and Mansuri 2020). "Positional rationing" is a descriptive label for that mechanism; the paper's contribution is the field-month measure of upstream and downstream demand on the irrigation network, and the directional design built on it.

[7] A "call on the river" is rooted in the prior appropriation doctrine. Senior rights holders can demand that upstream junior appropriators curtail diversions until the senior allocation is met (The Nature Conservancy Arizona

over the flow and no legal recourse. If the upstream senior extracts their full allocation, or slightly more without detection, the junior receives only the residual. The undetected excess is the *leak*, a behavioral margin distinct from the physical seepage losses treated below. The wedge bites hardest when junior rights and a downstream position coincide. Priority reorders the entitlement queue, not the physics. The leak falls on whoever sits downstream in either configuration, because no legal instrument recovers water already diverted. The *cushion* is the gap between what would reach a farmer if no one over-diverted and that farmer's own entitlement. When water is abundant, the cushion absorbs upstream leaks invisibly. When supply tightens, the accumulated leak lands on whoever sits last in line. Where water is allotted per acre and delivered by order, the seasonal gradient can vanish. The gap then moves into the months a canal is at capacity (Section 3.3). I make the accounting explicit below for a canal serving N users.

3.2 Residual Supply and the Positional Wedge

Consider a single irrigation season t and a directed canal serving users $i = 1, \ldots, N$ ordered from head to tail. System supply S_t is exogenous, and drought corresponds to a low realization of S_t. Each user holds a formal entitlement w_i^* and is either senior or junior in rights, a distinction that matters for the priority-queue comparative static of Section 3.4. Each upstream diversion carries a non-negative leak $\varepsilon_{jt} \geq 0$: *undetected over-diversion*, extraction above entitlement that monitoring cannot detect, whether because meters have finite precision or because turnouts are unmetered. Monitoring intensity μ governs its scale: the leak is decreasing in μ, with $\varepsilon_{jt} \to 0$ as $\mu \to \infty$.[8] The empirical counterpart of μ varies widely across Arizona's districts, from metered Bureau of Reclamation turnouts to smaller systems that allocate per acre with no volumetric measurement at all (Appendix A1).[9] Applied water is not fully consumptive: a fraction $\rho \in [0, 1]$ returns to the hydrologic system, and $\lambda_{ij} \in [0, 1]$ is the share of user j's return flow recoverable at downstream user i.

The residual supply physically available at user i is

$$A_{it} \;=\; S_t \;-\; \sum_{j<i} \big(w_j^* + \varepsilon_{jt}\big) \;+\; \rho \sum_{j<i} \lambda_{ij} \big(w_j^* + \varepsilon_{jt}\big), \tag{1}$$

and delivery is $w_{it} = \big[\min\{w_i^*, A_{it}\}\big]_+$. Let A_{it}^0 denote residual supply under perfect monitoring ($\varepsilon_{jt} = 0$ for all j). The gap between the two is the *positional wedge*,

$$\Delta_{it} \;=\; A_{it}^0 - A_{it} \;=\; \sum_{j<i} \big(1 - \rho\lambda_{ij}\big)\, \varepsilon_{jt}, \tag{2}$$

the accumulated upstream leak net of return-flow recovery. The wedge harms user i only once it

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and Friends of the Verde River 2020; Feldman 2004).

[8] The leak is modeled as an exogenous shock rather than a strategic choice; its one-sidedness ($\varepsilon_{jt} \geq 0$) already encodes that users exploit, rather than fall short of, the detection margin. A fully strategic treatment of over-diversion would require equilibrium monitoring machinery without changing the sign predictions used here. Appendix A1 records this and the other parameterization choices.

[9] Delivery losses from evaporation and seepage are not modeled separately; they would enter equation (1) as a deterministic depletion term that compounds along the canal, reinforcing the head-to-tail gradient without changing any sign prediction.

exhausts the cushion $\ell_i(S_t) = A^0_{it} - w^*_i$: the realized shortfall is $q_{it} = \left[\Delta_{it} - \ell_i(S_t)\right]_+$.[10] Crop outcomes follow $Y_{it} = Y^0_i - \beta\, q_{it}/A^0_{it} + u_{it}$ with $\beta > 0$, where Y_{it} stands in for water use and the yield proxy, Y^0_i is field i's baseline, and u_{it} is mean-zero. The shortfall is scaled by the water that would have been available rather than by the entitlement, since the same acre-foot shortfall costs more when little water is on the canal. Normalizing by w^*_i would bound the ratio. It would also remove the second amplification channel noted below.

Four sign predictions follow. *First*, a head-to-tail **level gradient**: the wedge in equation (2) cumulates over upstream users $j < i$. Expected shortfall $E[q_{it}]$ rises toward the tail under maintained assumptions on return-flow recovery and the entitlement profile (Appendix A1). *Second*, **drought amplification**: a low draw of S_t lowers residual supply A^0_{it} and with it the cushion $\ell_i(S_t)$. The *same* wedge then binds more often and weighs more heavily against the water that remains. Holding participation and supplemental water fixed, upstream demand lowers a field's water use and yield proxy more in drought years than outside them. *Third*, **directional asymmetry**: the wedge sums only over upstream users $j < i$. Diversions by downstream users cannot enter field i's residual supply through the physical delivery channel. Section 6.3 tests it with position, upstream minus downstream demand at a fixed canal total. A shortage that the canal shares among all its fields, as when a canal cannot carry its peak, moves the canal total instead. The test bounds the physical delivery channel only. Factors that propagate in the same direction as water, such as branch releases or head-to-tail scheduling, remain possible. *Fourth*, **canal-size scaling**: $E[\Delta_{it}]$ grows with the number of upstream extractors, so the wedge should be larger in bigger canal systems.[11]

The timing of shortfalls matters for which institutions bite. When shortfalls are anticipated before delivery begins, prior appropriation operates as designed. Junior rights holders are curtailed first, regardless of position, and districts apply rotational schedules and pro-rata reductions. When shortfalls are realized *during* delivery, upstream users can maintain planned extraction, and the tail absorbs the residual shock. A delivery scheduled for ten hours that ends after nine leaves downstream users whatever remains of the final hour. Equation (1) formalizes that residual-claiming regime. Appendix A1 shows that the wedge and its predictions survive under faithful pro-rata rationing too. Residual claiming is a transparent polar case, not a necessary assumption. The distinction also seeds the enforcement asymmetry behind the seniority prediction. Section 3.3 takes the identity to the months of a season.

3.3 Within the Season

Water reaches fields month by month, and a canal falls short in particular months. Appendix S8 writes equation (1) for one month and works out four cases. They can hold together, and three of them sign patterns that the seasonal identity cannot.

(a) A canal limit that binds at the peak. Schedules follow crop need, so a canal reaches its capacity in the peak months. Below capacity, the district releases what the farms order, and no field goes short. At capacity, a farm upstream that takes more than its schedule leaves less for the

[10] Where the cushion is negative, q_{it} pools the leak-driven wedge with the shortfall entitlement claims alone would produce; both cumulate over upstream users $j < i$, and the empirical measures do not separate them (Appendix A1).

[11] The model signs the *level* of the wedge in the number of upstream extractors. The design of Section 6 does not test it.

fields below it in that month. The upstream slope is then negative only in months at capacity, and the downstream slope is zero. A cut in the canal's water brings more months to capacity, which is drought amplification within the season.

(b) Releases routed by orders. When the flow at a junction falls short of the orders below it and the district splits it by orders, more orders on a field's branch send more water down that branch. A field that is short takes some of it as it passes. Demand downstream of a field can then raise its water. The contrast between the two directions stays as in case (a).

(c) District wells along the canal. A district well that pumps into the canal between upstream land and a field, and runs below its capacity, replaces what that land diverts. A field with such a pump above it should lose less to upstream demand.

(d) Seasonal allotments per acre. Central Arizona and Maricopa–Stanfield charge each farm's deliveries against a seasonal allotment per acre (Section 8.5). A farm that takes more in one month takes less later. If every farm uses its whole allotment, the season's wedge sums to zero, and the fixed head-to-tail gradient can disappear. The monthly wedge remains.

Table 1 adds the patterns of cases (a) to (d): water lost in the months a canal falls short with no gradient over the season, a downstream slope that can be positive, and protection by a canal pump above a field. Section 8 tests cases (a), (b), and (d) together, and case (c) through the district wells of Maricopa–Stanfield.

3.4 Institutions in the Model

In the stylized model, two extreme mechanisms would remove the wedge. One is frictionless trade, in which downstream farmers buy allocations from upstream users who value the water less, the standard Coasian solution. The other is perfect detection of diversions beyond entitlement. Arizona meets both benchmarks only partly.

Markets remain thin. Volumes are low, and cross-district transfers occur at the district level instead of the individual level. Conservation-based trading often compensates below opportunity cost (Harris 2024). Rights historically operated under "use it or lose it" provisions, giving an incentive to extract the full entitlement even at low marginal value (Brocious 2021). The 2021 reform relaxing that rule covers only the sample's final three years. The same measurement margins that generate the leak also limit enforcement. Institutions therefore enter the model through its ingredients (the leak ε, the cushion ℓ_i, or the crop-relevant cost of the shortfall q_{it}).

- **Groundwater access** caps the crop cost of a surface-water shortfall. A farmer with a well replaces undelivered surface water by pumping at marginal cost c_g, bounding the shadow cost of q_{it}. A slower recharge channel, upstream return flows raising the recovery share λ_{ij}, works in the same direction with multi-year lags. Both push the crop-outcome wedge toward zero where groundwater is a feasible substitute. In Harquahala and New Magma, a field's canal shortfall lowers its water use where the field has no well. It does not where the field has one (Section 8.4). In Maricopa–Stanfield, district wells on the canals mark which fields keep water, and I cannot show that a field's own well does (Section 8.5).

- **Volumetric pricing** forces measurement. Charging per acre-foot requires metering deliveries, which raises monitoring intensity μ and shrinks the leak. It also puts a marginal cost on every unit diverted, counteracting "use it or lose it" incentives. Pricing that also reduces runoff and recharge lowers the recovery term $\rho\lambda_{ij}$. The net effect on the wedge is ambiguous in theory, and district records are too sparse to estimate it here.

- **Water rights seniority** operates through the priority queue. Where enforced, the queue fills senior entitlements before juniors receive residual delivery, pushing senior shortfalls toward zero ($q_{it}^{\text{sen}} \approx 0$). Juniors compete over a thinner residual pool. The queue reorders who absorbs the leak instead of removing it. Two assumptions qualify the queue. Enforcement is asymmetric, so realized mid-season shocks can reach seniors before curtailment operates. Seniors also must not sit systematically downstream of juniors. Appendix A1 formalizes the queue and records the Arizona enforcement mechanics. The binding queue runs at the river level, with districts as its users.

- **Irrigation district membership** bundles coordinated scheduling, maintenance, and administrative oversight. It could work through any of the three parameters, tightening μ, smoothing the cushion ℓ_i, or capping the crop cost of a shortfall. The model leaves the three channels unseparated, so district membership enters empirically instead of as a derived comparative static.

Together, the comparative statics predict the largest position effect in junior-rights districts with surface water. There, priority protects less and pumping need not replace the whole shortfall.

3.5 Predictions and Scope

Table 1 maps each prediction to an estimate. The estimates of Sections 7.1 and 7.2 are drought interactions. Section 8 tests the cases within the season, in the irrigation districts whose canal water is recorded by month. I estimate junior-rights districts and senior-rights districts separately and do not test the differences between them. Rights and policies are recorded for irrigation districts, not fields, so I cannot test the model's predictions about priority within a canal or about pricing. The design does not test its prediction about canal size either.

The model has four limits. It signs directions, not magnitudes, so the dollar figures (Section 7.2.3) lie outside it. It takes fallowing as given, and its regressor marks demand, not observed diversion. It has no spatial correlation of shocks; Section 6.1 handles it in the data. And its canal is a line. The irrigation network branches. Fields on sibling branches share the water of the trunk without lying upstream of each other. The design leaves such pairs out (Section 6.2), so upstream demand misses some competition, whose sign I leave open.

Table 1: Model Predictions and What the Data Show

Prediction	Predicted sign	Estimate	Where
Drought amplification	Upstream demand lowers water use more in drought	Upstream demand × drought, water use: −0.021 ($p = 0.014$)	Sec. 7.1, Table 3
Direction	Downstream demand does not lower it	Downstream demand × drought: +0.012 ($p = 0.048$). Position × drought: −0.017 ($p < 0.001$). This year's change in position × drought, yield proxy: −0.042 ($p = 0.002$)	Sec. 7.1, Table 3
Level gradient	Fields low on the canal lose more in drought	Fallowing on usual position × drought, junior-rights districts: +1.54 percentage points ($p < 0.001$). Yield proxy on usual position × drought, all fields: +0.001 ($p = 0.79$)	Sec. 7.2.2, Table 3
Seniority	Position matters less where rights are senior	Position × drought, water use: junior −0.020 ($p = 0.008$), senior +0.002 ($p = 0.85$). Canal total × drought, senior: −0.034 ($p = 0.015$)	Sec. 7.2.1, Fig. 5
Groundwater	A field with its own well loses less	Yield proxy, this year's change in position × drought: no own well −0.042 ($p = 0.003$), own well −0.023 ($p = 0.12$). Canal shortfall lowers water use only on fields without a well in Harquahala and New Magma	App. Table A8; Sec. 8.4
Within the season (cases a, d)	A field loses water in the months its canal falls short; no fixed gradient over a season of allotments	Maricopa–Stanfield, fields without a well, monthly shortfall from 0 to 1: water use −18.2 percent (5.2). Junior-rights planted fields, change in crop-season water use in drought, tail minus head third: −0.001 log points (95 percent interval −0.013 to 0.011)	Sec. 8.4; Sec. 7.2.5
Routing by orders (case b)	Downstream demand can raise a field's water	Downstream demand × drought, junior-rights districts: +0.018 ($p = 0.044$). Equal sharing at canal splits rejected (score −7.1, $p = 0.005$)	Sec. 7.2.1; Sec. 8.4
Canal pumps (case c)	A field with a district well above it on the canal loses less	Maricopa–Stanfield, fields without a well, slope on the pooled shortfall: 1.0 (10.3) with a canal pump within 2 km above, −27.6 (11.0) without; difference with the district's shortage by group 22.8 (14.5)	Sec. 8.5
Canal size	Larger wedge with more upstream users	Not tested	—

Notes: Estimates of Sections 7.1 and 7.2: 2SLS, 3 km, district-year drought, coefficients on drought terms. Water use and the yield proxy in log points per log point of demand; fallowing in percentage points per log point of usual position. Junior-rights and senior-rights districts are estimated separately; I do not test the differences between groups. Rows on cases (a) to (d): Section 3.3; the estimates of Section 8 average over all rules at a take of 0.70, with standard errors in parentheses. Appendices A1 and S8 give the derivations and the model objects behind each prediction.

4 Data

4.1 Field Panel and Geographic Setting

I assemble a field-level panel from nine data blocks, and the delivery model of Section 8 adds two more. The overview below lists each block, its main source, and its role. The supplement

documents each source, variable and construction step (Appendices S1–S6).

Data Inputs and Role in the Empirical Design

Data block	Main source	Role in the paper
Field panel	USDA Crop Sequence Boundaries	Defines 54,946 fields, 2016–2023.
Outcomes	OpenET and Sentinel-2	Measures monthly water use, fallowing, and the yield proxy (GCVI).
Crops	USDA Cropland Data Layer	Sets the crop-by-year effects and predicted demand.
Drought	SPEIbase	Measures drought (SPEI-24 and SPEI-3).
Canal infrastructure	NHDPlus and DEM	Builds the irrigation network (Section 5).
Water access	ADWR rights, groundwater rights, well registry, USGS withdrawals	Measures priority, groundwater access, wells, and water-source dependence.
Institutions	Irrigation-district records	Measures pricing, seniority, shortage protocols, rotation schedules, and district governance.
Soils	SSURGO	Describes field quality (salinity, clay, and other soil properties).
Geography	OpenStreetMap; protected-area, tribal, state-trust, and federal-land layers; community-water-system	Describes market access and land status.
Canal water	CAP delivery reports; Reclamation decree accounting; Gila Water Commissioner and San Carlos project records	Measures the water that enters each district's canals each month (Section 8.1).
Owners and wells	2025 CoreLogic tax roll; ADWR well registry; Maricopa–Stanfield well map; Pinal County ditch layer	Drops same-owner pairs; sets which fields have a well and where canal pumps stand (Section 8.1).

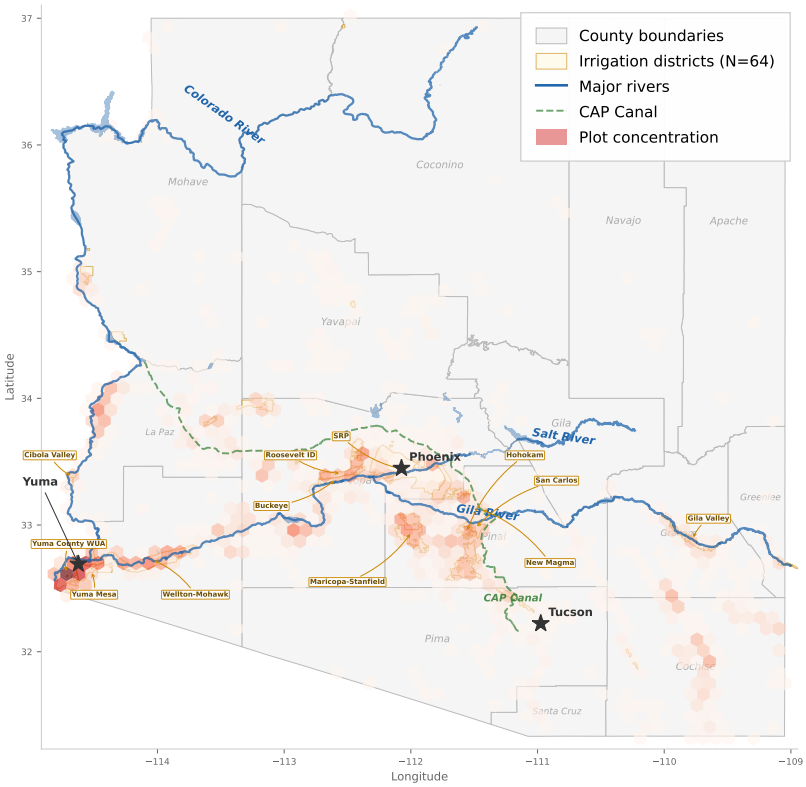
Note: A few further inputs feed controls within these blocks. District-institution variables draw on ADWR records and Central Arizona Project allocation documents alongside district records. Census and watershed (HUC12) boundaries enter only as base layers for mapping and spatial joins. The supplement gives the complete source list and construction steps.

The unit of observation is a field in a month for water use, and in a crop year for the yield proxy and fallowing. The fields come from the USDA Crop Sequence Boundaries, which define stable field boundaries from the satellite record. The same 54,946 Arizona fields are observed in every year of 2016–2023. The irrigation network links 28,427 of them to canals (Section 5). The estimation samples hold the fields with upstream or downstream fields beyond 3 km (Table 2).

The USDA Cropland Data Layer records each field's crop in each year. The crop sets the crop-by-year effects and each field's predicted demand, its acres times the typical water use of its crop (Section 6.3). A field is fallow in a crop year when the Cropland Data Layer records no crop or its monthly water use shows none (Section 4.2). About a quarter of the field-years in the fallowing sample are fallow (Table 2).

Section 5 describes the canal maps and how I join them into the irrigation network. Arizona's canal network saw no major construction or realignment during the sample period, so the network is fixed. What moves is the water its fields use, as fields change crops or go fallow and as the season and drought change their demand.

Figure 1 shows the setting. Irrigated agriculture concentrates along three surface-water corridors: the Colorado River and its diversion canals in the western border counties (Yuma, La Paz),



54,946 agricultural plots (2016–2023); 24,923 (45%) inside an irrigation district, 30,023 (55%) outside

Figure 1: Study area: Arizona agricultural fields and canal infrastructure. Field locations (dots) cluster along the Colorado River corridor (west), the Salt–Gila system (central), and the upper Gila Valley (east). Major irrigation districts and the Central Arizona Project aqueduct are shown.

the Salt–Gila system feeding Phoenix (Maricopa, Pinal), and the upper Gila Valley (Graham). The 540 km Central Arizona Project aqueduct connects the Colorado to central Arizona. Field density tracks the canal network closely. Surface-water infrastructure, not rainfall, determines where agriculture occurs here.

Section 5 describes the irrigation network built on these maps, and Section 6.2 measures each field's upstream and downstream demand on it.

4.2 Outcomes and Drought

All three outcomes come from satellite data. Water use is the field's monthly evapotranspiration from the OpenET ensemble, in millimetres, from January 2016. The models use its log. It measures the water the crop uses, not the water delivered. The yield proxy is GCVI, a chlorophyll index from Sentinel-2 imagery. I take its peak over the crop year, from November to September, and use its log. Satellite vegetation indices track harvested yield in ground data (Lobell et al. 2020). The yield-proxy coefficients are log changes in GCVI, not percent changes in yield. A field is fallow in a crop year when the Cropland Data Layer records no crop or its monthly water use shows none. A crop year starts in November. Appendix S2 gives the formulas and the windows.

Drought comes from the Standardized Precipitation Evapotranspiration Index (SPEI), which counts both rainfall and evaporative demand. I use the 24-month index, SPEI-24. It follows

reservoir storage and the reliability of surface water over several years, which is what sets canal deliveries. A 3-month index would mark short dry spells. I average each field's SPEI-24 over May–August, the months of peak irrigation demand. Section 6.2 turns it into one drought indicator for each irrigation district and year. Drought covers 19.9 percent of field-months in junior-rights districts and 10.4 percent in senior-rights districts (Table 2). SPEI marks the years when the canals run short. It does not measure a cut. Appendix A16 reports the CAP cuts, and Appendix S3 gives the construction.

4.3 Water Access, Rights, and Institutions

The rights groups of Section 7.2.1 rest on district classifications verified against district records. I code water source, priority, and mapped boundaries from each district's own documentation, not from proxies. The district set is fixed at the 30 districts accounting for 95.6 percent of service-supported member fields. I classify physical supply from district evidence as groundwater only, surface water only, mixed, or unresolved. A broader-system definition resolves six cases where a district belongs to a larger encompassing system as mixed, without imputing a groundwater share. The resulting comparison contains 3 groundwater-only, 5 surface-water-only, and 22 mixed systems.

I measure groundwater fallback access by grandfathered groundwater rights and the count of registered wells serving each field, an alternative supply during surface-water shortage. The well counts come from the ADWR Wells55 registry and measure predetermined infrastructure. Appendix A16 uses direct depth-to-water readings from the ADWR Groundwater Site Inventory, which measure aquifer conditions, not pumping volume. The water-right and well variables enter the yield and fallowing models interacted with drought (Section 6.4).

I classify priority within each district's own allocation system, not on a common 1–7 ladder. Of the 30 districts, 6 are senior, 10 junior, 5 mixed, 6 unknown, and 3 not applicable.[12] Pricing and governance records distinguish mechanisms documented at some point in 2016–2023 from mechanisms documented for the whole period. Unknown, conflicted, and period-ineligible entries remain missing. District boundaries define the inside/outside geography. The junior-rights and senior-rights groups of Section 7.2.1 hold the fields inside the junior and senior districts. An outside field carries a nearest-system label for spatial comparison without being coded as a member or customer.

Surface-water rights come from the Arizona Department of Water Resources. Roughly 72% of fields match no recorded right and are assigned the lowest possible seniority. The seniority control is a coarse proxy, not a precise legal ranking. State-law priority, district entitlement, CAP contract position, and grandfathered groundwater rights are distinct legal objects. None is a measured annual delivery entitlement for a given field. They enter as controls and in observational comparisons. Appendix S4 details the rights-matching procedure, the groundwater and well registries, and the

[12]Within a district, the district usually holds surface priority, not individual farms. Among the 30 audited districts, systems where one district-held priority covers all member farms' surface deliveries contain 77.8% of audited fields. Reclamation, CAP, district, and decree records verify 63.9%, and 14.0% are inferred. Farm-level priority differences, at least on paper, are documented in nine districts with 15.8% of fields. They include the Salt River Project (Kent Decree land classes), San Carlos, and five Upper Gila canal companies (Globe Equity decree schedules). They also include Mohave Valley (individual present perfected rights) and Buckeye (the Benson–Allison decree). The remaining 6.4% are in three groundwater-only systems and one single-farm entity. Where a rate schedule was located, the district bills members under one schedule. The Salt River Project, Mohave Valley, and Unit B instead use differentiated charge classes, and many retail schedules were not located.

Table 2: Three Samples, All Fields and by Rights Group

	All fields		Junior rights		Senior rights	
	Mean	(SD)	Mean	(SD)	Mean	(SD)
A. Water use, field-months						
Field-months	1,786,219		418,362		724,565	
Fields	23,545		5,803		8,524	
Irrigation districts	42		11		7	
Drought	0.133	(0.339)	0.199	(0.399)	0.104	(0.305)
Water use (mm per month)	83.9	(54.4)	81.7	(56.7)	80.4	(50.9)
Upstream demand (log), side present	4.360	(1.781)	4.723	(1.472)	3.929	(1.640)
Downstream demand (log), side present	4.808	(2.024)	4.912	(1.824)	4.249	(2.028)
No upstream fields beyond 3 km	0.134	(0.341)	0.116	(0.321)	0.133	(0.340)
B. Yield proxy, planted field-years						
Field-years	92,926		21,128		39,157	
Fields	22,609		5,569		8,454	
Drought	0.178	(0.383)	0.241	(0.428)	0.142	(0.349)
ln(GCVI), crop-year peak	2.001	(0.496)	2.050	(0.441)	2.012	(0.494)
Usual position	3.252	(4.884)	2.942	(5.384)	3.781	(4.297)
This year's change in position	0.148	(1.177)	−0.263	(1.185)	0.424	(1.322)
Usual canal total	8.553	(4.555)	10.399	(3.873)	6.896	(3.949)
This year's change in canal total	0.034	(1.210)	−0.432	(1.237)	0.461	(1.296)
Field acres	23.4	(39.0)	31.5	(43.6)	14.2	(14.6)
C. Fallowing, field-years						
Field-years	123,530		31,050		43,245	
Fields	24,706		6,210		8,649	
Drought	0.187	(0.390)	0.222	(0.416)	0.141	(0.348)
Fallow	0.248	(0.432)	0.320	(0.466)	0.095	(0.293)
Usual position	3.224	(4.921)	2.943	(5.363)	3.746	(4.337)
Field acres	21.8	(36.0)	28.3	(40.1)	13.8	(14.3)

Notes: Samples of equations (4), (5) and (6): fields with upstream or downstream fields beyond 3 km on the irrigation network. Junior and senior rights: fields in junior-rights and senior-rights districts. Drought: share of field-months or field-years in irrigation districts in drought. Water use: monthly evapotranspiration, OpenET ensemble. Upstream and downstream demand: log of the water used by the fields on each side in the month, over field-months with fields on that side. Usual position and canal total: the field's 2017–2018 mean, in log points. This year's change: the crop year's value minus the usual one. Fallow: no crop in the field's monthly water use.

district-level construction of each institutional variable. Appendix S5 documents the market-access controls.

4.4 Summary Statistics

Table 2 describes the three samples of Sections 7.1 and 7.2, for all fields and for the junior-rights and senior-rights districts. The water sample has 1,786,219 field-months of 23,545 fields in 42 irrigation districts, 2016–2023. The yield sample has 92,926 planted field-years of 22,609 fields in crop years 2019–2023. The fallowing sample has 123,530 field-years of 24,706 fields in the same years. Fields in junior-rights districts are larger than fields in senior-rights districts, 31.5 against 14.2 acres in the yield sample. They are left fallow far more often, in 32.0 against 9.5 percent of field-years. In 13.4 percent of field-months a field has no upstream fields beyond 3 km. It then enters with zero upstream demand and its own indicator (Section 6.2).

5 The Irrigation Network

The description of the irrigation network is being updated and will appear here in the next version.

6 Identification Strategy

6.1 Neighbours Fallow Together

Fields near each other go fallow together. They share weather, pests, soils and markets. Figure 2 shows the pattern in the fallowing sample. Between a field and the fields within 500 m, the correlation of fallow status is 0.70, averaged over crop years 2019–2023. It falls with distance, to 0.35 at 3–5 km and 0.20 at 30–50 km, while shuffled fallow status gives a correlation near zero. The pattern is not one owner's decisions. With the field's own owner's fields left out, the correlation is 0.65 within 500 m and almost the same as before from 2 km on. When a nearby upstream field goes fallow, upstream demand falls, and the shock that made it go fallow may also hit the field itself. Demand from nearby fields would then mix competition for water with shared local shocks. The design answers in three ways. It counts only upstream and downstream fields more than 3 km away (Section 6.2; Appendix A4 reports cutoffs from 2 to 10 km). The yield model controls for the fallow shares within 3 km. And it compares upstream with downstream demand (Section 6.3). A local shock reaches fields on both sides of a field, but water flows only one way. Beyond 3 km the correlation falls but does not vanish, so the last comparison matters most.

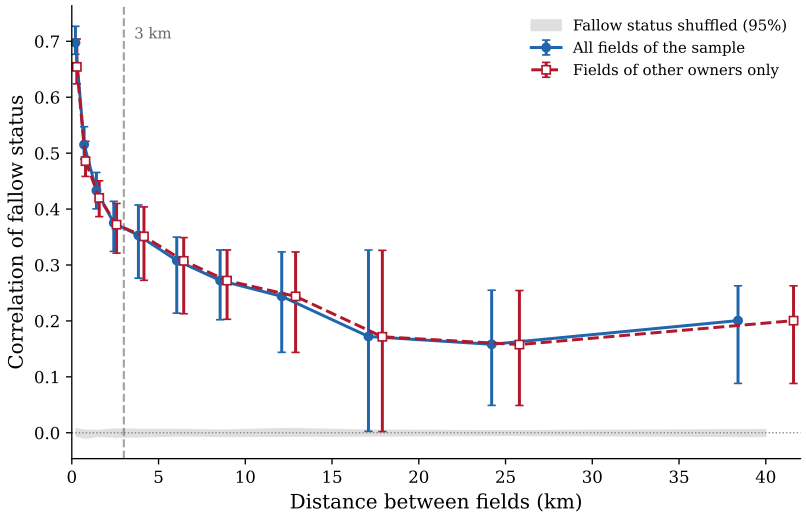


Figure 2: Fields Near Each Other Go Fallow Together

Notes: Correlation between a field's fallow status and the share of fallow fields at each distance from it, in the fallowing sample (Table 2, panel C: 123,530 field-years of 24,706 fields, crop years 2019–2023). A field-year is fallow when the crop maps record no crop or its monthly water use shows none. Solid: the neighbours are all fields of the sample in that crop year. Dashed: the same, with the field's own owner's fields left out, as the design leaves out same-owner pairs; a field without a matched owner is its own owner. Each point is the mean of the five crop years' correlations, and bars run from the lowest to the highest year; the two versions are set slightly apart. Distance bands: 0–0.5, 0.5–1, 1–2, 2–3, 3–5, 5–7.5, 7.5–10, 10–15, 15–20, 20–30 and 30–50 km, plotted at their midpoints; a field enters a band with at least three fields in it. Shaded: the 95 percent range of the same mean with fallow status shuffled across the year's fields, 20 draws. Vertical line: the 3 km cutoff of Section 6.2.

6.2 Upstream and Downstream Demand on the Irrigation Network

Sections 7.1 and 7.2 ask whether, in drought, a field gets less water when the fields upstream of it use more. To answer, I need to know which fields lie upstream and downstream of each field and how much water they use. I also need a change in that use that the field's own conditions do not drive.

The irrigation network. I trace the canals from the water source to every field, so each field has one route. A field's *upstream fields* lie on its route before it. Its *downstream fields* are the fields whose route passes it. I leave out fields on other branches, because the water of one never passes the other. I keep only fields of the same irrigation district, because I treat each irrigation district as a closed system. Arizona distributes water the same way. The state allocates it to irrigation districts, and each irrigation district divides its share among its own fields. An irrigation district here is whoever delivers the water, so a reservation or a private system counts as one. I also drop pairs of fields with the same owner in 2025 parcel records. An owner can move water between their own fields and makes their other decisions for all of them together. Upstream demand would then partly measure the owner's own choices. Same-owner pairs are 5.3 percent of all pairs. Owners are matched for 81 percent of fields, so some same-owner pairs remain. Section 5 describes the irrigation network.

Upstream and downstream demand. Upstream demand is the log of the water that a field's upstream fields use in the month. Each upstream field adds its acres times its water use per acre. Water use comes from the OpenET ensemble. Only fields more than 3 km away count. For field i in month m of year t,

$$U_{itm} = \ln\Big(\sum_{j \in \mathrm{up}(i),\ d_{ij} > 3\,\mathrm{km}} \mathrm{acres}_j \, w_{jtm} \Big), \tag{3}$$

where w_{jtm} is field j's water use per acre. Downstream demand D_{itm} is the same log over downstream fields. A side with no use in the month, or no fields beyond 3 km, enters as zero with its own indicator. I leave out nearby fields because they share weather and pests. They also share markets, and their cropping moves together (Section 6.1). Appendix A4 reports cutoffs from 2 to 10 km.

Position and canal total. I write the two sides as the field's *position*, $P_{itm} = U_{itm} - D_{itm}$, and the *canal total*, $T_{itm} = U_{itm} + D_{itm}$. Position is high when upstream fields use much water and downstream fields use little, as at the tail of a busy canal. The canal total is high when the whole canal draws much water. A model written with U and D and one written with P and T give the same estimates (Appendix Table A7).

Drought. An irrigation district is in drought in a crop year when the mean of its fields' May–August SPEI-24 (Section 4.2) is below -0.6. The measure is the same for every field of an irrigation district in a year. Drought only marks the years when the canals run short. Farmers learn the year's cut in advance. In winter, the irrigation district sends each farm a letter with its limit, and the farmer then decides what to plant and what to leave fallow.

6.3 Downstream Demand as the Test of Direction

Downstream demand tests the direction. Water flows one way, so upstream fields can take water before a field and downstream fields cannot. If drought lowers a field's water use because upstream fields take the water first, upstream demand lowers it and downstream demand does not. Drought can instead lower water use wherever the canal draws more. A cut shared across the canal does so, and so does a canal too small for its peak. Demand on either side then lowers a field's water use alike. The position coefficient is half the difference between the coefficients on the two sides, so it is zero in the second case. Both sides enter one regression, so the test compares them on the same fields and months.

The instrument. The water use of upstream and downstream fields responds to the same month's heat and the same delivery cuts as the field's own. Ordinary least squares mixes "the fields upstream took the water" with "everyone got less." I instrument P, T, and their drought terms with the same terms built from *predicted demand*. A field's predicted demand is its acres times the typical water use of its crop in its county and month in other years. It is zero when the field is fallow that year. Predicted demand leaves out the current year's water use. It moves only when fields change crops or go fallow. The instrument assumes that planting, decided before the season, does not respond to the shortage of a particular month. The comparison with ordinary least squares in Section 7.1 shows where the instrument matters.

6.4 Three Outcomes, Decided at Different Times

Each outcome is decided at a different time, so each model compares what can change then. Water is used month by month, so the water model compares months within a field's year. A crop grows over its whole season, so the yield model compares years. Planting is decided before the season, so the fallowing model uses where a field usually sits on its canal, which a farmer knows then.

Water use. The water model is

$$\begin{aligned} y_{itm} = b_P \left(P_{itm} \times \mathrm{Dry}_{g(i)t}\right) + b_T \left(T_{itm} \times \mathrm{Dry}_{g(i)t}\right) + a_P\, P_{itm} + a_T\, T_{itm} \\ + \alpha_{im} + \mu_{it} + \gamma_{c(i)tm} + \delta_{k(i)tm} + \lambda_{g(i),\mathrm{Dry},m} + e_{itm}, \end{aligned} \tag{4}$$

where y_{itm} is the log of field i's water use in the month, over all months of 2016–2023. The term $\mathrm{Dry}_{g(i)t}$ marks drought in the field's irrigation district $g(i)$. The effects are field by calendar month, field by crop year, crop by year by month, and county by year by month. The model also includes irrigation district by drought by month effects. Field by crop year effects absorb the field's level of water use in each year. They also absorb its crop, drought, weather, nearby fallowing, rights, and wells. So the model compares months within a year. It compares months when the upstream fields use more water, relative to the downstream fields, with months when they use less. The smallest first-stage F is 2,016. Standard errors are clustered by field and by crop by year by month.

Yield proxy. The yield model splits position into the usual position and this year's change in position. The yield proxy is GCVI, the satellite index of Section 4.2. It is measured once per crop year, so the yield model is annual. The model uses the log of GCVI at its crop-year peak, on planted field-years of crop years 2019–2023. I sum position and canal total over the crop season and split each in two. The *usual position* H_i^P is the field's mean position in crop years 2017 and

2018, before the sample.[13] *This year's change in position* is $W^P_{it} = P_{it} - H^P_i$. The terms H^T_i and W^T_{it} split the canal total the same way. A field low on its canal has a high usual position. This year's change in position is positive when the field's upstream neighbours planted more than in 2017–2018. It is also positive when the field's downstream neighbours left more land fallow. The model is

$$\begin{aligned} y_{it} = {} & \beta_W \left(W^P_{it} \times \mathrm{Dry}_{g(i)t}\right) + \beta_H \left(H^P_i \times \mathrm{Dry}_{g(i)t}\right) + \theta_W \left(W^T_{it} \times \mathrm{Dry}_{g(i)t}\right) + \theta_H \left(H^T_i \times \mathrm{Dry}_{g(i)t}\right) \\ & + c_P\, W^P_{it} + c_T\, W^T_{it} + X'_{it}\pi + \alpha_i + \gamma_{c(i)t} + \lambda_{g(i),\mathrm{Dry}} + e_{it}, \end{aligned} \tag{5}$$

with field, crop by year, and irrigation district by drought effects. The controls X_{it} hold the paper's control set and the fallow shares within 3 km. They also hold the water-right and well variables interacted with drought. Field effects absorb H itself. Predicted demand, split the same way, instruments the four drought terms and the two changes. The smallest first-stage F is 1,943. Standard errors are clustered by field and by crop by year.

Fallowing. The fallowing model uses only the usual levels. A field is fallow in a crop year when its monthly water use shows no crop. The model is a linear probability model on the usual position and the usual canal total, each interacted with drought. It has field, year, and irrigation district by drought effects and the same controls:

$$F_{it} = \psi_H \left(H^P_i \times \mathrm{Dry}_{g(i)t}\right) + \psi_T \left(H^T_i \times \mathrm{Dry}_{g(i)t}\right) + X'_{it}\pi + \alpha_i + \tau_t + \lambda_{g(i),\mathrm{Dry}} + e_{it}. \tag{6}$$

I leave out this year's changes in position and canal total, because neighbours decide the year's planting at the same time as the field. Year effects replace crop by year effects, since a fallow field has no crop. I also leave out the nearby fallow shares, since they measure fallowing too. Usual levels of predicted demand instrument the two terms. Standard errors are clustered by field and by irrigation district by year.

Every coefficient I report is on a drought term. It measures how much a slope shifts in drought years. The units are log points of the outcome per log point of demand. For fallowing, they are percentage points per log point of demand.

7 Results

In a drought, the canals of a junior-rights district can carry less water than their fields order, and someone goes without. The results show who. Fields lose water when the fields upstream of them use more, and not when the fields downstream do. The loss shows in two ways. Before the season, fields that usually sit low on junior canals are left fallow more often. During the season, a planted field grows less when its upstream neighbours use more water than usual. Position matters in junior-rights districts. In senior-rights districts, fields lose water when the whole canal draws more. Position matters when the canals run short, and not in the drier years of 2008–2015, when CAP deliveries were not cut.

Section 6.2 builds the measures, and Section 6.4 builds the models. I follow each canal from its source to every field. A field's *upstream demand* is the water used by the fields between it and

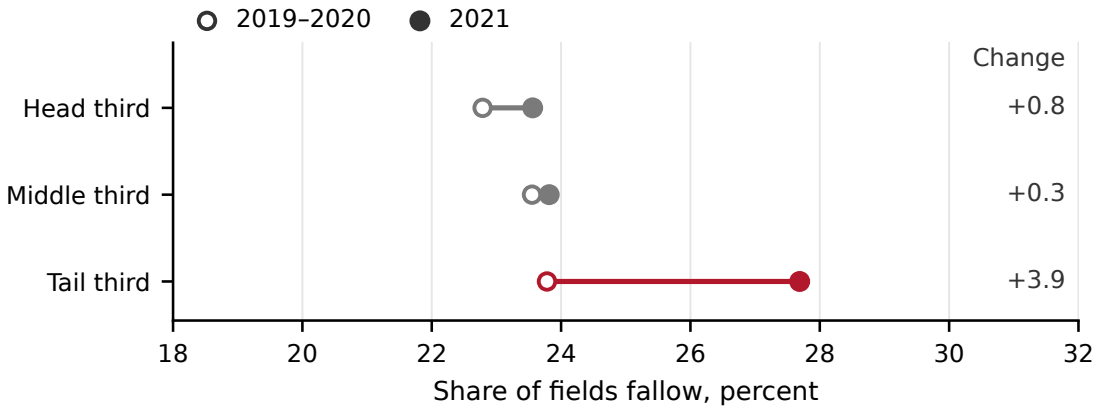
[13] Crop year 2016 has no crop-season position. A crop year starts in November, and the monthly water-use data start in January 2016.

the source. Its *downstream demand* is the water used by the fields below it. Both count only fields more than 3 km away. Its *position* is upstream minus downstream demand, and the *canal total* is their sum. I measure drought for each irrigation district. Predicted demand instruments the water that upstream and downstream fields use. It moves only when fields change crops or go fallow. I measure water by month and the yield proxy by season. I relate fallowing to the usual position, the field's mean position in 2017–2018, which a farmer knows before planting.

7.1 In Drought, Fields Lose Water to the Fields Upstream of Them

A first look. The raw data already show that in 2021, fields at the tail of junior canals were left fallow more often. If a junior canal runs short in drought, its tail should run short first. I start without a model, in the junior-rights districts, which lose surface water first. In each irrigation district, I rank fields by usual position and split them into thirds. Figure 3 compares the share left fallow in 2019–2020 with 2021, when 97 percent of junior fields were in irrigation districts in drought. The share rose 0.8 percentage points in the head third, from 22.8 to 23.6 percent. It rose 3.9 points in the tail third, from 23.8 to 27.7 percent. The gap of 3.1 points has a 95 percent interval of 1.1 to 5.2 points, from a bootstrap over fields. The question is whether the water itself follows the direction of the canal.

Figure 3: In the 2021 Drought, Tail Fields of Junior-Rights Canals Fallowed More



Notes: Share of fields left fallow, by third of usual position within each junior-rights district: open circles 2019–2020, filled circles 2021. Usual position: the field's 2017–2018 mean of upstream minus downstream demand on the irrigation network, beyond 3 km. Head third: least upstream demand relative to downstream; tail third: most. 6,210 fields with upstream or downstream fields beyond 3 km. Raw shares, no controls.

Water use. In a drought month, upstream demand lowers a field's water use and downstream demand does not. A short canal could share water in two ways. If it shared equally, water used anywhere on the canal would lower a field's use alike. If fields took water in the order it flows, only water used upstream would. The monthly water model tells the two apart (Table 3, Panel A, and Figure 4(a)). Upstream demand lowers a field's water use (-0.021 log points per log point, $p = 0.014$). Downstream demand raises it ($+0.012$, $p = 0.048$). Downstream fields cannot take water first. The position coefficient compares the two sides at a fixed canal total. It is -0.017 (standard error 0.005), on 1,786,219 field-months of 23,545 fields. If upstream fields use 10 percent more and downstream fields 10 percent less, water use in a drought month falls by about 0.3 percent.

The instrument changes the canal total and leaves position alone. The water use of upstream and downstream fields responds to the same heat and cuts as the field's own. With ordinary least squares, the canal total is -0.029 ($p < 0.001$) and position is -0.022. With the instrument, the canal total shrinks to -0.004 ($p = 0.47$) and position is -0.017. Shared shocks lower everyone's use together. They do not sort fields by the direction of the canal. Ordinary least squares puts them on the canal total.

The position effect holds at each of seven cutoffs from 2 to 10 km, between -0.013 and -0.035, each with $p < 0.01$ (Appendix Figure A1).

Table 3: Position on the Canal and Drought: Water Use, Yield Proxy, and Fallowing

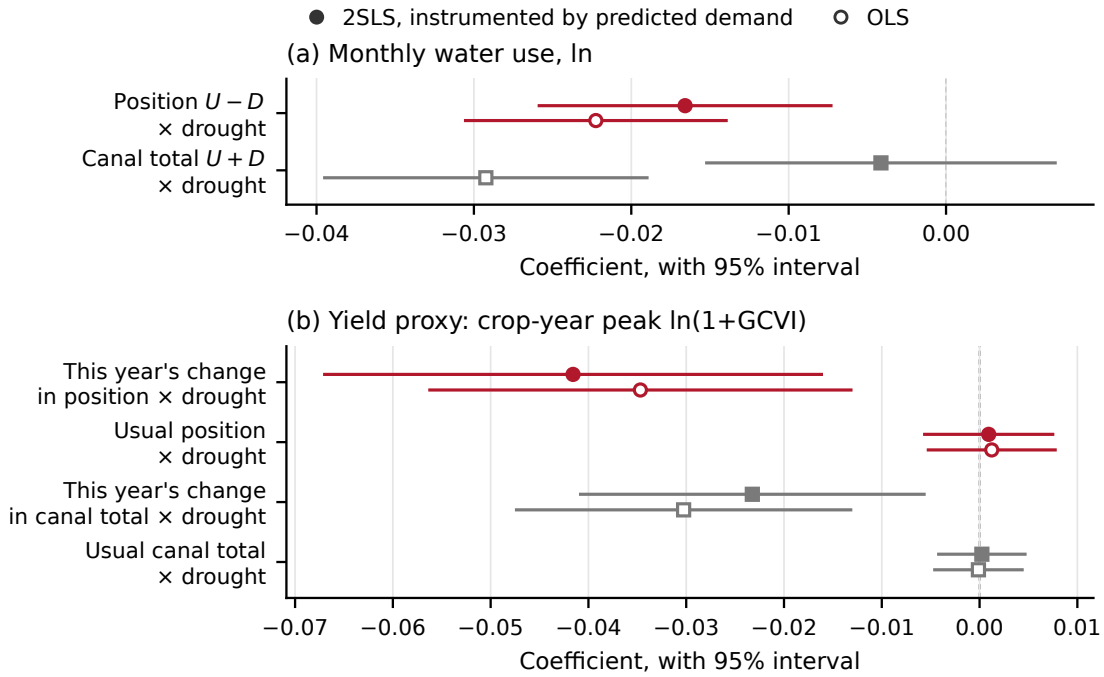
	(1) All fields	(2) Junior rights	(3) Senior rights
Panel A. Monthly water use, ln			
Position $U - D$ × drought	-0.017^{***}	-0.020^{***}	0.002
	(0.005)	(0.007)	(0.009)
Canal total $U + D$ × drought	-0.004	-0.002	-0.034^{**}
	(0.006)	(0.008)	(0.014)
Field-months	1,786,219	418,362	724,565
Fields	23,545	5,803	8,524
Smallest first-stage F	2,016	1,471	458
Panel B. Yield proxy, ln(1+GCVI) at its crop-year peak			
This year's change in position × drought	-0.042^{***}	-0.031^{**}	-0.034
	(0.013)	(0.016)	(0.027)
Usual position × drought	0.001	-0.009^{**}	0.001
	(0.003)	(0.004)	(0.004)
This year's change in canal total × drought	-0.023^{**}	0.004	-0.015
	(0.009)	(0.016)	(0.030)
Usual canal total × drought	0.000	-0.004	0.005
	(0.002)	(0.006)	(0.007)
Field-years	92,926	21,128	39,157
Fields	22,609	5,569	8,454
Smallest first-stage F	1,943	4,307	537
Panel C. Fallowed, percentage points			
Usual position × drought	0.48	1.54^{***}	-0.15
	(0.30)	(0.42)	(0.58)
Usual canal total × drought	0.45	0.44	0.75^{*}
	(0.45)	(0.49)	(0.44)
Field-years	123,530	31,050	43,245
Fields	24,706	6,210	8,649
Smallest first-stage F	146,347	59,598	109,030

Notes: 2SLS; the drought terms of position and canal total are instrumented by the same terms of predicted demand. Upstream and downstream fields beyond 3 km on the irrigation network, same-owner pairs removed; a side without such fields counts as zero demand. Drought: the irrigation district's May–August SPEI-24 below -0.6. Junior and senior rights: fields inside the boundaries of irrigation districts with junior and senior rights, each estimated separately. Panel A: equation (4), all months 2016–2023; clustered by field and by crop by year by month. Panel B: equation (5), crop years 2019–2023; usual level: 2017–2018 mean; clustered by field and by crop by year. Panel C: equation (6), crop years 2019–2023, coefficients times 100; clustered by field and by irrigation district by year. Smallest first-stage F: the smallest cluster-robust Wald F of the excluded instruments across the instrumented terms. Standard errors in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Yield proxy. The lost water shows in the crop through this year's change in position. In drought, a planted field has a lower yield proxy when its upstream neighbours planted more this year than in 2017–2018. It also has a lower yield proxy when its downstream neighbours left more land fallow (Table 3, Panel B, and Figure 4(b)). This year's change in position has a drought coefficient of -0.042 (standard error 0.013, $p = 0.002$), on 92,926 field-years of 22,609 fields. In drought, one log point more upstream demand relative to downstream demand lowers the yield proxy by 0.042 log points. Over all fields, the usual position does not lower the yield proxy ($+0.001$, $p = 0.79$). A

field low on its canal every year grows as well in drought as a head field, relative to its own usual level. Its owner knows where it sits and plans for it. The change in canal total also lowers the yield proxy (-0.023, $p = 0.011$). The usual total does not ($p = 0.92$). The yield model compares years, so a canal that draws more all season shows up here, unlike in the water model.

Figure 4: In Drought, Upstream Demand Lowers a Field's Water Use, and This Year's Change in Position Lowers Its Yield Proxy



Notes: All fields with upstream or downstream fields beyond 3 km. Panel (a): equation (4), 1,786,219 field-months of 23,545 fields. Panel (b): equation (5), 92,926 field-years of 22,609 fields. Filled markers: 2SLS, instrumented by predicted demand; open markers: ordinary least squares on the same sample. Red: position $U - D$; grey: canal total $U + D$. Bars are 95 percent intervals with the clustering of Table 3.

The yield-proxy result holds over fewer cutoffs than the water result. From 2 to 5 km, this year's change in position is between -0.027 and -0.042, with $p < 0.05$. From 6 km on, the estimates are similar, -0.022 to -0.037. They are imprecise (p from 0.12 to 0.19; Appendix Table A6).

A field with an irrigation well of its own can pump when the canal falls short, so it should lose less. On fields without one, this year's change in position gives -0.042 ($p = 0.003$, 21,336 fields). On the 1,273 fields with one, it gives -0.023 ($p = 0.12$; Appendix Table A8).

7.2 Junior Rights, Fallowing, and Delivery Shortages

Water law decides which irrigation districts lose surface water first. Position then decides which fields inside them go short.

7.2.1 Position Decides Water in Junior-Rights Districts, and the Canal Total in Senior Ones

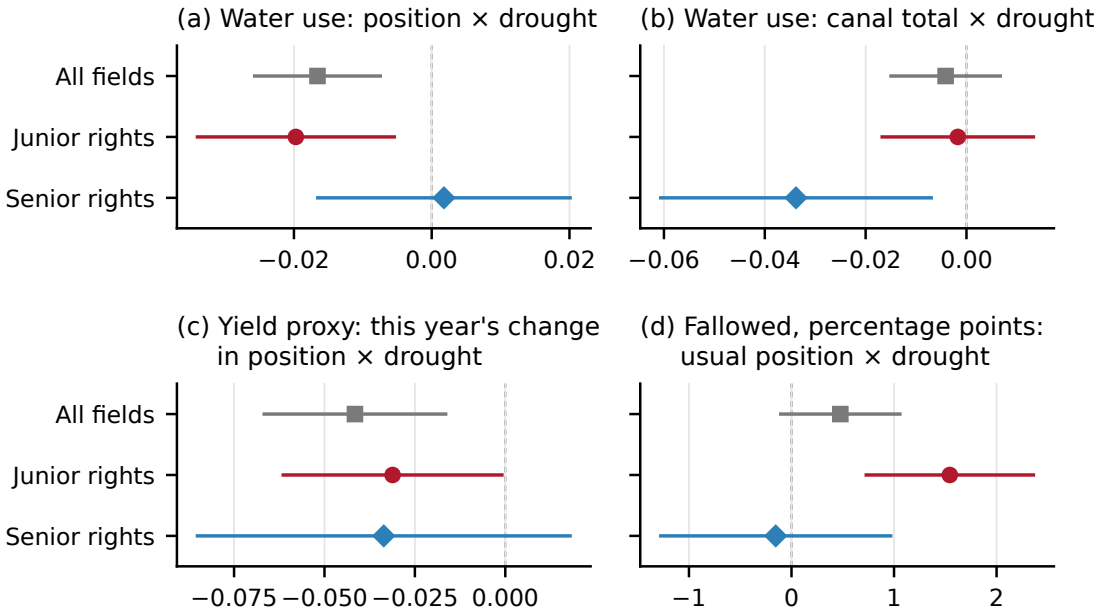
Almost all junior-rights fields lie inside nine irrigation districts. Some are the CAP districts Central Arizona, Maricopa–Stanfield, and Hohokam. Another is Cibola Valley on the Colorado River. The rest are the Upper Gila canals of Fort Thomas, Graham, Montezuma, San Jose, and Union. The

senior-rights fields lie inside Wellton-Mohawk, Yuma County Water Users, Yuma Mesa, Yuma Irrigation, North Gila Valley, and Unit B. I estimate each group separately, with all the effects of the main models. Columns (2) and (3) of Table 3 and Figure 5 report the estimates.

Junior and senior fields lose water in different ways. In junior-rights districts, upstream demand lowers a field's water use in drought (-0.021, $p = 0.083$). Downstream demand raises it ($+0.018$, $p = 0.044$). The position coefficient is -0.020 ($p = 0.008$, 5,803 fields), and the canal total is near zero (-0.002). In senior-rights districts, demand on either side lowers a field's water use in drought. Upstream demand lowers it by -0.032 ($p = 0.076$), and downstream demand by -0.036 ($p = 0.021$). The canal total is -0.034 ($p = 0.015$, 8,524 fields), and the position coefficient is near zero ($+0.002$, $p = 0.85$). A junior field loses water to the fields upstream of it. A senior field loses water when the whole canal draws more, wherever the demand sits. A limit on what the canal carries at its peak would produce that pattern.

In the yield proxy, this year's change in position gives -0.031 in junior-rights districts ($p = 0.049$). In senior-rights districts, the estimate is as large, -0.034, but imprecise ($p = 0.21$). On junior fields without an irrigation well of their own, it gives -0.034 ($p = 0.045$; Appendix Table A8). In junior-rights districts, the usual position also lowers the yield proxy a little, by -0.009 ($p = 0.036$).

Figure 5: Position Matters in Junior-Rights Districts; the Canal Total in Senior Ones



Notes: 2SLS estimates of Table 3, each group estimated separately. Panels (a) and (b): equation (4). Panel (c): equation (5). Panel (d): equation (6), in percentage points per log point of usual position. Bars are 95 percent intervals.

7.2.2 Fields That Usually Sit Low on Junior Canals Are Left Fallow

Fields that usually sit low on junior canals are left fallow in drought. A farmer decides before the season which fields to plant (Appendix S9). Take a field that usually sits low on a junior canal. The farmer can expect it to run short when its irrigation district is in drought. If the canal will not finish a crop, the farmer can leave the field fallow. In equation (6), one log point more of usual position raises a junior field's chance of fallowing in drought by 1.54 percentage points ($p < 0.001$,

6,210 fields; Table 3, Panel C). Ordinary least squares gives 1.57. In senior-rights districts the coefficient is −0.15 ($p = 0.79$). Over all fields it is 0.48 ($p = 0.12$). The usual canal total does not predict fallowing at the 5 percent level in any group.

The fallowing result comes from 2021. Relative to 2019, the junior fallowing slope on usual position is 1.48 percentage points steeper in the 2021 drought ($p = 0.013$; Figure 6(c)). In 2020 the difference from 2019 is near zero (+0.37). It reverses to −0.44 in 2022 ($p = 0.39$) and to −1.23 in 2023 ($p = 0.029$). Section 7.2.4 returns to 2022 and 2023.

Position works on two margins. The usual position decides which fields are planted. This year's change in position decides how the planted fields grow. A field that usually sits low on its canal shows its loss mostly in fallowing. A field whose upstream neighbours planted more this year shows its loss in its yield proxy.

7.2.3 Junior Fields Lose Mostly Through Fallowing

I put the two margins in dollars. Table 4 reports both losses over the drought years of 2019–2023. I apply the yield-proxy estimate $\hat{\beta}_W$ to planted drought field-years whose change in position this year, W_{it}^P, is positive. Each such field-year loses its crop revenue times $0.8\,(1 - e^{\hat{\beta}_W W_{it}^P})$. I assume that revenue falls by 0.8 times the proportional fall in the yield proxy. I apply the fallowing estimate to fields lower on their canals than a field at the 25th percentile of usual position. A fallowed acre saves its variable costs, so I value it at its net return, \$192, the mean of the cotton and alfalfa budgets. I also value it at \$1,886, the mean crop revenue, an upper bound. A planted field has already spent its costs, so both totals count its yield loss at revenue.

Position costs junior fields mostly through fallowing. It costs them \$3.4 million at net return over the drought years (95 percent range \$1.9–4.9 million). It costs them up to \$26.0 million at revenue (\$13.0–39.6 million). Fallowing is 76 to 97 percent of the total. The fallowing loss comes from 13,364 extra fallowed acres, 12,399 of them in 2021. The yield loss is \$0.8 million. Over all fields, the total is \$10.2 million at net return (\$4.6–15.8 million), mostly yield loss. Senior fields show no clear loss in either margin.

The dollar figures count what fields lower on their canals lose, so they measure who carries the shortage.

Table 4: Losses from Position in Drought: Yield and Fallowing Together

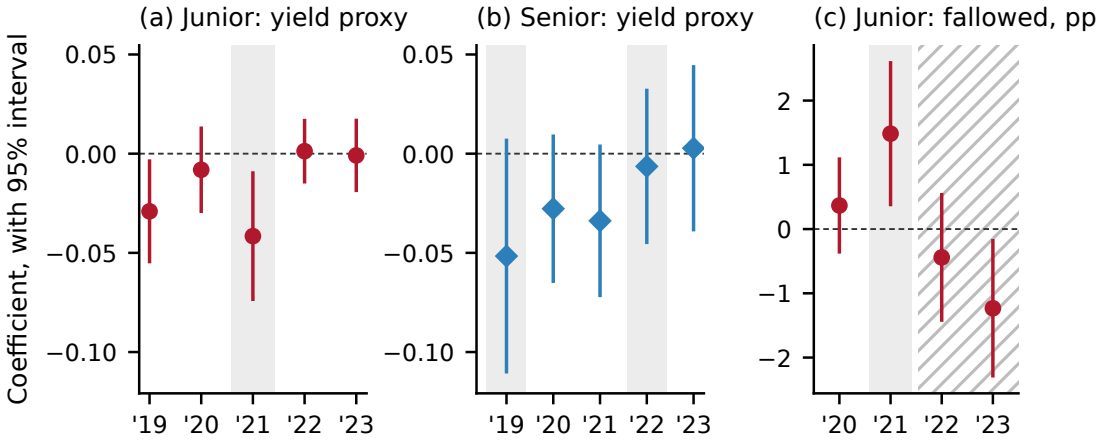
	Yield	Extra	Fallowed acre at net return (\$192)		Fallowed acre at revenue (\$1,886)	
	loss	fallowed acres	Fallowing	Total	Fallowing	Total
All fields	8.3	9,822	1.9	10.2	18.5	26.8
	[3.3, 13.1]	[−2,853, 22,371]	[−0.5, 4.3]	[4.6, 15.8]	[−5.4, 42.2]	[2.8, 51.3]
Junior rights	0.8	13,364	2.6	3.4	25.2	26.0
	[0.0, 1.6]	[6,337, 20,394]	[1.2, 3.9]	[1.9, 4.9]	[12.0, 38.5]	[13.0, 39.6]
Senior rights	11.1	−430	−0.1	11.0	−0.8	10.3
	[−7.0, 26.5]	[−3,597, 2,806]	[−0.7, 0.5]	[−7.1, 26.4]	[−6.8, 5.3]	[−8.5, 26.9]

Notes: Million dollars over all drought field-years of 2019–2023 (drought in 2019, 2021, and 2022 for some irrigation districts; 2021 and 2022 for the junior group), except the acres column. Yield loss: planted drought field-years whose position this year was worse than their 2017–2018 average; revenue is acres times the NASS revenue per acre of the crop and year; revenue falls by 0.8 times the proportional fall in the yield proxy. Extra fallowed acres: the fallowing coefficient times the usual position above its 25th percentile, times acres. A fallowed acre is valued at the mean net return of the cotton and alfalfa budgets (returns over total costs) or at mean crop revenue (an upper bound). Totals add yield and fallowing; the yield loss falls on planted field-years and the fallowing loss on fallowed ones, so the two do not overlap. Brackets: 95 percent of joint draws of the two coefficients, drawn independently. Gains of fields whose position improved and of fields that received the water are not netted out.

7.2.4 Position Matters When the Canals Run Short

The yield-proxy result appears in the drought year of junior fields. If position matters because the canal runs short, it should matter in drought years when the canal could not meet demand. It should not matter in dry years when deliveries were full. In Figure 6(a) and (b), I replace the drought term on this year's change in position with one term per crop year. In junior-rights districts, the coefficient is −0.042 in 2021, their drought year ($p = 0.014$). The junior coefficient is also negative in 2019 (−0.029, $p = 0.032$), not a drought year for junior fields. In senior-rights districts, the largest coefficient is in 2019 (−0.052, $p = 0.089$), a drought year for 46 percent of senior fields.

Figure 6: Junior Fields Lose Most in 2021, Their Drought Year

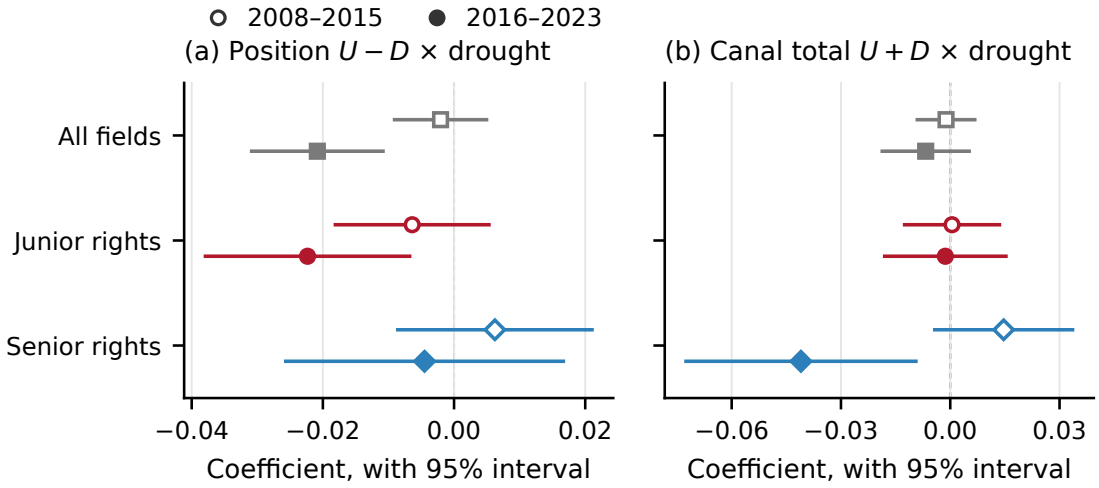


Notes: Panels (a) and (b): equation (5) with this year's change in position interacted with each crop year 2019–2023 in place of drought, 2SLS, 5,569 junior and 8,454 senior fields. No year is left out: each point is that year's own slope, read against zero, and the change is measured from the field's 2017–2018 usual position. Panel (c): equation (6) with usual position interacted with each crop year 2020–2023, in percentage points. Field effects absorb the usual position itself, so one year is left out: each point is the slope relative to 2019, the first year of the fallowing sample and not a drought year for junior fields. Grey: years with at least 30 percent of the group's fields in irrigation districts in drought (junior 2021; senior 2019 and 2022). Hatched: 2022–2023, when the CAP districts received little surface water. Bars are 95 percent intervals.

The drier years of 2008–2015 show no position effect, and CAP deliveries were not cut then. Monthly water use and crop maps go back to 2008. I estimate the water model on the same fields in 2008–2015 and 2016–2023, with one drought measure and one instrument. The sample keeps 20,051 fields cropped in at least two years of each period. By SPEI, the earlier period was drier, with 59 percent of field-months in drought against 13 percent later. The position coefficient is −0.002 ($p = 0.58$) in 2008–2015 and −0.021 ($p < 0.001$) in 2016–2023, a difference of −0.019 ($p = 0.004$; Figure 7). In junior-rights districts it moves from −0.006 to −0.022 ($p = 0.006$ for 2016–2023; the difference has $p = 0.115$). The difference follows CAP deliveries. In 2008–2014, CAP agricultural deliveries stayed within 1 percent of schedule and no shortage was declared. The 2015 reduction was voluntary and compensated (Appendix Table S2). With full deliveries, the canals could absorb the drought, and position had little to decide. In senior-rights districts, the canal total moves instead. It is near zero in 2008–2015 (+0.015) and −0.041 ($p = 0.012$) in 2016–2023.

The deepest cut, in 2022–2023, does not fit the design, because the water entered along the canals. In those years, the junior yield-proxy coefficients are near zero, and the junior fallowing slope reverses. The CAP agricultural pool fell by about 88 percent in 2022 (Appendix Table S2)

Figure 7: No Position Effect in 2008–2015, When Deliveries Were Not Cut



Notes: Equation (4) with each term interacted with the period, 2SLS, on 20,051 fields cropped in at least two years of each period (3,526,174 field-months). Activity and crop come from the crop maps in both periods; predicted demand uses each crop's mean over 2008–2023, leaving out the year. Open markers: 2008–2015; filled markers: 2016–2023. Drought covers 59 percent of field-months in 2008–2015 and 13 percent in 2016–2023. Effects as in equation (4), with field by month and irrigation district by drought by month effects for each period. Bars are 95 percent intervals.

and received nothing in 2023. Maricopa–Stanfield and Central Arizona kept farming on district wells, which pump into their canals along their length. In Maricopa–Stanfield, CAP brought 29,754 acre-feet in 2022 and the district's wells brought 169,764 (Section 8.5). Water then entered the canals below many fields as well as at the head. Position counted from the canal's source no longer says who is first in line.

7.2.5 What the Design Cannot Show

The design shows that water follows the direction of the canal. It does not show who moves the water. Operators who ration or schedule deliveries in flow order produce the same pattern as upstream farms that take water first. The design measures demand, not water, so it cannot say how much water a field lacked. The instrument assumes that planting does not respond to the shortage of a particular month. I cannot test the assumption directly. This year's change in position is measured from a fixed 2017–2018 mean. It holds lasting shifts as well as one-year changes, so I do not read it as a surprise.

Within the rights groups, the yield-proxy results are less precise than the water results. In junior-rights districts, this year's change in position is clear only at 3 km and turns positive from 6 km on (Appendix Table A6). The junior result rests mostly on water use and fallowing. The yield proxy measures cotton and alfalfa yield poorly. Cotton's yield is not its GCVI peak. Alfalfa is cut several times a season, so its GCVI has several peaks. Appendix A14 builds a crop-specific proxy from each crop's own weeks, for Maricopa–Stanfield only. I do not split the yield-proxy results by crop. A field's own well comes from registry points, which are often placed from the legal description of the land. A well at a field's edge may fall on its neighbour. I do not test differences between groups, whether between junior and senior fields or between fields with and without an own well. The split between junior and senior fields is clearest at 3 km. At wider cutoffs the junior canal total turns negative too, and the senior position is negative but imprecise

(Appendix Table A5).

The 2021 junior estimate leans on Central Arizona. Without it, the estimate is −0.020 ($p = 0.40$). Without Cibola Valley, it is −0.034 ($p = 0.056$; Appendix Table A9). The negative junior coefficient in 2019 stays between −0.031 and −0.021 when any one irrigation district or reservation is left out. I have no explanation for it. Of the 13,364 extra fallowed acres of junior fields, 965 are in 2022, a year the design reads poorly. The junior group pools irrigation districts that divide water in different ways. Farms in Maricopa–Stanfield order water against a seasonal allotment. Central Arizona runs its own wells first, and the Upper Gila canals rotate their water.

The first look has no controls. Fields in the tail third may differ from those in the head third in soils or crops that respond to drought. The pattern also appears in senior-rights districts, where position does not decide water. In their own drought years, the share left fallow rose 2.3 points in the tail third and fell 1.8 points in the head third. It fell most in the middle third, by 2.9 points. In junior-rights districts, planted fields in the head and tail thirds changed their crop-season water use alike (a gap of −0.001 log points, 95 percent interval −0.013 to 0.011). The water result comes from timing within the season, which a yearly comparison cannot show. The dollar figures do not net out the gains of the fields that received the water. They do not show what an irrigation district loses as a whole.

Upstream and downstream demand show that, when the canals run short, the direction of the canal decides who goes short in junior-rights districts. Section 8 turns to the water records. It carries each irrigation district's recorded water down its canals, month by month, and asks which fields run short. The records can show how much water a field lacked and where the water entered. They can also show how each irrigation district divides its water.

8 Who Gets Canal Water When It Falls Short

Monthly canal records show how the losses of Sections 7.1 and 7.2 arise. I carry each district's recorded water down its canals and ask which fields run short, month by month, before any regression on water use. One model covers the districts that hold 85 percent of the acres in Arizona's large surface-water districts, and ten Upper Gila canals. Fallowing rises only in the districts it says run short. Inside them, water use follows the model's shortfall where the records measure the water that reaches fields. In Maricopa–Stanfield, a field without a well uses 18 percent less water in a month when the canal meets none of its claim than in a month when it meets all of it. New Magma, a district the model had not seen, repeats the pattern. Equal sharing fits neither, and it fits best in the Upper Gila, which rotates its water. In the 2022–2023 cut, the same shortfall predicts which fields lose water and which go fallow. Where district wells stand on the canals predicts which fields keep water. A shortage added only on paper in four senior-rights districts produces no such result in three, and the fourth may have a real limit.

8.1 A Monthly Model of Canal Delivery

The irrigation network of Section 5 says where a field sits. It does not say how much water reaches the field in a given month. Most large surface-water districts record the water that enters their canals each month, and a delivery model can carry that water down the canals to the fields. The model has five parts: the water that enters, the crop demand of each field, the canal network, the wells, and the rule that divides the water. I built it on Maricopa–Stanfield, whose records and

maps are the fullest, and I run the same recipe in the other districts. It leaves three parameters free: how much a field with a well claims from the canal, and two that say how water is divided.

Canal water. A district's water V_t in month t is its own monthly record. In the CAP districts it is the district's CAP delivery, Agricultural Settlement Pool and agricultural in-lieu water, from CAP's delivery reports. In Wellton-Mohawk and the three Yuma districts it is the diversion in Reclamation's decree accounting. In San Carlos it is the district's share of the project's gravity diversion under the Globe Equity decree, plus its Agricultural Settlement Pool water. In the Upper Gila it is each canal's monthly diversion from the Gila Water Commissioner's records (Gila Water Commissioner, n.d.). The model places a district's fields on its canal network and gives them their share of that water, $V_t \times D_t^{\text{placed}}/D_t^{\text{all}}$, where D is the crop demand of the placed and of all district fields. Crop demand, below, is what the crops consume, so a field needs more water at its turnout than its crops use: with irrigation efficiency e, it needs $1/e$ acre-feet for each acre-foot of crop demand. I take e from each district's own water balance over 2017–2019, the years of the crop calibration below: 0.72 in Maricopa–Stanfield, 0.73 in Central Arizona (2017, the only year with its district-well total), and 0.75 in Hohokam. Harquahala, without measured pumping in those years, takes their pooled value, 0.73, and the districts added later take 0.70 (Appendix A2). Most of the difference is water applied but not consumed by the crop: CAP's recorded deliveries to Maricopa–Stanfield exceed the district's own delivery figures by only 3.5 to 5.5 percent a year (Maricopa–Stanfield Irrigation and Drainage District 2024). Water lost along a canal enters only through e, so no loss in the model grows with distance.

Crop demand. A farmed field's irrigation demand in month t is

$$d_{it} \;=\; A_i \times \max\bigl(0,\; \widetilde{K}_{it}\,\mathrm{ETo}_{it} \;-\; P^e_{it}\bigr), \tag{7}$$

and a fallow field's demand is zero. A_i is the field's acreage, ETo_{it} reference evapotranspiration, and P^e_{it} effective rain. $\widetilde{K}_{it} = c_{gm}K_{it}$ is the field's *locally calibrated crop coefficient*, built in two steps from different data. First, the FAO-56 coefficient K_{it} follows the field's own crop calendar, the timing of its crop in the 2017–2019 Sentinel-2 imagery: Sentinel-2 sets when the crop needs water. Second, the factor c_{gm} scales the coefficient so that each crop's need in 2017–2019, the years of the Sentinel-2 calendars, equals what the same crop consumed according to OpenET: OpenET sets how much. Without the factor, the FAO-56 coefficients put crop need 30 to 50 percent above measured use in every year, the years of full supply included. There is one factor for each of two regions (Maricopa–Stanfield, Central Arizona, Hohokam, and Harquahala together, or five Upper Gila canals), crop group g, and calendar month m, common to all fields of the group; Appendix A2 gives its formula and the other inputs. The districts added later take these factors unchanged. In Wellton-Mohawk, which played no part in the calibration, demand comes to 0.93 to 1.03 of OpenET less rain by year. What moves a field's demand from year to year is whether it is farmed, its crop, the weather, and rain.

The network. Each district's canals form a tree below the gates through which its water enters. The tree uses the canal lines of the irrigation network of Section 5, below the district's documented intake and above its fields, and it crosses the gaps a canal passes under: roads, siphons, and pipes. Water leaves only at laterals, so a field linked to a main canal moves to the nearest lateral within

2 km. A field may lie at most 1 m above the canal where it takes water.

Where a field takes its water comes from the best evidence at hand, in a fixed order. First comes evidence of the field's own connection: an intake I marked on imagery, a lateral within 30 m, or a mapped concrete ditch followed upstream for at most 3 km to a lateral. Where such a ditch shows no turnout, the intake is inferred at the lateral nearest its highest point, within 1 km. Next comes evidence from the fields around it: the turnout of its ditch system, or that of a touching neighbor with a connection. Last, a field with no such evidence projects onto its canal line, and such fields form blocks cut at branch points and every 800 m. Fields at one point of the tree, or in one block, have no order among them, because the data cannot place them more finely. The mapped ditches exist only in Pinal County, so elsewhere the tree rests on contact, neighbors, and projection. Of Maricopa–Stanfield's 2,465 fields, the model places 2,383: 1,782 on evidence of their own connection, 404 on their neighbors', and 197 by projection (Appendix A2).

Wells. A pump is a canal pump or a local pump by how its water reaches fields. A *canal pump* is a district well that puts its water into a canal. In Maricopa–Stanfield these are the 130 district wells that pumped in 2019, lie within 250 m of a tree line or 120 m of a mapped ditch, and are rated at 1,000 gallons per minute or more or have no rate. They pumped 85.7 percent of the district's 2019 well water. Each year they pump their share of the district's reported well water, or what the canals still need after CAP if that is less. The months follow that need, up to a share of capacity set by the year's total. Within a month each pump fills the claims below it, starting from the pumps furthest down, and water that no claim can take is not pumped. At a take of 0.70 that leaves out less than 1 percent (Appendix A2).

A *local pump* serves its owner's fields, and a field *has a well* if a local pump serves it. Well access goes by ownership and not by distance to the nearest well. Sections 7.1 and 7.2 use a narrower own well, a registered well inside the field (Appendix Table A8). A registered well rated at 100 gallons per minute or more belongs to the owner of the field it sits on, or of the nearest field within 200 m, and it serves all of that owner's fields. In Maricopa–Stanfield, where the ditches show what a well can reach, it serves the owner's fields down its ditch, or within 400 m if the well is not on a ditch. District and federal wells are no field's private water, so in the other districts they do not give a field a well. Only Maricopa–Stanfield and Central Arizona have canal pumps in the model, since the other districts have no such wells or no record of their pumping. Central Arizona's records give each well's capacity and not which wells ran, so I also build a version from its documents, in which the district runs all 434 wells and no field has a well of its own (Appendix A12). In the Upper Gila all wells stay local, since no record says which pump feeds which canal.

Claims and the take. A field without a well claims its whole crop demand from the canal. A field with a well claims only a part of it, the *take* τ, and its well serves the rest. The take is the model's first free parameter. Maricopa–Stanfield's water record gives an upper bound without using any data on water use: one minus e times the local district wells' water over the need of fields with a local pump, which is 0.737.

The rule. Canal pumps serve the fields below them first, with no order among those fields. The canal's main supply then meets what is still claimed. That water, eV_t, enters the tree at its heads and moves downstream; scaling the water by e gives the same shares as scaling every claim by $1/e$. Two parameters say how it is divided. At a point that receives a flow F, with claims c from its

own fields and claims b on the branches below it, the fields there take

$$o \;=\; \kappa \min(c,\, F) \;+\; (1-\kappa)\, F\, \frac{c}{c+b}, \tag{8}$$

and the rest moves on. At $\kappa = 1$ they take all they need before water moves on. At $\kappa = 0$ every field on the line gets the same share of its claim. Where a side line leaves a canal, let R be the water left and N_s and N_c the claims on the side line and on the line that continues. If R is short of those claims, the side line takes

$$\kappa_J \min(N_s,\, R) \;+\; (1-\kappa_J)\, R\, \frac{N_s}{N_s + N_c}, \tag{9}$$

and the line that continues takes the rest. At $\kappa_J = 1$ the side line comes first, and at $\kappa_J = 0$ the two share by claims. Elsewhere branches share by their claims. Each pair (κ, κ_J) is a *rule*. Figure 8 shows the corner rules on a small canal, and Appendix A2 writes out the recursion. A field's *delivered share* is the part of its claim that the canal meets:

$$r_{it} \;=\; \frac{w_{it}}{c_{it}}, \qquad 0 \le r_{it} \le 1, \tag{10}$$

where w_{it} is the water the model delivers to field i in month t and c_{it} its claim. Fields at one point of the tree share one delivered share, which is 1 in a month when that point has no claim. At $\kappa = 1$ on a single line, the rule is the residual supply of equation (1) with each upstream user's entitlement set to its claim and no leak. A field falls short when the claims upstream leave it a residual supply below its own claim, which is the negative cushion of Section 3.2. The model therefore needs no over-diversion to produce a shortfall: in a month when the canal carries less than the crops need, upstream claims alone exhaust the water before it reaches the tail.

The Upper Gila. I treat the ten Safford Valley canals that took water as one system, each with its own tree and its own water. Reported diversions depend partly on what farmers ordered, so beside them I run the river itself: the flow at the Solomon gage, passed down the decree's list of rights to each canal. A rotation gives each field its turn in proportion to its demand. Over a month that is the rule at $\kappa = 0$: fields on a canal get the same share of their claims, wherever they lie. So the model needs no separate equation for a rotation, and the search of Section 8.2 can find one if the data show it (Appendix A3).

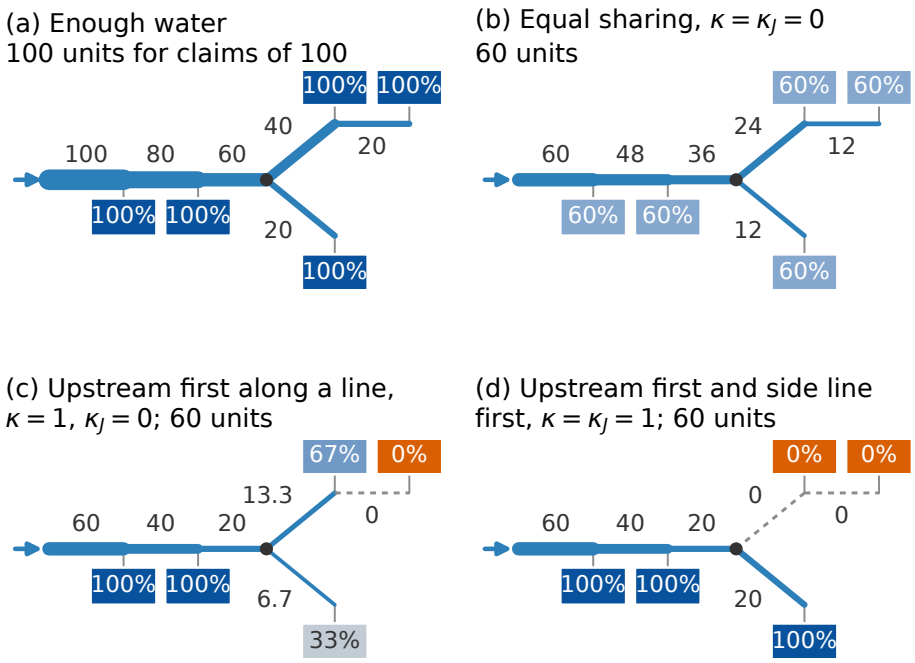
8.2 The Within-Season Design

The delivery model gives each field a canal shortfall for each month, the share of its crop demand that the canal does not meet:

$$\ell_{it} \;=\; \begin{cases} 1 - r_{it} & \text{for a field without a well,} \\ 1 - \tau\, r_{it} & \text{for a field with a well,} \end{cases} \tag{11}$$

where r_{it} is the delivered share of equation (10) and τ the take. The shortfall is a prediction of the model, not a regression estimate: the model computes it from deliveries, demand, and the network before the ET regression is run, and that regression takes it as data. The design asks whether a field's water use falls with its shortfall within the season. The shortfall, not the share, enters so

Figure 8: The Division Rule on a Small Canal



Notes: An illustration of the rule, not data; the shares come from the model's code. Five fields, each with a claim of 20 units, on a canal that divides once; the lower branch is the side line. Numbers on the canal: the water flowing in each reach, also shown by the width of the line; dotted: no flow. Squares: the fields, shaded by their delivered share (equation (10)); orange: no water. Panel (a): the water equals all claims, 100, and every field is served under any rule. In the other panels it is 60. Panel (b): every field gets the same share. Panel (c): the two fields above the junction take their claims first. The remaining 20 splits two to one between the branches, by the claims along each (40 and 20); the first field on the upper branch takes all that reaches it, and the field after it receives nothing. Panel (d): the side line comes first and takes all of the remaining 20.

that a larger value means less water, as a larger upstream demand does in Sections 7.1 and 7.2. The estimating equation is

$$y_{it} \;=\; \beta_0\,\ell_{it}\,(1-W_i) \;+\; \beta_1\,\ell_{it}\,W_i \;+\; \phi\,S_t\,(1-W_i) \;+\; X_{it}'\gamma_m \;+\; \alpha_{ikm} \;+\; \alpha_{iy} \;+\; \alpha_{kym} \;+\; u_{it}, \quad (12)$$

where y_{it} is 100 times the log of the field's monthly ET from OpenET, its *water use*; $W_i = 1$ if the field has a well; and k, m, and y are the field's crop, the calendar month, and the year of month t. S_t is the district's *shortage*, the share of the month's crop demand that its water cannot meet, $S_t = 1 - \min\bigl(1,\, e \times \text{water}_t / D_t\bigr)$. It counts the main supply and any district well water, and it takes one value a month for a district, or for the whole valley in the Upper Gila. It holds what fields without a well share in a month, so that β_0 rests on how shortfalls differ among fields. The controls X_{it} are the 3- and 24-month drought indices (SPEI), each with its own coefficient in each calendar month. The sample is the farmed field-months of March to October on the tree, after cells with a single observation drop out. It runs over 2016–2021 in Maricopa–Stanfield, Central Arizona, Harquahala, and Hohokam, where the cut has its own design below, and over 2016–2023 elsewhere. Standard errors are clustered by owner and by year-month; a field with no known owner is its own cluster.

β_0 is approximately the percent change in the monthly water use of a field without a well when its shortfall goes from 0 to 1, that is, from a canal that meets its whole claim to one that meets none of it. The prediction of Section 3 is $\beta_0 < 0$, and β_1 closer to zero if a field's own well replaces the water, since groundwater caps the crop cost of a shortfall.

The fixed effects leave only variation within the season. Field by crop by month effects remove each field's usual water use in each month of its crop's season. Field by year effects remove what changes for the field from year to year: its planting, its allotment, and its pumping. Crop by year by month effects remove what a crop's fields share in a month, such as heat and prices. What remains is how a field's shortfall moves against its own usual months as the month's water and the demand upstream change. I leave out owner effects. An owner's fields sit together on a canal, so their position differs little, and comparing them removes the variation the design uses. A zero within owners would say nothing about delivery.

The search over rules. I do not choose the rule by hand. A search covers the take, κ, and κ_J, each from 0 to 1 in steps of 0.02, which gives 132,651 cells and 2,601 rules at each take. A cell's *score* is how much its shortfall improves the prediction of water use in a year left out of the estimate. For each year in turn I fit the fixed effects and coefficients on the other years. The prediction for the year left out carries the field by crop by month effects of those years; its own field by year and crop by year-month effects are then removed. The score is the fall in squared error against a baseline with $S_t(1-W_i)$ and the controls, summed over the years, in basis points of the total. The search uses a compressed version of the score, and I score the best 200 cells exactly.

The *bar* allows for the size of the search. It is the 95th percentile of the best score in 200 draws, each of which flips the sign of the baseline's residuals owner by owner and repeats the whole search. The score is blind to sign, so a cell can clear the bar with a slope of the wrong sign, and I report each with its slope. The score also covers all fields of a district, so it tests prediction across the district and not the slope of one group. To test a named rule I invert the same comparison on a coarser grid, with rules in steps of 0.1 and takes from 0.50 to 0.90: a rule stays in the set when the gain of that grid's best cell over it is below the 95th percentile of that gain in sign-flip draws

around the rule's own fit. The best cell can rest on chance, so the main estimate picks no single rule. It averages the shortfall over all 2,601 rules at the district's best take, so no rule is picked on water use.

Reading β_0. Three features of the shortfall bear on the reading of β_0. First, the shortfall moves with demand as well as with supply: with the month's water, and with the crops grown upstream and at the field. Weather moves demand, but weather that raises demand raises the shortfall and raises water use at the same time, so it biases β_0 toward zero. Second, I do not observe deliveries to individual fields, so the design judges the model by what it predicts. Randomization asks whether the model's inputs could produce $\hat{\beta}_0$ on their own. Each test permutes one input and reruns the whole model at the take. The date test moves each calendar month's water across the years. The field test moves fields, with their own demand, to other fields' points on the tree. The owner test moves whole owners' fields along the tree and keeps them neighbors; I prefer it, because moving fields one by one scatters an owner's fields and makes the draws too narrow. The one-sided p-value is $(1+\#\{b : \hat{\beta}_0^{(b)} \leq \hat{\beta}_0\})/(1+B)$ over $B = 1{,}000$ draws. Third, the CAP districts deliver on order. Within a calendar month, Maricopa–Stanfield's CAP moves with the demand the model computes (correlation 0.63 over the 48 month-years), so a month's shortfall can fall because growers irrigated, not only because CAP arrived. The randomizations do not separate a shortage of supply from a pattern of orders.

The cut. Farmers decide before the season which fields to plant (Appendix S9), so the 2022–2023 cut gets an annual design. I hold each field's demand at its 2016–2021 calendar-month mean, so that only water moves the shortfall. A field's *predicted rise* is its demand-weighted mean shortfall in 2022–2023 minus that in 2016–2021, averaged over all rules at the take. I regress the change in a field's outcome, 2022–2023 against 2016–2021, on its predicted rise, its alfalfa share in 2016–2021, its log acres, and its kilometers from the head. The outcomes are water use and the farmed share. The same regression of 2020–2021 against 2016–2019 gives the earlier contrast. I stack the two and report the difference, per standard deviation of the predicted rise, with standard errors clustered by owner. The Upper Gila had no 2022 cut, so there I report 2022–2023 against 2016–2021.

Growers know the year's delivery before they decide what to fallow, and they can read this year against last year. A field's *total stress* therefore adds two shares of its own year's demand: the share left short last year, at the demand it had, and the share predicted short this year, at its usual demand. A month with no demand of its own adds nothing, and the sum runs from 0 to 2. I regress fallowing, times 100, on total stress among fields farmed the year before, 2017–2023. One version has year effects and the controls above, and one has field and year effects. Standard errors are clustered by owner.

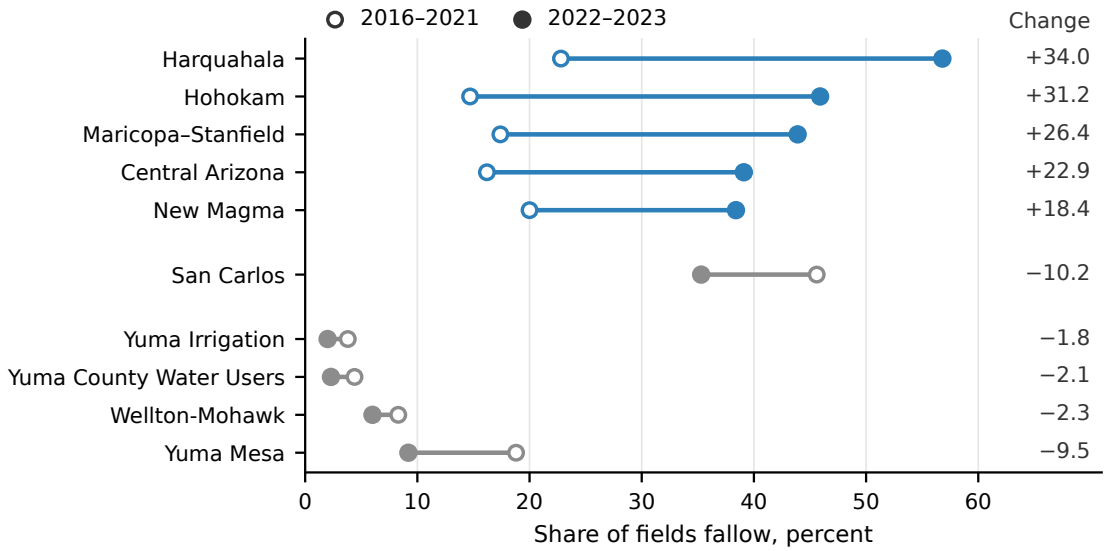
8.3 Fields Went Fallow Where the Model Says Water Fell Short

I do not observe deliveries to individual fields, so I judge the model by what it predicts. The model runs on the districts of at least 10,000 acres with a monthly water record that can show a shortage. Figure 9 shows the share of fields left fallow in each. The five CAP districts run short when CAP falls, and their fallow share rose 18 to 34 percentage points from 2016–2021 to 2022–2023. The three Yuma districts have no short month under any rule, and Wellton-Mohawk has two. Their fallow share fell by 2 to 10 points. San Carlos can meet 0.78 of demand in both periods, so the model predicts no rise, and its fallow share fell 10 points. Fallowing also rose in three other

districts where the model finds no shortage, two of them mostly near cities (Appendix A13). A rise in fallowing alone does not show a shortage.

The divide also follows water law. CAP holds a junior right on the Colorado River, and the Yuma districts and Wellton-Mohawk hold senior ones. A reader could predict which districts ran short without the model. I take it as the setting. The model's test lies inside the districts, where rights are the same and position and wells differ.

Figure 9: Fallowing Rose in the Districts the Model Says Ran Short



Notes: Share of a district's fields left fallow. Blue: the five CAP districts. Grey: San Carlos and the four Colorado River districts with senior rights. These districts hold 390,143 of the 460,999 acres in surface-water districts of at least 10,000 acres; the Upper Gila's canals, each smaller than that, come in Section 8.6. Left out: Roosevelt and Buckeye, with no monthly supply record; Salt River Valley, whose farms kept full allocations; Cortaro–Marana, where wells are the main source (Appendix A13). Yuma Mesa's level may reflect crop labels.

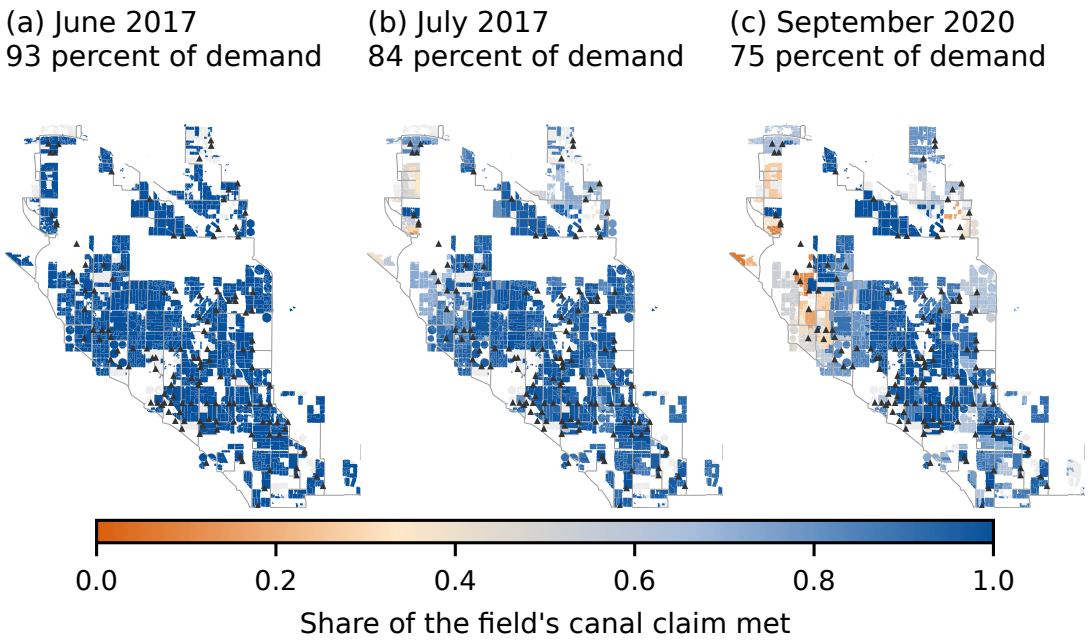
8.4 Inside a District, Water Use Follows the Shortfall

CAP water enters each district at a few turnouts, the gates through which it leaves the CAP canal, and flows through main canals to the laterals that serve the farms. Maricopa–Stanfield and Central Arizona give each farm a seasonal allotment per acre, and farms order water as their crops need it (Maricopa–Stanfield Irrigation and Drainage District 2024). Both also pump district wells into their canals (Irrigation Leader, n.d.). In Maricopa–Stanfield in 2016–2021, CAP and the district's wells each supplied about 130,000 acre-feet a year. Together they came close to crop demand over a year, but not in every month. Central Arizona wrote that shortages "usually occur during periods of above normal temperatures in the growing season when there is high demand" (CAIDD 2022).

Figure 10 shows who gets water in the model in three summer months. In June 2017, every farmed field receives its full claim. In July 2017, 6 percent do, and 2 percent receive less than half. In September 2020, 12 percent receive less than half. They gather in the northwest of the district, where a quarter of fields are that short, against 1 percent in the southeast.

The estimates use only variation within the season (equation (12)). A field's *water use* is its crop's monthly evapotranspiration from OpenET. The regression also holds the district's *shortage* for fields without a well, the share of the month's crop demand its water cannot meet. I leave out owner effects, because an owner's fields sit together on a canal and comparing them removes

Figure 10: In Short Months the Canal Stops Meeting Some Fields' Claims, Maricopa–Stanfield



Notes: The share of each farmed field's claim delivered, averaged over all 2,601 rules at a take of 0.70, in the June–September months of 2016–2021 with the highest coverage, the coverage closest to the midpoint of the range, and the lowest. Coverage in the titles: delivered water as a percent of crop demand. Light grey: fields not farmed that month. Triangles: the 130 canal pumps. Grey lines: the canal tree.

the variation the design uses. The sample in Maricopa–Stanfield has 67,064 field-months of 2,020 fields of 462 owners, and 1,656 fields have no well.

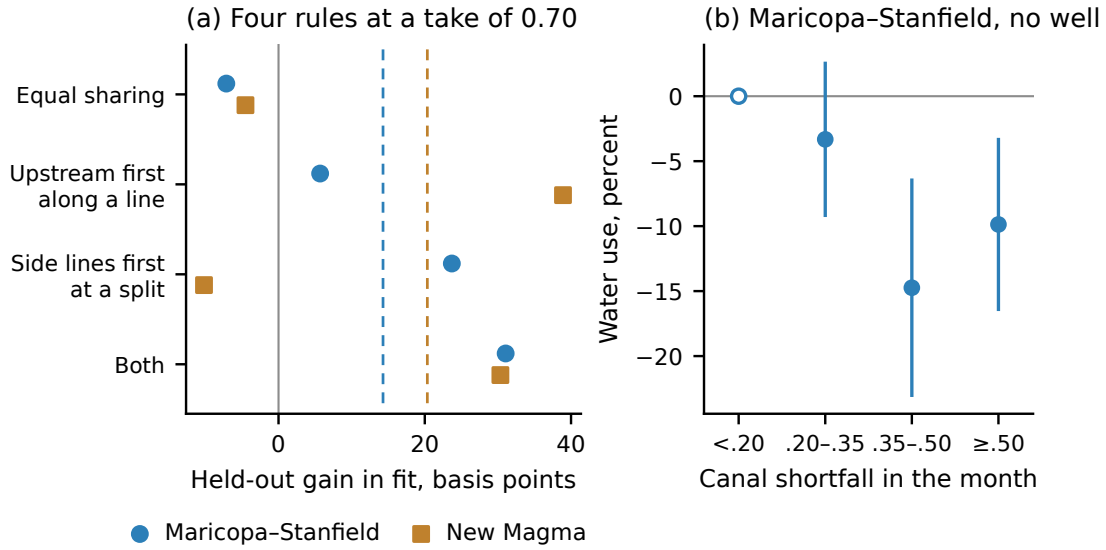
I do not choose the rule by hand. A search covers 51 values of κ, 51 of κ_J, and takes from 0 to 1. Each rule's *score*, in basis points, is how much its shortfall improves the prediction of water use in a year left out of the estimate. The *bar* is the 95th percentile of the best score in 200 draws that flip the sign of the residuals owner by owner. The score is blind to sign, so I report each with its slope. In Maricopa–Stanfield, the best rule scores 32.6 against a bar of 14.3, and no draw reaches it. It sits at a take of 0.70, κ of 1.00, and κ_J of 0.98, with a slope of -12.3 for fields without a well (standard error 3.8).

The main estimate picks no single rule and averages the shortfall over all 2,601 rules at that take. Raising the averaged shortfall from 0 to 1 lowers the monthly water use of a field without a well by 18.2 percent (-18.2, standard error 5.2). A typical move within the season lowers it by 1.7 percent. The loss grows with the shortfall up to about 0.5 and then holds (Figure 11, panel b).

The monthly slope rests on each month's actual demand. With demand held at its usual level, it falls to -1.7 (standard error 4.4). A field's own need does not drive the slope. CAP orders follow actual demand, so usual demand misdates the short months. The cut design holds demand fixed and still finds a slope, because supply moved most in the cut.

No year can serve as a placebo, because water falls short in some month of every year. I randomize the model's inputs instead. Reassigning each calendar month's water across the years reaches the real slope in 1 of 1,000 draws ($p = 0.002$) and the best rule's score in 2. Moving fields to other fields' positions reaches neither in 1,000 draws, and neither does moving whole owners' fields.

Figure 11: Equal Sharing Fits Neither CAP District, and Water Use Falls in Steps With the Shortfall



Notes: Panel (a): each rule's score at a take of 0.70, Maricopa–Stanfield 2016–2021 and New Magma 2016–2023. Dashed lines: each district's bar. Equal sharing: $\kappa = \kappa_J = 0$. Upstream first along a line: $\kappa = 1$, $\kappa_J = 0$. Side lines first at a split: $\kappa = 0$, $\kappa_J = 1$. Both: $\kappa = \kappa_J = 1$. Panel (b): water use by size of the shortfall in the month, fields without a well, all rules at a take of 0.70, against months below 0.20; bars are 95 percent confidence intervals.

The data reject equal sharing and do not pin down the exact rule. At a take of 0.70, the test rejects equal sharing (score −7.1, $p = 0.005$) and upstream first only along a line (score 5.7, $p = 0.005$). It keeps side lines first at a split (score 23.7, $p = 0.453$) and both together (score 31.1, $p = 0.692$). The test keeps takes of 0.62 to 0.78, κ_J of 0.24 to 1, and every κ. Water is not shared in proportion to demand, and side lines take some priority at the splits. The data cannot say what happens at each turnout along a line.

Fields with a well also use less water when their shortfall rises (−47.6, standard error 21.7). I cannot show that a field's own well shields it here. Such a field with a large shortfall may lie far from other water. At the same location I cannot tell the two groups apart (−9.7, standard error 12.1). A well on the canal predicts which fields keep water (Section 8.5).

I built the model on Maricopa–Stanfield's data in steps, each for an institutional reason, so its results there are not a test on unseen data. New Magma comes closer to one. I fixed the model on Maricopa–Stanfield and ran it unchanged on New Magma's 524 fields of 249 owners in 2016–2023, of which 430 have no well. The best rule scores 39.1 against a bar of 20.4, and no draw of 200 reaches it. Over all rules the slopes are −12.6 (standard error 5.4) without a well and 1.5 (standard error 8.3) with one. The two districts share their best take of 0.70 (Figure 11, panel a). Equal sharing fits poorly in both, at −7.1 and −4.5. Upstream first along lines and at splits fits well in both, at 31.1 and 30.3, above each bar. They do not share a rule, as κ_J is 0.98 in one and 0 in the other.

Harquahala's farms pump their own wells, and only 79 of its 648 fields have no well. The best score across all fields, 37.9, stays under the bar of 49.2. The slope still has the sign the model predicts, −34.8 without a well (standard error 12.6) and −2.2 with one (standard error 9.3). No shuffled date or position in 1,000 reaches that slope, and it lies below 97 percent of the draws

that move whole owners' fields ($p = 0.029$). The district's shortage in the same regression has the wrong sign (58.4, standard error 28.4). CAP follows demand there (correlation 0.86), so the shortage term may partly measure high demand. I have not tested that reading.

8.5 Losses in the Announced Cut Followed the Model, and Canal Pumps Mark Which Fields Kept Water

Reclamation announced the 2022 cut on the Colorado River in August 2021 (Reclamation 2021), and the districts rationed it by rule. Maricopa–Stanfield set each turnout's volume by its acreage. The usual 4 acre-feet per acre became 1.7 to 2.8 "depending on where the grower's fields are in the district" (Evancho 2022, 12). In 2023 it sent the little CAP that remained to the areas with no wells (Maricopa–Stanfield Irrigation and Drainage District 2022a). CAP fell from about 131,700 acre-feet a year in 2016–2021 to 29,754 in 2022 and 4,000 in 2023. The district's wells rose from about 129,300 to 169,764 and 183,246.

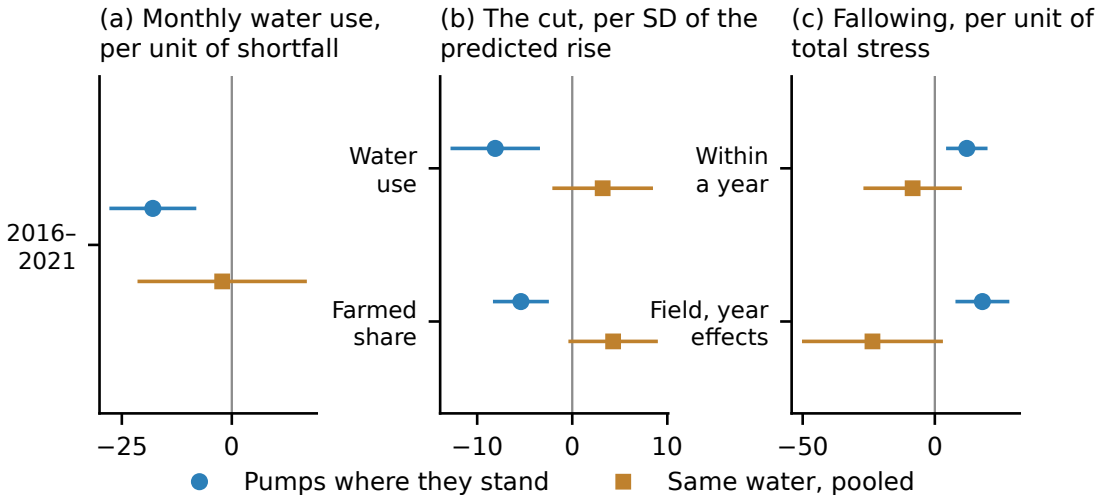
Farmers decide before the season which fields to plant (Appendix S9), so the cut shows in annual outcomes. The design holds each field's demand at its 2016–2021 level, so that only water moves the shortfall. It sorts fields by the rise in shortfall the model predicts. Among fields without a well, a predicted rise one standard deviation larger lowers water use in 2022–2023 by 8.5 percent (standard error 2.4). That estimate is net of the same contrast in 2020–2021, which is zero (−0.0, standard error 2.0). The farmed share falls by 5.9 points (standard error 1.4), and its earlier contrast, 2.6 (standard error 1.2), accounts for part of that fall. The loss runs mostly through land left out of production. On fields farmed in all eight years, the estimate is −0.9 (standard error 2.0).

Growers know the year's delivery before they decide what to fallow, and they can read this year against last year. A field's *total stress* adds the share of its own demand left short last year, at the demand it had, to the share predicted short this year, at its usual demand. It runs from 0 to 2. Among fields without a well farmed the year before, one unit raises the chance of fallowing by 10.7 points within a year (standard error 3.8). With field and year effects it raises it by 16.7 points (standard error 5.0), or 4.3 points per standard deviation. The response comes from the cut years (12.8, standard error 4.5, against 0.5 in 2017–2021, standard error 8.0).

Where the pumps stand produces most of the variation in the predicted rise in shortfall. Its standard deviation is 0.232 with the pumps where they stand, and 0.063 with the same pumped water pooled over all fields. Figure 12 enters the two side by side. Pumps where they stand predict water use in the cut (−8.1, standard error 2.4) and the farmed share (−5.4, standard error 1.5). The pooled version has the opposite sign on each. Fallowing divides the same way, as does the monthly slope in ordinary years (−18.0, standard error 5.0, against −2.2, standard error 9.8). Competition on the canal matters little for a field with a pump above it. The slope on the pooled shortfall is 1.0 for the 1,066 fields with a canal pump within 2 km above (standard error 10.3). It is −27.6 for the 590 without one (standard error 11.0). With the district's shortage by group, the difference keeps its sign, though I cannot tell it from zero (22.8, standard error 14.5). A pump below a field cannot deliver to it and shows nothing (−4.1, standard error 22.1). The district chose where its pumps stand, so the estimates show what pump locations predict, not what moving a pump would do. The cut cannot tell one rule from another, because every rule keeps the pumps in place.

New Magma's fallowing divides the same way. The gap between its fields without and with a

Figure 12: Pumps Where They Stand Predict Who Lost Water; the Same Water Pooled Does Not, Maricopa–Stanfield



Notes: Each panel enters two shortfalls in one regression on fields without a well: the model's, with canal pumps where they stand, and one with the same pumped water pooled over all fields each month, so that no field gains from its location. All rules at a take of 0.70. Panel (a): monthly, 2016–2021. Panel (b): 2022–2023 net of 2020–2021, per standard deviation of the predicted rise in each shortfall, fixed demand. Panel (c): chance of fallowing, in points, among fields farmed the year before, 2017–2023. Bars are 95 percent confidence intervals.

well widens from nothing in 2016 to 24 points in 2022 and 26 in 2023. Within a year the link to total stress is weaker, at 9.9 points in 2017–2021 (standard error 4.5). With field and year effects it is −0.6 (standard error 7.0). Its cut shows no slope on water use (1.5, standard error 3.2). The water at its fields fell to 0.68 and 0.40 of usual demand, so fields at every position may have gone fallow together.

8.6 Where the Rules and the Records Differ, the Result Changes

The model stays the same across districts, while the rules that divide water and the records that measure it do not. Table 5 sets what each district's rules and records predict beside what I find. Before any regression, the model's sort into short and not short holds in all nine CAP and senior-rights districts (Figure 9). Inside districts, nine of the eleven match, I cannot tell one, and one goes wrong. Four of the nine have senior rights and almost no short month, so water law predicts their result as well. Among the seven districts that run short, five match.

The rows fit together. Water use follows the shortfall where water falls short and the record shows that water. Equal sharing fails where water moves by order along a canal, and it fits best where a canal rotates. Position shows nothing where water is allotted per acre, and nothing moves where water is not short. The two districts that do not match are the two whose records miss the water their fields use.

Rotation on a river: the Upper Gila. The Upper Gila canals run on the Gila River's natural flow, which the Gila Water Commissioner divides by a 1935 court decree (Doyle 2022). Inside a canal, farms may take the water in rotation (Arizona Supreme Court 1955, 217–218), and wells supply more than half of the farm water (Gila Water Commissioner, n.d.). A rotation gives each

Table 5: What Each District's Rules and Records Predict, and What I Find

District	Its rules and records	What they predict	What I find	Match
Water falls short, and the record shows the water				
Maricopa–Stanfield	CAP by order along canals; the district reports its canal pumps	Fields without a well lose water in short months; equal sharing fails; losses in the cut follow the model	Slope −18.2 (5.2). Equal sharing rejected. Cut −8.5 (2.4). Fallowing 16.7 (5.0).	Yes; own wells no
New Magma	CAP by order along canals; farms barely replaced lost CAP with wells	The same	Slope −12.6 (5.4). Equal sharing −4.5, upstream first 38.9, bar 20.4. Cut 1.5 (3.2). Fallowing −0.6 (7.0).	Yes; cut and fallowing no
Harquahala	CAP, but farms pump: 79 of 648 fields have no well	Those fields lose water and the rest do not; too few to pick a rule	Slope −34.8 (12.6) without a well, −2.2 (9.3) with; gap −32.6 (15.4). Search under its bar (37.9, 49.2).	Yes; shortage term no
Upper Gila	Rotation inside canals, decree across them; each canal reports its diversions	Equal sharing fits best; fields without a well lose farmed land where the river gives least	Equal sharing 57.3, upstream first 20.3, bar 10.9. Slope −6.3 (2.2). River, farmed share: −6.3 (1.7) without a well, −1.1 (1.6) with.	Yes; river by month no
Water is allotted per acre				
San Carlos	Stored water by decreed acre, set each January	Position does not matter; fallowing follows the year's water and each field's calendar	Search 4.3, bar 57.3. Slope 2.4 (3.6). Fallowing 45.3 (7.3), mostly calendar.	Yes
Senior rights: water does not fall short				
Wellton-Mohawk	Water 1.30 to 1.42 times demand	No slope; no rise in fallowing	Search 1.5, bar 457.8. Slope 16.6 (9.2), the opposite sign. Fallowing 0.3 (7.0).	Yes
Three Yuma districts	Water 1.4 to 2.5 times demand	No short month; no rise in fallowing	No short month. Fallow share −2.1, −9.5, −1.8 points.	Yes, all three
Water falls short, and the record misses water				
Central Arizona	District wells first; records give capacity, not which wells ran	A record of that kind hides the slope: Maricopa–Stanfield's falls from about −18 to about −1	Search 4.7, bar 10.2. Slope 7.9 (5.6). Cut −3.7 (2.4), fallowing 6.5 (4.8): right sign.	Cannot tell
Hohokam	Duty to "equitably deliver"; half of the acres without a well farmed through 2023 on water the record does not show	A flat cut; no rule predicts a positive slope	Cut −1.0 (6.8). Slope 18.1 (7.3), the wrong sign, between parts of the district.	No
Count, 11 districts: yes 9 (4 with senior rights), cannot tell 1, no 1.				

Notes: A district counts as a match when its main prediction holds; the last column names any part that does not. Slope: 100 times log monthly ET per unit of the shortfall, fields without a well, over all rules, at a take of 0.70 in Maricopa–Stanfield and New Magma, 0.62 in Harquahala, 0.64 in Central Arizona, and 1.00 elsewhere; 2016–2021 in Maricopa–Stanfield, Central Arizona, Harquahala, and Hohokam, 2016–2023 elsewhere; standard errors in parentheses, clustered by owner and by year-month. Search and rule scores: held-out gain in fit, in basis points, against the district's bar. Cut: water use in 2022–2023, or farmed share where stated, per standard deviation of the predicted rise in shortfall. Fallowing: points per unit of total stress, fields without a well, with field and year effects; in Central Arizona, the version built from its documents. Fallow share: 2022–2023 against 2016–2021. Three Yuma districts: Yuma County Water Users, Yuma Mesa, Yuma Irrigation. I built the model on Maricopa–Stanfield, and earlier Upper Gila estimates were known. I wrote the predictions for New Magma, Harquahala, Wellton-Mohawk, and the Yuma districts before their estimates. San Carlos's, Central Arizona's, and Hohokam's rows I read from their documents and records after the estimates; my design for San Carlos had expected a negative slope.

field its turn in proportion to its demand, so the search should find equal sharing here, and it does. On each canal's reported diversions, over 43,200 field-months of 932 fields of 310 owners on ten canals, the best rule scores 57.5 against a bar of 10.9. It sits at a take of 1.00 and κ of 0. Equal sharing almost ties it at 57.3, and upstream first along a line falls to 20.3. Over all rules the slope is −6.3 without a well (standard error 2.2) and −6.2 with one (standard error 1.5).

Reported diversions depend partly on what farmers ordered; the water the decree assigns each canal from the river does not. The monthly search finds nothing on the river's water (best score −4.5, bar 15.1). A daily river list may not time water inside a season, as reported diversions do. I have not tested that reading. I run the annual design on the river's water, comparing each canal with its own 2016–2021, because the Upper Gila had no 2022 cut. Per standard deviation, fields without a well lose 2.9 percent of water use (standard error 1.8) and 6.3 points of farmed share (standard error 1.7). Fields with a well lose 1.2 and 1.1 (standard errors 1.4 and 1.6). Reported diversions give nothing in that design (3.7, standard error 2.2).

Equal shares of stored water: San Carlos. San Carlos allots its stored water by the decreed acre, set each January, so position along a canal should not decide who gets it.[14] The search finds nothing (best score 4.3, bar 57.3), and the slope is near zero (2.4, standard error 3.6). Fallowing rises with total stress (45.3 points per unit, standard error 7.3). That stress is mostly each field's crop calendar against the year's water, not its position. The calendar explains 58.5 percent of it, and position by year 0.8 percent.

District wells first: Central Arizona. Central Arizona runs its own wells, which supply about half its water, and, by accounts of its operations, fills the remaining orders with CAP.[15] The search finds little (best score 4.7, bar 10.2). The slope has the wrong sign, and I cannot tell it from zero (7.9, standard error 5.6). Fallowing and water use in the cut have the right sign, and I cannot tell there either (6.5, standard error 4.8, and −3.7, standard error 2.4). Its records give each well's capacity but not which wells ran. On such a record, Maricopa–Stanfield's slope falls from about −18 to about −1, so the estimates may say more about the records than the rule.

The exception: Hohokam. Hohokam is the one district where the model goes wrong. Its search clears the bar (best score 39.3, bar 28.8), but at the edge of the grid and with the wrong sign. Fields the model calls short use more water (18.1, standard error 7.3). At a take of 0.70 I cannot tell that slope from zero (11.5, standard error 7.6). Its fallowing does not follow total stress (−13.5, standard error 9.7).

The reason is partly in the records. They miss water. Half of the acres without a well were still farmed in 2023. Their water use above fallow fields, 2,959 acre-feet, exceeds all the CAP water that could have reached any field that year (1,861). So a shortfall here need not mean a shortage. The slope is a contrast between parts of the district (35.5, standard error 13.5), not a loss along a canal

[14] San Carlos Irrigation and Drainage District, Water Policy (undated), and the Globe Equity decree, Art. VII. Natural flow goes by each account's priority date, and the CAP water of short years reached only some farms. The model sees neither split.

[15] The district's annual water user agreements of 2016 and 2022 reserve to it the choice of which wells to run, and its 2022 application for federal funds describes its supply (CAIDD 2022). The order, district wells first and CAP for the rest, comes from accounts of the district's operations; I have not found it in a district document.

(1.7 within parts, standard error 10.3). The district's stated duty is to "equitably deliver" water,[16] and I find no document that gives it an order along its canals. The flat cut (−1.0, standard error 6.8) is what such rules predict. No rule predicts a positive slope. Why it is positive and not zero, I do not know.

8.7 A Shortage on Paper Does Not Produce the Result

For the senior-rights districts I run the same satellite data through the same code and network rules, on water that is not short. In Wellton-Mohawk, water covers crop demand in every March–October month but two. The search finds nothing (best score 1.5, bar 457.8). The slope has the opposite sign from the short districts (16.6, standard error 9.2). Fallowing does not follow total stress (0.3, standard error 7.0).

I then remove water on paper. I lower the share of diverted water reaching crops from 0.7 to as low as 0.3 in the four senior-rights districts, 15 runs in all. The fields' real water does not change, so a model that finds a slope in any shortage it is given would find one here. In three of the four districts no rule clears the bar with the right sign. Wellton-Mohawk and Yuma Mesa stay under it in every run, and Yuma County Water Users clears it twice with the wrong sign. Two slopes turn negative without clearing it. Wellton-Mohawk's is −7.4 to −10.0, which I cannot tell from zero, and Yuma Mesa's at one share is −6.7 (standard error 2.0). Fallowing does not rise. Of 12 regressions with enough stress, 11 are negative and the largest is 6.4 (standard error 5.7), against 16.7 in Maricopa–Stanfield.

Yuma Irrigation is the exception. At a share of 0.6, the highest that makes any month short, the search clears the bar (40.5 against 34.5). The slope has the right sign (−51.8, standard error 16.7). The result fades at lower shares. Over all rules at a take of 1.00, the slope goes from −88.7 at 0.6 (standard error 31.9) to 11.8 at 0.3 (standard error 12.3). Before the runs I listed a real limit below 0.7 among three possible outcomes and named Yuma Irrigation and Yuma Mesa as its likeliest places. Reclamation's accounting puts their consumptive use at 0.53 to 0.56 and 0.52 to 0.70 of diversions, against about 0.7 in the other two. Yuma Irrigation's short months at 0.6 fall in the peak months, as such a limit would. Yuma Mesa clears no bar, so the limit appears in one district of the two. I read it as a possible limit, not a finding.

8.8 What Shortfalls Cost in Ordinary Years, and What a Remedy Would Need

The slopes put a price on the shortfalls. Each allocation below changes the shortfalls, and the slope turns that change into a change in water use. Each crop's yield response to water and Arizona revenue per acre turn it into gross revenue, with slopes and pumping held as estimated (Appendix A15). In Maricopa–Stanfield in 2016–2021, giving fields without a well all they claim would add $213,000 a year (range $91,000–339,000). The gain is $7 per acre on 31,434 acres, or 0.5 percent of their water use and gross revenue. The gain is small because canal pumps and CAP together nearly met demand, and the mean shortfall is 0.03. The loss is uneven, at $17 per acre in the third of fields with the largest shortfall and $1 in the smallest third. The cut is where the larger sums lie. A predicted rise in shortfall one standard deviation larger lowers the farmed share

[16] Hohokam Irrigation and Drainage District, "About Us" page, on its 1985 contract with Reclamation.

by 5.9 points, about $75 of gross revenue an acre a year.

The lost water shows in the crops once the cut years are included. Each crop's yield proxy rises within a field by 1.0 to 1.8 percent for each percent of its seasonal water use relative to demand (Appendix A14). In 2017–2023, a field without a well whose seasonal shortfall is 0.1 above its usual level has a yield proxy 3.7 percent lower (standard error 1.5). In 2017–2021 alone the seasonal shortfall hardly moves within a field, and I cannot tell (slope 34.4, standard error 32.5).

The same water can be divided differently. Sharing it equally changes almost nothing ($3,000 a year). Serving fields without a well first, as Maricopa–Stanfield did in 2023, gives them all they claim. It moves 3,364 acre-feet a year away from fields with a well and brings the others the same $213,000, or $63 per acre-foot moved. Whether it pays depends on what fields with a well do when they lose that water. If their wells replace all of it, the change pays whenever pumping costs less than about $63 an acre-foot, a cost I do not observe. If they lose water at the slope of fields without a well, the gain nearly cancels ($15,000). At their own estimated slope they lose $490,000 (range $57,000–856,000), and the district loses $277,000. My estimates do not show own wells replacing canal water, so I cannot say that it pays.

Investments do not repay their capital in ordinary years. A new well with pipeline costs $163 per acre a year before energy (Maricopa–Stanfield Irrigation and Drainage District 2024). The gain is $7 per acre, or $17 in the third with the largest shortfall (4 and 11 percent of cost). Each acre-foot filled is worth about $80, more than the $36 of capital per acre-foot at full use, but the fields without a well lack only 0.09 acre-feet per acre. Since CAP nearly vanished in 2023, the case for a well rests on years the calculation does not price. Lining the Upper Gila canals would not pay either. The model has no loss that grows with distance, so I express lining as a rise in the share of diverted water that reaches crops, and the size of that rise is an assumption. A rise from 0.7 to 0.8 adds $58,000 a year (range $28,000–87,000), or 1 to 4 percent of capital. A rise to 0.9 gives 2 to 7 percent, before any seepage that returns to the aquifer.

9 Conclusion

This paper asks whether farms with the same water rights share the shortage from a long drought equally. Arizona is a hard case for finding a gap, because its water rules are well developed. Water rights and contracts decide which users lose water first, and the 2022 cut in Colorado River water was carried out as the rules required. To answer the question, I trace Arizona's canals from their sources to every field and measure, for each field and month, how much water the fields upstream and downstream of it use.

Farms with the same rights do not share the shortage equally. In a drought month, a field uses less water when the fields upstream of it use more, and not when the fields downstream of it do. The water follows the direction of the canal. In junior-rights districts, position decides who goes short. A field that usually sits low on its canal is left fallow more often in drought. A planted field grows less when its upstream neighbours plant more than usual. In senior-rights districts, a field loses water when the whole canal draws more, wherever the demand sits, as on a canal that cannot carry its peak. Over the drought years of 2019–2023, position costs junior fields $3.4 million at net return, and up to $26.0 million at crop revenue, mostly through fallowing.

Position matters when the canals run short. Junior fields lose most in 2021, the year their irrigation districts were in drought. The drier years of 2008–2015 show no position effect, and CAP deliveries were not cut then.

Districts with monthly water records show how the gap arises. In months when canal supply fell short of crop demand, fields with more crop demand upstream received less canal water, and fields without a well used less water. Where district wells pump into the canals marks which fields keep water. Shuffled supply dates and shuffled positions do not reproduce the pattern. The districts' rules decide which position matters. In the Upper Gila canals, which rotate water, a field's canal matters and its place within the canal does not. In the senior districts, where water does not fall short, nothing moves. Where the records miss water, as in Central Arizona, I cannot tell. In the 2022 cut, announced in advance, the same shortfall predicted which fields lost water and went fallow.

The findings do not show who causes the gap. A district may serve the upstream farms first when water is short, or farms upstream may use more water than the rules intend. Both would produce the same pattern. So would a canal that cannot carry what every field needs at peak demand. A shortage shared in equal proportion at every turnout would leave no gap. The study also has limits. The crop measures come from satellites and are proxies for yield. I do not observe the water delivered to each field, so I cannot separate delivery schedules, losses along the canal, and gate operations. District maps stand in for the areas each district serves. Most fields in the sample sit in the largest canal systems of one state and are observed over one main drought. The revenue figures assume a link between the satellite index and revenue, and they measure lost revenue, not lost profit or welfare.

These results matter for how shortage rules are judged. In 2022 Arizona's rules worked as written at the source. Inside the canals, farms with the same claims did not lose the same amount. Programs that pay farms to leave land fallow might consider where along the canal the fallowed land lies, although water saved upstream does not always reach the fields below. Inside a district, the cheapest remedy may be the allocation rule. In Maricopa–Stanfield, serving fields that cannot pump first would pay only if fields with wells replace the water they give up, which my estimates do not show, and new wells and lined canals do not repay their cost in ordinary years. The method in this paper makes such questions possible to study. It uses only public data on canals, land elevation, and field boundaries, so it can be applied to other irrigated regions and to other networks that deliver a resource to users one after another.

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Appendix A. Supplementary Results and Model Details

Appendix A1 gives the model's derivations. Appendices A2 and A3 give the details of the delivery model of Section 8.1. Appendices A4–A9 support Sections 7.1 and 7.2, and Appendices A10–A15 support Section 8. Appendix A16 reports the CAP curtailment schedule and the outcomes in the districts it cut.

A1 Model Details

This appendix collects the derivations and auxiliary assumptions behind the model of Section 3. Notation follows Section 3.2: S_t is system supply, w_i^* the formal entitlement, ε_{jt} the upstream leak with monitoring intensity μ, ρ the return-flow fraction, λ_{ij} the recovery share, Δ_{it} the positional wedge of equation (2), $\ell_i(S_t) = A_{it}^0 - w_i^*$ the cushion, and $q_{it} = [\Delta_{it} - \ell_i(S_t)]_+$ the realized shortfall.

Monotonicity of the shortfall in position. The head-to-tail level-gradient prediction of Section 3.2 rests on two maintained assumptions: recovery λ_{ij} constant or weakly decreasing in hydrologic distance (the standard case for tailwater and shallow-percolation pathways) and entitlements that do not fall toward the tail faster than adjacent return-flow recovery, $w_{i+1}^* \geq \rho\lambda_{i+1,i}\, w_i^*$ (automatic when entitlements are equal or rising, since $\rho\lambda_{i+1,i} \leq 1$, and mild when they decline, because only a small share of applied water is recoverable at the immediately downstream turnout). Together the two assumptions sign the cushion increment,

$$\ell_{i+1}(S_t) - \ell_i(S_t) \;=\; \rho\lambda_{i+1,i}\, w_i^* \;-\; w_{i+1}^* \;-\; \rho \sum_{j<i} \big(\lambda_{ij} - \lambda_{i+1,j}\big)\, w_j^* \;\leq\; 0,$$

so the cushion falls, rather than rises, along the canal. Without these assumptions, monotonicity depends on primitives. Comparing adjacent users, $q_{i+1,t} \geq q_{it}$ holds whenever

$$\big(1 - \rho\lambda_{i+1,i}\big)\, \varepsilon_{it} \;+\; \rho \sum_{j<i} \big(\lambda_{ij} - \lambda_{i+1,j}\big)\, \varepsilon_{jt} \;\geq\; \ell_{i+1}(S_t) - \ell_i(S_t), \tag{13}$$

where the left-hand side is the wedge increment $\Delta_{i+1,t} - \Delta_{it}$ and the right-hand side is the cushion increment; the condition is sufficient rather than necessary (when neither user's constraint binds, both shortfalls are zero and the comparison holds trivially). Two limits clarify the primitives: as monitoring intensity $\mu \to \infty$, all ε terms vanish and the left-hand side collapses; when S_t is high, the cushion is large and $q_{it} = 0$ regardless of the wedge. The left-hand side shrinks (and monotonicity can fail) when ρ is large or when recovery $\lambda_{i+1,j}$ rises with hydrologic distance, because returned water is recaptured before it compounds into a downstream wedge; the right-hand side turns positive (and monotonicity can likewise fail) when entitlements fall toward the tail faster than adjacent recovery, because a small own entitlement enlarges the next user's cushion. Under the maintained assumptions the left-hand side of (13) is non-negative and the right-hand side non-positive, so the inequality holds; this is the prediction behind the fallowing of fields that usually sit low on junior canals (Section 7.2.2). Groundwater substitution can still flatten the wedge in theory. Evidence for buffering comes from wells in the CAP districts: in Harquahala and New Magma a field's canal shortfall lowers its water use where the field has no well and not where it has one (Section 8.4).

Drought amplification. Define the binding margin $x_{it} = \Delta_{it} - \ell_i(S_t)$ and the adjacent-user increment

$$\delta_{it} \;=\; \big(\Delta_{i+1,t} - \Delta_{it}\big) \;-\; \big(\ell_{i+1}(S_t) - \ell_i(S_t)\big),$$

which is non-negative under the maintained assumptions of the monotonicity paragraph. Then

$$q_{i+1,t} - q_{it} \;=\; [x_{it} + \delta_{it}]_+ - [x_{it}]_+,$$

which is non-decreasing in x_{it} for fixed $\delta_{it} \geq 0$. Since Δ_{it} does not depend on S_t and the cushion moves one-for-one with supply ($d\ell_i/dS_t = 1$ from equation (1) at $\varepsilon = 0$), we have $dx_{it}/dS_t = -1$: a lower supply draw raises the binding margin for every user, so the positional shortfall gap $q_{i+1,t} - q_{it}$ weakly widens as S_t falls. The outcome gradient is amplified further, on the domain where $A^0_{it} > 0$, because the denominator A^0_{it} in the outcome equation also falls with S_t. The increment δ_{it} is invariant to S_t under the maintained assumptions (both the Δ-increment and the ℓ-increment are supply-invariant), so the amplification operates purely through the binding margin. This is the comparative static behind the drought interactions of Section 7.1 (Table 3).

Pricing. Index usage-based pricing by τ and let monitoring intensity rise with it, $\mu'(\tau) > 0$, so the leak shrinks: $\partial\varepsilon_{jt}/\partial\tau < 0$. The direct effect on the wedge is

$$\frac{\partial \Delta_{it}}{\partial \tau} \;=\; \sum_{j<i} \big(1 - \rho\lambda_{ij}\big)\, \frac{\partial \varepsilon_{jt}}{\partial \tau} \;\leq\; 0,$$

strict for $i > 1$ whenever $\rho\lambda_{ij} < 1$ for some $j < i$. If pricing also reduces runoff and recharge, $\rho'(\tau) < 0$ adds an offsetting positive term through the recovery share, and the net theoretical effect is ambiguous. The district pricing records do not resolve this ambiguity empirically, and the two channels are not separately identified by the design (Section 7.2.1).

Canal size: level versus interaction. Taking the leak as i.i.d. across upstream users, $E[\Delta_{it}] = \sum_{j<i}(1 - \rho\lambda_{ij})\, E[\varepsilon]$ scales with the number of upstream extractors: the wedge is mechanically larger in bigger systems. A test on the drought *interaction* by canal size would require more: the wedge must also *bind* more readily in larger systems during drought. A sufficient auxiliary assumption is that the cushion per user is tighter in larger systems. Holding season supply S_t and the entitlement distribution fixed, this follows within the model: upstream net extraction $\sum_{j<i}(1 - \rho\lambda_{ij})\, w^*_j$ is mechanically larger where more users sit upstream, so residual supply A^0_{it}, and with it the cushion, is lower for a larger share of users in big systems. Because supply itself tends to scale with system size, the cross-system version is an assumption rather than a derivation: it requires that cushions per user not scale up one-for-one with capacity. Under it, a low S_t pushes a larger share of users past the binding threshold, and the wedge itself is larger. The assumption is auxiliary to the core identity, and the design of Section 6 tests neither version.

Priority queue and enforcement. The seniority comparative static of Section 3.4 rests on asymmetric enforcement. Under an enforced queue, the entitlement margin is served in priority order, and the leak margin retains the physical ordering. Abstracting from return-flow recovery of entitlement claims for readability (the general case nets those claims by the same recovery logic),

senior user i's residual supply for entitlement purposes is

$$A_{it}^{\text{pri}} \;=\; S_t \;-\; \sum_{j \in P(i)} w_j^* \;-\; \sum_{j<i} \big(1 - \rho\lambda_{ij}\big)\, \varepsilon_{jt},$$

where $P(i)$ is the set of users senior to i by priority date, not position; the middle sum is entitlement claims served ahead of i in the queue; and the last sum is the physically accumulated leak net of recovery, which no legal queue can reorder. Delivery is $w_{it} = \big[\min\{w_i^*, A_{it}^{\text{pri}}\}\big]_+$. For senior users, $P(i)$ is small, so $A_{it}^{\text{pri}} \geq w_i^*$ except for the leak term, giving $q_{it}^{\text{sen}} \approx 0$; the approximation fails exactly through the leak sum and mid-season shocks, which is the asymmetric-enforcement point. Juniors have large $P(i)$ and compete over the residual pool. The priority queue is enforceable ex post through curtailment authority (the "call on the river" of Section 3.1, with the Bureau of Reclamation as watermaster, curtailing Central Arizona Project deliveries first under the 2007 Interim Guidelines). The leak is not detectable in real time within a season. Because curtailment operates only ex post, realized mid-season supply shocks can reach senior users before the queue reorders delivery, which is why the senior shortfall is approximately rather than identically zero. The empirically relevant queue runs at the river level: the Bureau administers Colorado River priorities across districts. Within districts the order of service is an operating matter; where parcel-level priorities exist (Salt River's Kent classes, the San Carlos and Upper Gila decree schedules), they attach to land, not to canal position. The comparative static therefore treats the river as the canal and districts as its users.

Parameterization choices. Five reduced-form choices are implicit in Section 3. First, λ_{ij} is treated as constant or weakly decreasing in hydrologic distance; if recovery instead rose steeply with distance, non-monotone wedges become possible (see the monotonicity discussion above). Second, the leak is additive rather than proportional to entitlements; a proportional alternative ($\varepsilon_j \propto w_j^*$) would generate additional cross-district predictions not pursued here. Third, the leak is exogenous undetected over-diversion rather than a strategic choice; endogenizing it would require equilibrium monitoring-and-sanctions machinery without changing the sign predictions the paper uses. The empirical counterpart of monitoring intensity μ is heterogeneous across districts: metered Bureau of Reclamation turnouts (e.g., Yuma, Wellton-Mohawk) report per-device accuracy near $\pm 3\%$ (Stringam, Sauer, and Pugh 2003). Smaller systems allocate per acre with no volumetric measurement, and Arizona imposes no statewide metering requirement. Fourth, equation (1) treats upstream diversions as $w_j^* + \varepsilon_{jt}$ regardless of upstream availability. It is exact when upstream constraints are slack ($A_{jt} \geq w_j^* + \varepsilon_{jt}$ for all $j < i$) and physically coherent at i when additionally $A_{it} \geq 0$, which caps the shortfall at the entitlement ($q_{it} \leq w_i^*$); $A_{it}^0 > 0$ is required separately for the outcome equation's denominator. When intermediate users are themselves rationed, realized diversions fall short of entitlements and the shortfall cascades toward the tail, preserving the level and directional predictions; the drought-amplification comparative static, by contrast, is derived on the domain where upstream diversions remain feasible. In deep enough drought, rationed upstream users can no longer divert entitlement plus leak, and the leak-attributable wedge itself shrinks. Fifth, the comparative statics hold the primitives (μ, ρ, λ_{ij}, and the leak distribution) fixed across drought and non-drought conditions. The scaffold is deliberately static and non-strategic: entitlements w_i^* do not adjust through transfers or rights markets within the model.

Pro-rata rationing. The institutions described in Section 3 include pro-rata shortage reductions, under which districts scale allocations by a common ration factor $\zeta_t \in [0,1]$ rather than letting the tail absorb the aggregate shortfall. Equation (1)'s residual-claiming rule is the opposite polar case, in which administration cannot adjust within the season. The model's predictions survive under faithful pro-rata rationing, through a slightly different channel. With scaled entitlements $\zeta_t w_j^*$, the leak is extraction beyond the scaled entitlement (its absolute scale is governed by monitoring, not by the entitlement), and the wedge retains the form of equation (2), so the level and directional predictions are unchanged. For drought amplification, write the cushion under rationing as $\ell_i^\zeta = S_t - \zeta_t C_i$, where $C_i = \sum_{j<i}(1-\rho\lambda_{ij})\, w_j^* + w_i^*$ cumulates claims toward the tail. A faithful ration ties the factor to supply, $\zeta_t = S_t/C_N$, so $\ell_i^\zeta = S_t\,(C_N - C_i)/C_N$, which shrinks proportionally with S_t for every user; under the entitlement condition of the monotonicity paragraph above, C_i rises toward the tail, so the cushion also falls toward the tail (reaching zero at the last user, where any leak binds immediately) and $C_N = \max_i C_i$, making the ration feasible for every user. Drought therefore pushes users past the binding threshold from the tail upward, and the wedge is unchanged. This is the same amplification logic as the main text, with the aggregate cut shared rather than concentrated at the tail. The residual-claiming rule of the main text thus sharpens, rather than creates, the positional predictions; the empirically relevant regime (administered scheduling ex ante, physical residual claiming for realized within-season shocks) lies between the two poles, as the timing discussion in Section 3.2 states. The CAP tier cuts of Appendix A16 are close to this pole. Announced tier cuts are anticipated shortfalls, so the aggregate reduction is shared, and its effect on a field may vary with surface dependence and groundwater conditions.

Model-to-empirics map. Table 1 maps the model's sign predictions to the estimates of Sections 7.1 and 7.2. Two refinements matter. First, the seniority row compares junior-rights and senior-rights districts estimated separately. I do not test the difference, and rights are recorded for irrigation districts, so the priority queue within a canal is not tested. Second, an administrative pool cut enters as a reduction in district headgate supply S_t, net of well injection (Appendix A16). In Maricopa–Stanfield and Central Arizona, district pumps supply that injection (Sections 8.5 and 8.6). The spatial exclusion of Section 6 has no counterpart in the model, which contains no spatial-correlation structure for the leak.

The design leaves out fields on sibling branches (Section 6.2). Their competition for the water of the trunk then acts as a correlated omitted regressor, not as classical measurement error. Its sign turns on the within-field covariance between demand on sibling branches and upstream demand.

A2 The Delivery Model: Details

The canal trees. Each tree holds the lines of the rebuilt canal network below a district's documented intakes and above its fields, with joins across roads, siphons, and pipes. Water leaves only at laterals, so a field linked to a main canal takes its water from the nearest lateral within 2 km. A field may lie at most 1 m above the canal where it takes water, at every level of evidence. Table A1 gives Maricopa–Stanfield's fields by level. Maricopa–Stanfield surrounds the Ak-Chin Indian Reservation, which is not part of the District (Maricopa–Stanfield Irrigation and Drainage District 2022c): the 38 fields on the reservation are left out of the District, and its tree does not run through the reservation's farm ditches (two links cut). Elsewhere the trees place 2,390 fields in Central Arizona, 817 of 845 in Hohokam (96.8 percent of demand), 745 of 896 in Harquahala (81 percent of demand), 594 in New Magma, 1,380 in San Carlos, 4,307 of 4,412 in Wellton-Mohawk (98.7 percent of demand), and 1,039 on the ten Upper Gila canals. Only Maricopa–Stanfield's tree has had my marks on imagery and the checks against decoys. In those checks, on images shown blind, a takeoff is visible at 11 of 29 clear turnouts (38 percent) and at 2 of 28 decoys (7 percent), points on the same laterals away from any ditch ($p = 0.006$). For the inferred intakes it is 6 of 19 against 0 of 20 ($p = 0.008$).

Table A1: Where Maricopa–Stanfield's Fields Take Water

Level of evidence	Fields	Share of demand, %
Evidence of the field's own connection		
Lateral within 30 m	441	30.8
Mapped ditch followed upstream to a lateral, or an intake I marked	714	31.0
Ditch with no detected turnout: intake inferred within 1 km of a lateral	627	23.4
Evidence from the fields around it		
Turnout of its ditch system	121	3.6
Turnout of a touching neighbor	283	4.3
No canal evidence		
Projection onto its line, in blocks	197	3.5
Ditches reach a registered well: off the canal	17	1.2
Line off the tree: not reachable	65	2.1

Notes: 2,465 district fields; a field takes the first rule that applies, with a marked intake first, and the first six levels place 2,383. The inferred link is the weakest: 42 percent of inferred gaps exceed 300 m. Share of 2016–2021 demand. Ditches: Pinal County's water-management layer, concrete ditches of 2 to 5 feet, read for direction from the local terrain. The canal search runs before any well label, so a well never cancels a visible canal.

Water. The placed fields receive the district's water in the month times their share of its demand, and that water, times e, enters at the head of the tree. In the CAP districts the water is Agricultural Settlement Pool and agricultural in-lieu water from CAP's delivery reports. New Magma's is in-lieu water, 96 to 98 percent of its farm deliveries in 2016–2022 and 86 percent in 2023. In Wellton-Mohawk and the Yuma districts it is the diversion in Table 4 of Reclamation's decree accounting, and every year's months sum to the printed totals. In San Carlos it is the decree's split of the project's gravity diversion at Ashurst-Hayden: stored water equally per decreed acre, natural flow by Article VI(3) on the month's mean flow, and Agricultural Settlement Pool water at the root. The split comes to 0.43 to 0.48 of the project. The project's wells and private wells are not in that series. The Upper Gila's water comes in Appendix A3.

The allocation. Let a point v of the tree receive a flow F_v, with claims c_v from its own fields and claims b_v on the branches below. Its fields take o_v (equation (8)), and their delivered share is $\min(1, o_v/c_v)$, or 1 when $c_v = 0$. A field with no demand of its own keeps its point's share. The rest, $R_v = F_v - o_v$, goes to the branches. At a side line with claims N_s and a continuing line with

claims N_c, the side line takes $R_v N_s/(N_s+N_c)$ when $R_v \geq N_s+N_c$ and the amount in equation (9) otherwise; the continuing line takes what remains. At other branch points each branch takes a share by its claims. A claim on a branch is net of any pumped water that enters inside the branch. Where a district has canal pumps, the model runs in two steps. In the first, only pumped water flows, at $\kappa = \kappa_J = 0$, and a point's claims are met by a share s_v. In the second, the main supply flows under the rule over the claims still unmet, $c_v(1-s_v)$, and meets a share m_v of them. The delivered share is $r_v = s_v + (1-s_v)\, m_v$. In districts with a separate water record for each canal, each canal's water enters at its own root.

Canal pumps. The 130 canal pumps of Maricopa–Stanfield come from its 2019–2020 well map. Of these, 63 lie within 250 m of the tree and 67 only near a mapped ditch, and 37 have no rate. They pumped 107,428 acre-feet in 2019 and have a capacity of 461 cubic feet per second at 115 points of the tree. Unrated pumps take a capacity from their 2019 volume at the median use of rated canal pumps, 0.32. In each year they pump their share of the district's reported well water. The months follow the canal claims at the head, net of CAP, up to a share α_y of capacity, set so that the year sums to that share; a year whose need falls below it pumps the need. Within a month the pumps fill the claims below them from the bottom of the tree up. Each pumps the least of its capacity, λ times its 2019 volume, and the claims below it that pumps further down do not cover, with λ set so that the month's total is pumped. Water that no claim below a pump can take is not pumped. At a take of 0.70, of the 775,978 acre-feet of district well water reported in 2016–2021, the canal pumps' share is 664,856, following the canal need leaves 641,610, the placed fields' part of that is 616,923, and the pumps dispatch 613,494 of it, or 99.4 percent. The well step delivers what is pumped (largest gap 3.1×10^{-11} acre-feet), and total deliveries equal the lesser of supply and claims (5.1×10^{-11}). Thirty-eight pumps, with 25 percent of the canal pumps' 2019 water, stand on ditch systems with no mapped turnout, and they enter at the highest point of the fields on their system. The map's 174 inventory wells, which did not pump in 2019, and 66 private wells with no rate stay out. Monthly pumping by pump is not observed.

Local pumps and the take. Maricopa–Stanfield has 163 local pumps: 24 district wells that are small or not connected, and 139 private wells rated at 100 gallons per minute or more. A local pump serves the field it sits on, or the nearest within 200 m, and the owner's fields it can physically reach: on its ditch system and not above the well by more than 1 m, or within 400 m if the well is not on a ditch. It serves 431 of the 2,383 placed fields. In the other districts a well rated at 100 gallons per minute or more serves all its owner's fields, at any distance: 686 of 817 placed fields in Hohokam, 701 of 1,039 in the Upper Gila, 417 of 1,380 in San Carlos, 101 of 594 in New Magma, and 1,221 of 4,307 in Wellton-Mohawk. From the water record alone, $\tau^* = 1 - e\times$ (local district well water reaching placed fields) / (the need of fields with a local pump) is 0.737 in 2016–2021, against a best take of 0.70 on water use. Private wells have no record, so τ^* is an upper bound.

Demand. A field's monthly irrigation demand is its acres times $\max(0, \widetilde{K}_{it}\,\mathrm{ETo}_{it} - P^e_{it})$, with the locally calibrated crop coefficient $\widetilde{K}_{it} = c_{gm}K_{it}$ (equation (7)). K_{it} is the FAO-56 crop coefficient on the field's crop calendar: the timing of its crop in the 2017–2019 Sentinel-2 imagery, with the FAO-56 stage lengths between onset and harvest and alfalfa's summer dormancy. Crops without a calendar, about 5 percent of field-years, take the mean coefficient of the district's calendar crops in that month. ETo_{it} is gridMET reference evapotranspiration in the field's 4 km cell, scaled each month to the Penman–Monteith value at the AZMET station nearest the district. P^e_{it} is effective rain, $0.6P - 10$ millimeters for monthly rain P up to 70 and $0.8P - 24$ above, and zero when

negative. The calibration factor is

$$c_{gm} \;=\; \frac{\sum A_i \,\mathrm{ET}_{it}}{\sum A_i \,K_{it}\,\mathrm{ETo}_{it}},$$

where ET_{it} is the field's OpenET consumption and the sums run over the farmed field-months of crop group g in calendar month m of 2017–2019, the years of the Sentinel-2 calendars, within the region (Maricopa–Stanfield, Central Arizona, Hohokam, and Harquahala together, or five Upper Gila canals). In those CAP districts each factor rests on 201 (sorghum) to 6,639 (alfalfa) field-years; Upper Gila crop groups with fewer than 50 field-years (corn, other hay, small grains) take the factor of both regions together. A fallow field demands nothing. New Magma, San Carlos, Wellton-Mohawk, and the Yuma districts take the CAP region's factors, and the Upper Gila's added canals those of the first five, with no estimation on their own fields. Demand over OpenET less rain comes to 0.93–1.03 by year in Wellton-Mohawk, 0.79–1.02 in Yuma County Water Users, 1.02–1.31 in Yuma Mesa, 0.84–1.05 in Yuma Irrigation, 0.91–1.15 in New Magma, and 0.95–1.13 in San Carlos.

Irrigation efficiency. Demand is crop consumption, so a field must receive $1/e$ acre-feet at its turnout for each acre-foot its crops consume. I take each district's e from its water balance: the ET of its farmed fields net of the ET met by effective rain (OpenET, all twelve months), over all the water applied, surface water at the turnouts plus groundwater as the district or the state reports it, over 2017–2019, the years of the crop-coefficient calibration, rounded to two decimals. The ratio is 0.72 in Maricopa–Stanfield (0.71 to 0.75 by year; district wells only, so private pumping is missing and the ratio is an upper bound), 0.73 in Central Arizona (2017, the only year with its district-well total), and 0.75 in Hohokam (private wells, from the state's annual water-use reports). Harquahala has no measured pumping in those years (its modeled pumping gives 0.60 in 2020), so it takes the pooled value of the other three, 0.73. Little of the rest is canal loss: CAP's recorded deliveries to Maricopa–Stanfield exceed the district's own delivery figures by 3.5 to 5.5 percent a year (Maricopa–Stanfield Irrigation and Drainage District 2024). The districts added later take 0.70. In New Magma, OpenET less rain is about 0.85 of the water in 2016–2021, so 0.70 likely overstates its shortage in those years. In Yuma Mesa and Yuma Irrigation, consumptive use over diversions in Reclamation's accounting is 0.52–0.70 and 0.53–0.56 (Appendix A12).

Owners. Owners come from the 2025 CoreLogic tax roll. A field takes the owner of the largest parcel whose point lies inside it, or else of the nearest parcel point within 800 m. I keep only integer identifiers. The owner of record in 2025 stands in for the operator in 2016–2023.

Shuffles. Each draw reruns all 2,601 rules at the district's take, with 1,000 draws of each kind. *Dates:* each calendar month's water moves across the sample's years, and S_t and pumping follow. *Fields:* placed fields move to other fields' points on the tree, each with its own demand and well status; pumps are redispatched and S_t stays real. *Owners:* the tree's attachment points are listed in order along the tree, owners are put in random order and laid end to end along that list, and each keeps its fields' real order from upstream to downstream. Everything else is as for fields. Moving fields one by one scatters an owner's fields, so the draws are too narrow (their standard deviation is 0.54 of the clustered standard error in Maricopa–Stanfield); moving owners gives 0.82. A faster version of the code reproduces the live code's scores and coefficients on the real data and on draws recomputed afterwards. The shuffles hold the take fixed, so they allow for the search over rules but not over takes; the sign-flip bar allows for both.

A3 The Upper Gila Model

Rules. The Globe Equity Decree of 1935 lets a canal pool its rights and rotate deliveries among its fields, at 1 cubic foot per second per 80 acres. The decree does not regulate pumping. The later settlement meters the wells and lets decreed land take at most 6 acre-feet per acre a year of river and well water combined. The Gila Valley Irrigation District's wells give 53 to 66 percent of its farm water, 69,000 to 86,000 acre-feet a year. Some pumps feed the canals, but no record says which. The registry's owner field matches 125 live wells near the canals to a canal company or the district, about 182 cubic feet per second, and they stay local in the model.

Canals and water. The model has ten canals: San Jose, Montezuma, Union, Graham, Fort Thomas, Brown, Fourness, Smithville, Dodge-Nevada, and Curtis. Colvin-Jones took no water in any month of 2016–2023. Monthly diversions come from the Gila Water Commissioner's annual plates, read from scans and checked against the plates' own totals. The canals match the valley total within 1 acre-foot in 80 of 96 months. Brown and Smithville have one offtake each, so κ_J has an effect only there. The river version takes the flow at the Solomon gage (USGS 09448500) and passes it down the decree's list of rights by canal and priority year. Over March–October the list gives 54,859 to 84,006 acre-feet a year, against 36,415 to 62,588 reported. An upstream-first version beside it lets each diversion, in river order, take its fields' need at the head. Table A2 compares the versions with reported diversions before any data on water use. The decree on the river comes closest, and equal water per acre next; upstream first is furthest, though it omits return flows from the upper canals.

Table A2: Dividing the River Among the Upper Gila Canals, Against Reported Diversions

	All months		Short months	
Rule across canals	Error, AF	Correlation	Error, AF	Correlation
Decree on the river	149.7	0.949	113.7	0.933
Equal per acre	169.9	0.937	147.5	0.896
Upstream first, up to need	202.0	0.891	152.2	0.884
Upstream first, up to the full right	227.7	0.895	199.9	0.856

Notes: Each rule gives a canal's share of the month's reported valley water. Error: mean absolute error per canal-month. 572 canal-months in 52 months, March–October 2016–2023. Short months: valley water below the same calendar month's 2016–2023 mean (275 canal-months).

Search and slopes. Table A3 gives the search on each supply. On reported diversions the best cell clears the bar and has the right sign, and equal sharing almost ties it, at 57.34 against 20.29 for upstream first along a line at a take of 1.00 (39.89 and 11.23 at 0.70). Almost all the variation in a field's delivered share is between canals: only 0.01 percent of its variance lies inside a canal and month under equal sharing, so a field's canal carries the result. On the river no cell clears the bar. At equal sharing and a take of 1.00 the decree on the river gives −5.62 without a well (standard error 4.79) and −4.80 with one (3.54), the right sign at about half the size of reported diversions, and I cannot tell.

The annual design and pumping. The Upper Gila had no 2022 cut: reported water averages 45,085 acre-feet in 2022–2023 and 46,245 in 2016–2021, and on the river 2022 is the wettest of the eight years. So the design asks which canals gained or lost against their own 2016–2021. Table A4 gives the estimates. On the river, fields without a well lose water and farmed land where the river gives least, and fields with a well do not move; no orders enter these numbers. Reported diversions show nothing in this design. The Gila's dry time is July–September of 2019, 2020, and 2023. In those three summers the canals ran lowest and the Gila Valley district pumped most, 47,000 to

Table A3: The Upper Gila: Search and Slopes by Supply, March–October 2016–2023

Supply	Best	Bar	Take, κ, κ_J	All rules, no well	All rules, with a well
Reported diversions, ten canals	57.45	10.85	1.00, 0, 0.64	−6.31 (2.20)	−6.22 (1.48)
Reported diversions, five canals	45.25	13.25	1.00, 0, 0		
Decree on the river, ten canals	−4.48	15.05	1.00, 1.00, 1.00	−2.25 (3.02)	−0.91 (2.12)
Decree on the river, five canals	2.03	19.45	1.00, 0, 0		
Upstream first on the river, ten canals	−2.73	6.44	0, 0.64, 0	−0.12 (2.82)	−0.38 (2.73)

Notes: Best and bar: held-out gain in fit, in basis points, over 200 sign-flip draws. Slopes at a take of 1.00: 100 times log monthly ET per unit of the shortfall; standard errors clustered by owner and by year-month. 43,200 field-months of 932 fields of 310 owners on ten canals; 295 fields without a well. At the best rule on reported diversions the slopes are −10.53 (2.97) and −9.96 (2.07). S_t times no well alone: −4.16 (1.97) on reported diversions.

56,000 acre-feet against 18,000 to 31,000 in the others. Fields without a well then fallowed more than fields with one, by 9.6 points against 3.3. Those are eight points, descriptive, and owners with wells may differ in other ways. Fallowing on total stress has the right sign on reported diversions, 8.8 (standard error 6.5) without a well and 13.9 (3.7) with one, and most of it comes from this year's water, where a canal that fallows also diverts less. On the river it is −3.1 (5.4) and −0.4 (4.8), and I cannot tell.

Table A4: The Upper Gila: 2022–2023 Against 2016–2021, Per Standard Deviation of the Predicted Rise

	No well		With a well		No well minus well	
Supply	Water use	Farmed	Water use	Farmed	Water use	Farmed
River, decree	−2.9 (1.8)	−6.3 (1.7)	−1.2 (1.4)	−1.1 (1.6)	−2.2 (2.3)	−5.7 (2.3)
River, upstream first	−4.0 (1.8)	−4.4 (1.8)	−0.4 (1.4)	0.0 (1.5)	−3.2 (2.2)	−4.1 (2.3)
Reported diversions	3.7 (2.2)	0.2 (2.0)	1.3 (1.4)	−0.6 (1.1)	4.2 (2.3)	1.8 (2.1)

Notes: All rules at a take of 1.00; fixed 2016–2021 demand; each canal's actual water at its root; owner clusters. 1,039 fields, 338 without a well; 334 and 690 in the design. Before, 2020–2021 against 2016–2019, no well: decree water use 1.7 (1.8), farmed 0.1 (2.0); upstream first 2.0 (1.8), −0.1 (2.2).

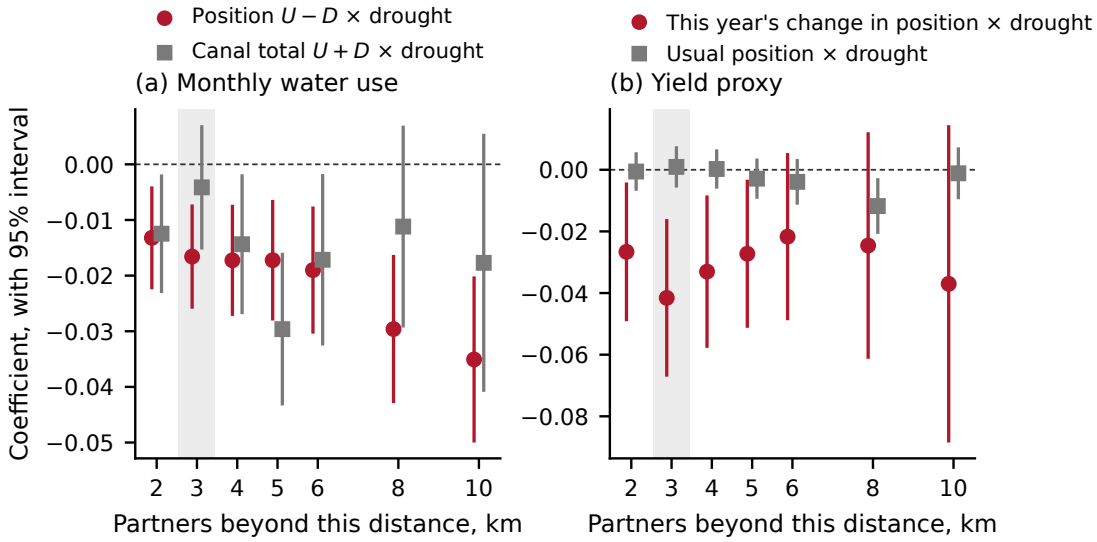
A4 Seven Cutoffs from 2 to 10 km

I repeat equations (4) and (5) with upstream and downstream fields beyond 2, 3, 4, 5, 6, 8, and 10 km. Figure A1 and Tables A5 and A6 report the estimates. Each cutoff changes which fields count on each side and which fields enter the sample. The sample falls from 25,389 fields at 2 km to 14,866 at 10 km.

In water use, the position coefficient is negative at every cutoff over all fields and in junior-rights districts. It is not significant at 5 km in junior-rights districts. In senior-rights districts it is near zero at 2 and 3 km and negative beyond, significant only at 8 km. The canal total is near zero at 3 km and negative at the other cutoffs. In junior-rights districts it is negative from 4 km on.

In the yield proxy, this year's change in position has $p < 0.05$ from 2 to 5 km over all fields. In junior-rights districts it is clear only at 3 km. It turns positive from 6 km on (+0.032, +0.053, and +0.056, on 4,916 to 3,714 fields). In senior-rights districts from 6 km on, this year's change in position and this year's change in canal total take large estimates of opposite sign. Few senior fields have fields more than 6 km downstream of them. The two changes move almost together there, so the model cannot separate them.

Figure A1: Position and Canal Total at Each Cutoff



Notes: All fields, 2SLS. Panel (a): equation (4); panel (b): equation (5). Shaded: the main cutoff, 3 km. Each cutoff is a separate regression on its own sample; the points are not joined for that reason. Bars are 95 percent intervals.

Table A5: Water Use at Each Cutoff, by Rights Group

		Position $U - D$ × drought			Canal total $U + D$ × drought		
km	Fields	2SLS	OLS	Reduced form	2SLS	OLS	Reduced form
All fields							
2	25,389	−0.013***	−0.018***	−0.014***	−0.012**	−0.036***	−0.013**
3	23,545	−0.017***	−0.022***	−0.017***	−0.004	−0.029***	−0.005
4	21,478	−0.017***	−0.024***	−0.017***	−0.014**	−0.033***	−0.016**
5	19,924	−0.017***	−0.023***	−0.018***	−0.030***	−0.044***	−0.033***
6	18,773	−0.019***	−0.023***	−0.019***	−0.017**	−0.034***	−0.019**
8	16,887	−0.030***	−0.032***	−0.031***	−0.011	−0.023***	−0.013
10	14,866	−0.035***	−0.037***	−0.036***	−0.018	−0.026***	−0.022*
Junior rights							
2	5,927	−0.015**	−0.017**	−0.017**	−0.005	−0.015**	−0.006
3	5,803	−0.020***	−0.022***	−0.021**	−0.002	−0.018***	−0.003
4	5,623	−0.016**	−0.020***	−0.016**	−0.016*	−0.026***	−0.019**
5	5,427	−0.011	−0.016**	−0.011	−0.032***	−0.038***	−0.037***
6	5,124	−0.013*	−0.016**	−0.012	−0.028***	−0.032***	−0.033***
8	4,566	−0.022**	−0.022***	−0.023**	−0.017	−0.024**	−0.017
10	3,862	−0.019*	−0.019*	−0.019*	−0.027*	−0.025*	−0.031*
Senior rights							
2	9,293	0.006	−0.008	0.006	−0.029**	−0.057***	−0.030***
3	8,524	0.002	−0.013	0.001	−0.034**	−0.053***	−0.034**
4	7,760	−0.012	−0.017*	−0.012	−0.030	−0.059***	−0.032
5	7,228	−0.015	−0.021*	−0.015	−0.049**	−0.065***	−0.054**
6	6,816	−0.032	−0.036*	−0.031	−0.004	−0.043*	−0.010
8	6,205	−0.077**	−0.082***	−0.079**	0.037	0.006	0.035
10	5,533	−0.056	−0.041	−0.056	0.026	−0.023	0.022

Notes: Equation (4). 2SLS: instrumented by predicted demand. OLS: realized demand, same sample. Reduced form: OLS on predicted demand. Fields: fields with upstream or downstream fields beyond the cutoff. Two-way clustering by field and by crop by year by month. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table A6: Yield Proxy at Each Cutoff, by Rights Group

		Position $U - D$				Canal total $U + D$	
		Change × drought			Usual × drought	Change × drought	Usual × drought
km	Fields	2SLS	OLS	Reduced form	2SLS	2SLS	2SLS
All fields							
2	24,368	−0.027**	−0.023**	−0.027**	−0.001	−0.025***	−0.000
3	22,609	−0.042***	−0.035***	−0.042***	0.001	−0.023**	0.000
4	20,649	−0.033***	−0.028***	−0.033***	0.000	−0.012	−0.005*
5	19,175	−0.027**	−0.018*	−0.027**	−0.003	−0.020	−0.004
6	18,074	−0.022	−0.006	−0.020	−0.004	−0.017	−0.001
8	16,274	−0.025	0.000	−0.023	−0.012**	−0.005	0.004
10	14,323	−0.037	−0.022	−0.036	−0.001	0.027	−0.003
Junior rights							
2	5,683	−0.020	−0.020	−0.019	−0.010**	−0.004	−0.002
3	5,569	−0.031**	−0.032**	−0.031**	−0.009**	0.004	−0.004
4	5,397	0.002	−0.000	0.002	−0.004	0.023	−0.008
5	5,206	−0.012	−0.015	−0.011	−0.004	0.022	−0.009
6	4,916	0.032	0.029	0.033	−0.015***	0.007	−0.013
8	4,389	0.053**	0.040	0.053**	−0.025***	0.028	−0.004
10	3,714	0.056	0.061	0.055	−0.006	−0.006	−0.014
Senior rights							
2	9,221	0.034*	0.040**	0.028*	0.003	−0.043*	−0.000
3	8,454	−0.034	−0.021	−0.034	0.001	−0.015	0.005
4	7,695	0.004	0.025	0.004	0.003	−0.016	0.005
5	7,164	0.033	0.073*	0.035	0.008	−0.036	0.001
6	6,757	−0.339	−0.108	−0.145	0.012	0.358	0.006
8	6,150	2.194	0.061	−0.122	0.009	−2.152	−0.012
10	5,481	−3.361	−0.037	−2.116	−0.072	3.438	0.043

Notes: Equation (5), crop years 2019–2023. Change: this year's value minus the field's 2017–2018 mean; usual: the 2017–2018 mean. 2SLS: instrumented by predicted demand, split the same way. Two-way clustering by field and by crop by year. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A5 One Model, Written Three Ways

Any two of upstream demand U, downstream demand D, the canal total $U + D$, and position $U - D$ hold the same information. 2SLS does not change when I rewrite the regressors and the instruments the same way. Let the drought coefficients be a on U and b on D. The coefficient on U at a fixed total is $a - b$. The position coefficient is $(a - b)/2$, and the canal-total coefficient is $(a + b)/2$. Table A7 reports each column as a linear combination of one fit. The position and canal-total columns match a direct fit to 10^{-8}.

Table A7: Water Use: One Model Written Three Ways

		U and D		U at a fixed total	Position and total	
km	Fields	U	D	U (with $U+D$)	$U-D$	$U+D$
All fields						
2	25,389	-0.026^{***}	0.001	-0.026^{***}	-0.013^{***}	-0.012^{**}
3	23,545	-0.021^{**}	0.012^{**}	-0.033^{***}	-0.017^{***}	-0.004
4	21,478	-0.032^{***}	0.003	-0.035^{***}	-0.017^{***}	-0.014^{**}
5	19,924	-0.047^{***}	-0.012	-0.034^{***}	-0.017^{***}	-0.030^{***}
6	18,773	-0.036^{***}	0.002	-0.038^{***}	-0.019^{***}	-0.017^{**}
8	16,887	-0.041^{***}	0.018	-0.059^{***}	-0.030^{***}	-0.011
10	14,866	-0.053^{***}	0.017	-0.070^{***}	-0.035^{***}	-0.018
Junior rights						
2	5,927	-0.020	0.010	-0.031^{**}	-0.015^{**}	-0.005
3	5,803	-0.021^{*}	0.018^{**}	-0.039^{***}	-0.020^{***}	-0.002
4	5,623	-0.032^{**}	-0.000	-0.031^{**}	-0.016^{**}	-0.016^{*}
5	5,427	-0.043^{***}	-0.021^{**}	-0.023	-0.011	-0.032^{***}
6	5,124	-0.041^{***}	-0.015	-0.026^{*}	-0.013^{*}	-0.028^{***}
8	4,566	-0.039^{**}	0.005	-0.044^{**}	-0.022^{**}	-0.017
10	3,862	-0.046^{**}	-0.007	-0.039^{*}	-0.019^{*}	-0.027^{*}
Senior rights						
2	9,293	-0.023	-0.035^{***}	0.012	0.006	-0.029^{**}
3	8,524	-0.032^{*}	-0.036^{**}	0.004	0.002	-0.034^{**}
4	7,760	-0.042^{*}	-0.019	-0.023	-0.012	-0.030
5	7,228	-0.064^{**}	-0.034	-0.029	-0.015	-0.049^{**}
6	6,816	-0.036	0.028	-0.064	-0.032	-0.004
8	6,205	-0.040	0.114^{*}	-0.155^{**}	-0.077^{**}	0.037
10	5,533	-0.029	0.082	-0.112	-0.056	0.026

Notes: Equation (4), 2SLS, drought terms. U and D: upstream and downstream demand in one model. U at a fixed total: the coefficient on U when the model is written with U and $U+D$. $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$.

A6 Fields With and Without Their Own Well

A field with an irrigation well of its own can pump when the canal falls short. I estimate equation (5) separately for fields with and without one (Table A8). A field has its own well when a registered well of the Arizona Department of Water Resources lies inside the field. The well must be non-exempt and not cancelled. It must have a casing of at least 8 inches. Six percent of fields have their own well.

Table A8: Yield Proxy on Fields With and Without Their Own Well

			Change in position × drought		Change in total	Usual position
	Fields	First-stage F	2SLS	OLS	× drought	× drought
All fields						
All	22,609	1,943	-0.042^{***}	-0.035^{***}	-0.023^{**}	0.001
No irrigation well on the field	21,336	1,911	-0.042^{***}	-0.034^{***}	-0.025^{***}	0.002
Irrigation well on the field	1,273	1,144	-0.023	-0.030^{*}	0.003	-0.004
Junior rights						
All	5,569	4,307	-0.031^{**}	-0.032^{**}	0.004	-0.009^{**}
No irrigation well on the field	4,821	3,749	-0.034^{**}	-0.033^{**}	-0.004	-0.008^{*}
Irrigation well on the field	748	780	-0.012	-0.023	0.045	-0.014
Senior rights						
All	8,454	537	-0.034	-0.021	-0.015	0.001
No irrigation well on the field	8,260	534	-0.033	-0.020	-0.014	0.001
Irrigation well on the field	194	13	0.170	0.129	-0.009	-0.024

Notes: Equation (5), 3 km, 2SLS except the OLS column; each row is estimated separately, and no test of the difference between rows is reported. For senior fields with a well (194 fields), the first-stage F is 13, too weak to read the estimate. $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$.

A7 Junior Yield Proxy by Year, One Irrigation District Left Out

Table A9 repeats the junior by-year estimates of Figure 6(a), leaving out one irrigation district at a time.

Table A9: Junior Yield Proxy by Year, Leaving Out One Irrigation District at a Time

Left out	Fields	2019	2020	2021	2022	2023
None (main estimate)	5,569	−0.029**	−0.008	−0.042**	0.001	−0.001
Central Arizona	3,547	−0.028*	0.003	−0.020	0.002	0.006
Cibola Valley	5,510	−0.031*	−0.014	−0.034*	−0.001	−0.005
Fort Thomas	5,441	−0.021**	−0.005	−0.041**	0.003	−0.004
Graham	5,466	−0.030**	−0.011	−0.044***	−0.001	−0.001
Hohokam	4,848	−0.023*	−0.002	−0.033**	0.009	0.007
Maricopa–Stanfield	3,464	−0.028*	−0.009	−0.057***	−0.008	−0.001
Montezuma	5,467	−0.027**	−0.008	−0.035**	0.002	0.001
San Jose	5,435	−0.031**	−0.008	−0.034**	0.003	0.001
Union	5,403	−0.029**	−0.009	−0.048**	0.003	0.001
Gila River Reservation fields	5,565	−0.029**	−0.008	−0.042**	0.001	−0.001
Ak-Chin Reservation fields	5,544	−0.029**	−0.008	−0.042**	0.001	−0.000

Notes: Equation (5) in the junior-rights districts, with this year's change in position interacted with each crop year, 3 km, 2SLS. The junior group also holds a few fields that the irrigation-district crosswalk assigns to the Gila River and Ak-Chin reservations; the last two rows leave them out. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A8 Standard Errors Clustered by Irrigation District

Drought and the delivery cuts it brings are set for each irrigation district and year. The effect I estimate works inside the irrigation district and year. Two fields of the same irrigation district face the same drought and the same cut in a year, but one draws on its canal before the other. The models include irrigation district by drought effects, so the drought terms compare fields within their own irrigation district.

The design has a limit here. Drought varies over only 42 irrigation districts, 11 with junior rights and 7 with senior rights. Clustering by irrigation district lets the errors of all fields of a district be correlated over all years. Each irrigation district then counts as one observation, although the effect works across the fields inside it. The paper's tables cluster by field and by crop and time instead. With so few clusters, I compute the district-clustered p-values with a wild cluster bootstrap that imposes the null (Cameron, Gelbach, and Miller 2008; Webb 2023).

Table A10 reports both sets of p-values for the estimates of Table 3 and Figure 5. On all fields, the results hold with district clusters. This year's change in position lowers the yield proxy in drought ($p = 0.003$), and so does this year's change in canal total ($p = 0.013$). Position lowers water use in drought ($p = 0.052$). The rights groups have 11 and 7 clusters, and their p-values rise. Junior position in water use goes from $p = 0.008$ to 0.143, and the senior canal total from 0.015 to 0.379. This year's change in junior position in the yield proxy goes from 0.049 to 0.345. The junior usual position in the yield proxy keeps $p = 0.033$.

Table A10: Water Use and Yield Proxy, With Standard Errors Clustered by Irrigation District

Term	Fields	Estimate	p, clusters		Districts
			Field and crop × time	Irrigation district	
A. Water use					
Position × drought	All fields	−0.017	<0.001	0.052	42
	Junior rights	−0.020	0.008	0.143	11
	Senior rights	0.002	0.851	0.869	7
Canal total × drought	All fields	−0.004	0.468	0.608	42
	Junior rights	−0.002	0.826	0.859	11
	Senior rights	−0.034	0.015	0.379	7
B. Yield proxy					
This year's change in position × drought	All fields	−0.042	0.002	0.003	42
	Junior rights	−0.031	0.049	0.345	11
	Senior rights	−0.034	0.207	0.254	7
Usual position × drought	All fields	0.001	0.785	0.790	42
	Junior rights	−0.009	0.036	0.033	11
	Senior rights	0.001	0.796	0.130	7
This year's change in canal total × drought	All fields	−0.023	0.011	0.013	42
	Junior rights	0.004	0.814	0.817	11
	Senior rights	−0.015	0.622	0.164	7
Usual canal total × drought	All fields	0.000	0.920	0.974	42
	Junior rights	−0.004	0.542	0.434	11
	Senior rights	0.005	0.471	0.210	7

Notes: Panel A: equation (4); Panel B: equation (5), crop years 2019–2023. Upstream and downstream fields beyond 3 km on the irrigation network, same-owner pairs removed, 2SLS. Field and crop × time: clusters by field and by crop by year by month (Panel A) or crop by year (Panel B), as in the paper's tables. Irrigation district: clusters by irrigation district, with reservations as their own districts; wild cluster bootstrap of the reduced form with the null imposed, Webb weights, 99,999 draws. Districts: irrigation districts in the sample. Junior and senior rights: fields in junior-rights and senior-rights districts.

A9 Documented Delivery Constraints in Non-CAP Systems

Table A11 collects documentary evidence from the Gila Valley, where Graham, Fort Thomas, Union, Montezuma, and San Jose are represented in the junior-rights sample. For a fixed irrigation-turn duration, a smaller stream delivers less water; additional delivery time or supplemental supply can compensate. The search was directed by the study geography and results. These cases document the existence of delivery constraints; they do not measure their prevalence or independently identify the crop-outcome mechanism.

Table A11: Non-CAP Delivery Constraints: Documentary Evidence and Sample Match

System and date	Documentary record and source	Match and interpretation
Upper Gila / Safford Valley, 2021	The Commissioner recorded Cosper Crossing as dry on March 11 and next wet on July 26. When dry, the Duncan Valley's Gila flow was issued to Duncan/Virden canals; flowing conditions placed Duncan and Safford diversions under the same priority year. (Doyle 2022, p. 3; PDF p. 7)	Dated inter-valley flow and administration evidence in the region containing the five focal Gila companies. It does not imply zero supply throughout Safford or measure unequal deliveries within a canal.
Graham Canal, testimony reproduced in 1955	During scarce-water rotations, the reported stream diminished from roughly 400 "inches" at the head to 250 at the old Oregon ditch's end. Water boss Emery B. Hubbard testified: "It affects more on the lower end." (Arizona Supreme Court 1955, pp. 217–218, Udall dissent)	Direct historical testimony about downstream conveyance losses in a named sample company. The account concerns a river-and-pump system and predates the study; it is reproduced in the dissent, not a contemporary delivery measurement.
Gila Valley Irrigation District, May 2019	The modernization plan recommended equalizing basins, turnout measurement and control, and operational changes to improve the flexibility and equity of field deliveries. (ITRC 2019, recommendations 6–8)	The plan's map includes the five focal Gila companies. A case-specific engineering assessment of delivery needs; the recommendations do not measure downstream deficits or completed improvements.
Gila Valley Irrigation District, February 8, 2022	Meeting minutes attribute to GVID representative Justin Layton losses from aging gates across the eleven-canal system: existing gates reportedly leaked 10–50 miner's inches, against a typical 300 miner's-inch stream. (AWPF 2022b, attached February draft minutes, p. 4; PDF p. 13)	Contemporary operator evidence of hydraulic losses at system scale, with no individual focal canal or tail farm identified. The revised gate project covered Smithville and Dodge-Nevada; Union had been removed.

Notes: Dates identify the reported event, plan, or court opinion; the 2021 Commissioner's report was submitted August 10, 2022. GVID-wide statements are not five separate canal observations. The detailed February 2022 account is reproduced as draft minutes in the June meeting packet; the approved scope revision is recorded separately (AWPF 2022a, PDF pp. 2–3, 27–28). Historical "inches" and modern miner's inches describe stream-flow rates, retained as reported without conversion. These records concern non-CAP systems; groundwater supplementation can still occur within them.

Table A12 places the dated account beside reported diversions. Safford Valley's May–June total was 2,112 acre-feet in 2021, compared with 8,675 in 2020 and 3,975 in 2022. In 2021, diversions recovered to 5,603 acre-feet in July and 12,135 in August; the annual total slightly exceeded 2020's. The record therefore locates scarcity within the growing season. Among the five focal companies, annual changes are heterogeneous: Fort Thomas reported much lower river diversion in 2021 than its 2016–2019 mean, whereas Union's reported diversion was higher. Calendar-year diversion totals cannot reveal how water was distributed among farms when supply was tight.

Table A12: Reported Gila River Diversions and Seasonal Flow Conditions

Panel A. Seasonal conditions, entire Safford Valley

Year	Cosper dry	Next wet	May (AF)	June (AF)
2020	June 5	Dec. 31	5,659	3,016
2021	March 11	July 26	1,096	1,016
2022	April 13	June 30	1,351	2,624

Panel B. Calendar-year diversions, focal Gila companies (AF)

Canal	2016–2019 mean	2020	2021	2022
Graham	5,598	5,927†	5,204	4,812†
Fort Thomas	3,874	2,406	1,554	2,747
Union	11,846	11,320	12,555	13,237
Montezuma	8,158	7,565	6,479	7,648
San Jose	8,900	8,992	8,499	8,897

Sources: Gila Water Commissioner annual reports (Gila Water Commissioner, n.d.). Year labels link to the original reports. Panel A: printed p. 3 (PDF p. 7), Cosper status changes; printed p. 5 (PDF p. 10), Safford monthly diversions. Panel B: Plate 3, "Gila River Decreed Acreages and Diversions," Safford Valley rows; PDF pp. 28, 30, 31, 31, 28, 31, and 32 in the 2016–2022 reports, respectively. The preperiod mean uses the 2016, 2017, 2018, and 2019 reports and is rounded to whole acre-feet.

Notes: Panel A covers the entire Safford Valley reporting area, not only the five junior companies. Dates are reported status changes, not continuous daily observations; 2021 had the earliest listed dry onset but not the longest interval. Panel B includes all five Gila companies in the junior sample. Values are published calendar-year river diversions, not farm receipts, orders, or a complete groundwater-inclusive water budget. San Jose, Montezuma, and Union share a described measuring flume with Fourness; separate published totals are not independently gauged measurements. † Graham's 2020 and 2022 annual totals are reproduced as published despite unresolved unit conflicts in their monthly records. No blank cell has been treated as zero.

Additional Evidence Outside the Junior-Rights Group

Additional searches covered senior and mixed-priority systems with negative coefficients in exploratory separate-district regressions, and systems with unresolved priority classifications and negative validation estimates. Table A13 reports both delivery constraints and a qualifying account of adequate supply. North Gila Valley and Yuma Irrigation District are near Yuma, distinct from the Upper Gila companies in Table A11.

Table A13: Additional Non-CAP Delivery Records: Senior, Mixed, and Unclassified Systems

System, priority, and date	District or operator record and source	Interpretation and limits
North Gila Valley, senior, July 2018	The McPherson Lateral application reports deteriorated lining, leakage, and seepage. Enlarging and relining the lateral was proposed to increase delivery capacity and shorten irrigation turns. (NGVIDD 2018, pp. 2–4)	A delivery constraint in a senior Colorado River system. Proposed gains are not measured outcomes. Earlier phases were completed in 2016–2017; the record does not establish unchanged conditions in later drought years.
Yuma Irrigation District, senior, May 2021	General manager Rex Green described year-round demand frequently testing operational capacity, while reporting no drought-induced diversion limitations yet. The district's undated ordering policy allows delayed service when system capacity is constrained. (Irrigation Leader 2021; YID, n.d.)	Capacity and scheduling pressure despite continued access to source water. Neither account measures a head-to-tail volume difference. The undated policy cannot establish when a particular delivery was delayed.
Roosevelt Irrigation District, unclassified, April 2017	The district reports summer prorating and difficulty matching supply to demand because of canal travel time. At an existing reservoir near the canal's downstream end, stored water is pumped back when downstream demand exceeds supply. (RID 2017, pp. 1, 3)	Direct downstream supply-demand mismatch and an operating response. The record does not establish a remaining seasonal deficit after compensation. This is Roosevelt Irrigation District, not Roosevelt Water Conservation District.
Buckeye, unclassified, July 2022	The district reports that changing main-canal levels alter farm delivery rates when manual gates remain fixed during an irrigation set. It also reports longer delivery durations associated with inadequate water levels and multiple users. (BWCDD 2022, pp. 15–16)	Contemporary farm-delivery rate and timing problems in a shared system. No head-to-tail comparison or drought attribution is provided; anticipated modernization benefits are not observed delivery gains.
Mohave Valley, mixed, 2019	The district's conservation plan states that, since formation in 1963, it had never been unable to provide a contract holder's requested amount. The district does not operate the private conveyance facilities. (MVIDD 2019, pp. 38–39)	A qualifying record against a district-wide supply-shortage interpretation. It does not audit distribution within individual private systems. A negative crop coefficient alone cannot establish unmet water orders.

Notes: Priority labels follow the study's audited classifications; "unclassified" is not a senior designation. The full-period North Gila and Yuma fits use nearest-district assignment; the stricter inside-boundary screen beyond 5 km lacks drought variation for both. Roosevelt and Buckeye have negative validation estimates but positive discovery estimates; Mohave's sign changes in the other direction. These exploratory matches guide source discovery, not independent causal validation. Grant applications report existing conditions alongside proposed improvements; the table distinguishes the two. No source here measures a dated, uncompensated seasonal water deficit at a named downstream farm.

A10 Estimates and Samples

Table A14 gives the samples, and Table A15 the numbers behind the scorecard of Table 5. Table A16 gives each named rule's score. In Maricopa–Stanfield, at a take of 0.70, test inversion rejects equal sharing ($p = 0.005$) and upstream first only along a line ($p = 0.005$), and it keeps side lines first at a split ($p = 0.453$) and both together ($p = 0.692$). Across the rules and takes it tests, it keeps takes of 0.62 to 0.78, κ_J of 0.24 to 1, and every κ. The take is sharp: the best score

is 15.8 at 0.60, 32.6 at 0.70, and 12.6 at 0.80. The gain comes from the years with shortage. Its share by year left out is −8, 21, 1, 29, 30, and 27 percent over 2016–2021, and 2016 and 2018 have the highest delivered shares. In New Magma the best score by take is 22.1 at 0.60, 39.1 at 0.70, 38.8 at 0.80, 33.9 at 0.88, and 26.4 at 1.00.

Table A14: The Monthly Samples

District	Years	e	Field-months	Fields	Owners	Fields without a well
Maricopa–Stanfield	2016–2021	0.72	67,064	2,020	462	1,656
New Magma	2016–2023	0.70	24,880	524	249	430
Harquahala	2016–2021	0.73	23,104	648	61	79
Upper Gila, ten canals	2016–2023	0.70	43,200	932	310	295
Central Arizona	2016–2021	0.73	76,408	2,099	626	1,488
Hohokam	2016–2021	0.75	27,176	767	135	117
San Carlos	2016–2023	0.70	38,832	1,111	285	763
Wellton-Mohawk	2016–2023	0.70	123,152	3,351	304	2,445
Yuma County Water Users	2016–2023	0.70	53,696	2,083	368	1,789
Yuma Mesa	2016–2023	0.70	23,928	564	239	457
Yuma Irrigation	2016–2023	0.70	14,992	553	77	228

Notes: March–October field-months on the tree with ln ET and the controls, after cells with a single observation drop out of the fixed effects. Owners count a field with no known owner as its own. The Yuma districts have no short month at $e = 0.70$, so their samples enter only the runs of Table A26. Central Arizona: the version built as in Maricopa–Stanfield, whose no-well fields are those without a local pump.

Table A15: The Numbers Behind the Scorecard

		Best score	Rule at the best	Slope on the shortfall, all rules		Fallowing on
District	Water	/ bar	(τ, κ, κ_J)	No well	With a well	total stress
Maricopa–Stanfield	CAP, canal pumps	32.56 / 14.29	0.70, 1.00, 0.98	−18.17 (5.22)	−47.57 (21.69)	16.7 (5.0)
New Magma	CAP	39.09 / 20.35	0.70, 0.96, 0	−12.58 (5.41)	1.53 (8.29)	−0.6 (7.0)
Harquahala	CAP	37.91 / 49.15	0.62, 0.46, 0.38	−34.79 (12.56)	−2.19 (9.25)	
Upper Gila	Reported diversions	57.45 / 10.85	1.00, 0, 0.64	−6.31 (2.20)	−6.22 (1.48)	8.8 (6.5)
Central Arizona	CAP, district wells	4.72 / 10.19	0.64, 0.98, 0.80	7.94 (5.59)	8.59 (10.61)	6.5 (4.8)
Hohokam	CAP	39.32 / 28.82	1.00, 0.26, 0.66	18.08 (7.32)	10.23 (7.79)	−13.5 (9.7)
San Carlos	Stored water	4.27 / 57.28	0, 0, 0.64	2.42 (3.63)	−6.68 (5.13)	45.3 (7.3)
Wellton-Mohawk	Colorado River	1.45 / 457.83	0.90, 0.02, 0	16.62 (9.17)	20.49 (18.55)	0.3 (7.0)
Three Yuma districts	Colorado River			No short month		

Notes: Best score and bar: held-out gain in fit, in basis points. Hohokam's best rule has the wrong sign. Slopes: 100 times log monthly ET per unit of the shortfall, over all rules, at a take of 0.70 in Maricopa–Stanfield and New Magma, 0.62 in Harquahala, 0.64 in Central Arizona, and 1.00 elsewhere; standard errors clustered by owner and by year-month. Fallowing: points per unit of total stress, fields without a well, with field and year effects, at a take of 1.00 in New Magma and San Carlos; in Central Arizona, the version built from its documents. Harquahala's fallowing uses a separate design on idle land (Table A23). Three Yuma districts: Yuma County Water Users, Yuma Mesa, Yuma Irrigation.

Table A16: The Score of Each Named Rule

District, take	Equal sharing	Upstream first along a line	Side lines first	Both	Best at any take	Bar
Maricopa–Stanfield, 0.70	−7.14	5.69	23.70	31.06	32.56	14.29
New Magma, 0.70	−4.52	38.90	−10.19	30.34	39.09	20.35
Upper Gila, reported, 1.00	57.34	20.29			57.45	10.85
Upper Gila, reported, 0.70	39.89	11.23				
Harquahala, 0.62		−4.98	0.03	−0.89	37.91	49.15
Central Arizona, 0.64	−0.26	−2.25	−1.21	3.23	4.72	10.19
Hohokam, 0.70	−1.33	−1.05	−24.98	−20.06		
Hohokam, 1.00	0.22	−13.20	17.76	3.04	39.32	28.82

Notes: Held-out gain in fit, in basis points. Equal sharing: $\kappa = \kappa_J = 0$. Upstream first along a line: $\kappa = 1$, $\kappa_J = 0$. Side lines first: $\kappa = 0$, $\kappa_J = 1$. Both: $\kappa = \kappa_J = 1$. The score is blind to sign; Table A15 gives the slopes. Blank in the Upper Gila's row: κ_J matters on only two canals, so I report the first two rules. Harquahala's equal sharing cannot be judged: its shortfall barely varies within a month, and S_t explains 72 percent of it.

Table A17 gives the estimates behind Maricopa–Stanfield's results in Section 8.4 and Section 8.5. The model predicts how short a field is as well as whether it is. Small shortfalls leave water use

unchanged and larger ones reduce it. The loss grows up to a shortfall of about 0.5 and then holds.

Table A17: Maricopa–Stanfield: The Shortfall, the Cut, and Fallowing

A. Monthly water use by size of the shortfall, fields without a well, against below 0.01	
0.01–0.05	0.55 (1.35)
0.05–0.10	−0.04 (1.60)
0.10–0.20	0.23 (2.33)
0.20–0.35	−3.32 (3.05)
0.35–0.50	−14.74 (4.29)
0.50 or more	−9.87 (3.40)

B. The cut, per standard deviation of the predicted rise, 1,917 fields without a well	Water use	Farmed share
Cut minus before	−8.5 (2.4)	−5.9 (1.4)
Without kilometers from the head	−8.3 (2.4)	−5.8 (1.4)
Before, 2020–2021 against 2016–2019	−0.0 (2.0)	2.6 (1.2)
The best rule alone	−7.9 (2.4)	−5.1 (1.5)
Fifteen variants of the build, equal sharing: range	−6.1 to −10.4	−4.4 to −8.7

C. Fallowing on total stress, fields without a well farmed the year before	Per unit	Per SD
Within a year, 2017–2023	10.7 (3.8)	2.8
Kilometers from the head by year beside it	11.2 (3.9)	2.9
Field and year effects	16.7 (5.0)	4.3
Both years at usual demand	9.4 (3.4)	3.1
Within a year, 2022–2023	12.8 (4.5)	4.7
Within a year, 2017–2021	0.5 (8.0)	
Next year's stress, 2017–2022	−7.6 (6.6)	

D. Fallowing by total stress, against below 0.01	Within a year	Field and year effects
0.01–0.05	1.8 (1.4)	0.3 (1.5)
0.05–0.10	4.4 (1.7)	3.3 (2.3)
0.10–0.20	1.6 (2.0)	0.4 (2.9)
0.20–0.35	3.5 (2.6)	4.4 (3.6)
0.35–0.50	6.0 (3.5)	4.1 (4.5)
0.50–0.75	3.0 (4.1)	3.6 (5.4)
0.75–1	7.8 (5.2)	9.8 (6.2)
1 or more	19.6 (6.1)	25.9 (7.6)

Notes: All rules at a take of 0.70 unless stated. Panel A: 2016–2021, specification of equation (12), clusters by owner and year-month. Panels B to D: owner clusters. Panel B, best rule's before: water use −0.9 (1.9). Panels C and D: mean fallowing 13.3; last year's and this year's stress in one regression sum to 20.7 (11.2), joint $p = 0.015$.

The 2022–2023 cut in more detail. In June 2022 Maricopa–Stanfield wrote that it had cut allotments to growers by half (Maricopa–Stanfield Irrigation and Drainage District 2022b). In the other districts CAP fell to 37 (Central Arizona), 24 (Hohokam), and 9 percent (Harquahala) of its 2016–2021 mean in 2022, and to 0 to 6 percent in 2023 (CAP 2023). In a balanced panel of monitoring wells, depth to groundwater in five CAP agricultural-pool districts rose 8.33 feet more than in comparison districts from 2021 to 2023 ($p = 0.008$; Appendix A16). By 2024 Maricopa–Stanfield wrote that limits in its infrastructure, and the coming removal of wells from the Santa Rosa Canal, left some farmers as little as 0.78 acre-feet per acre (Maricopa–Stanfield Irrigation and Drainage District 2024). Other measures of who the cut reached point the same way, per standard deviation in water use: the model's CAP share in 2016–2021, −7.9 (standard error 2.3; before 0.5, 1.8), and its well share in the cut, 7.8 (2.5; before −1.0, 1.9).

A11 Robustness

Pump locations. Table A18 keeps the water and moves only its location: delivered at the pumps, or pooled over all claims each month. Delivered water matches by year (0.997 and 0.997 in 2016; 0.651 and 0.654 in 2022 at fixed demand). The monthly slope needs both location and rule. Equal sharing at the pumps and all rules pooled, in one regression, give −15.15 (standard error 7.29) and −16.86 (9.60). In the ordinary years the model separates rules from equal sharing. In the cut and in fallowing it cannot, because every rule keeps the pumps in place (fallowing on the two together: 1.6, standard error 7.8, and 10.5, standard error 8.6; correlation 0.93).

Table A18: Pumps Where They Stand Against the Same Water Pooled, Maricopa–Stanfield

	One at a time		Together	
Fields without a well	At the pumps	Pooled	At the pumps	Pooled
Monthly slope	−18.17 (5.22)	−16.47 (9.47)	−17.95 (5.04)	−2.19 (9.82)
Water use in the cut, cut minus before	−8.5 (2.4)	4.6 (2.8)	−8.1 (2.4)	3.2 (2.7)
Farmed share in the cut, cut minus before	−5.9 (1.4)	5.2 (2.3)	−5.4 (1.5)	4.3 (2.4)
Fallowing on total stress, within a year	10.7 (3.8)	0.1 (8.8)	12.1 (4.0)	−8.4 (9.5)
Fallowing on total stress, field and year effects	16.7 (5.0)	−14.4 (12.9)	18.0 (5.2)	−23.6 (13.6)
Slope on the pooled shortfall, by a canal pump within 2 km above the field				
	Pump above	None	Difference	
Plain	1.01 (10.34)	−27.59 (11.02)	28.61 (10.14)	
S_t times kilometers from the head beside it	15.49 (10.87)	−13.05 (9.52)	28.54 (9.94)	
S_t times group beside it	−1.71 (10.87)	−24.48 (12.86)	22.78 (14.50)	

Notes: All rules at a take of 0.70. Monthly: 2016–2021, clusters by owner and year-month. Cut: per standard deviation of each predicted rise, owner clusters. Lower panel: 1,066 fields with a canal pump within 2 km above and 590 without. A pump below a field cannot deliver to it: −4.08 (22.07), against 28.14 (10.78) for a pump above. By kilometers to the nearest pump above, up to 0.5, 0.5–1, 1–2, 2–5, and beyond 5: 11.15 (14.75), −9.00 (11.79), −2.10 (12.30), −30.55 (13.65), −26.16 (13.98).

Which wells count as local. Table A19 keeps the 130 canal pumps and widens the local pumps from 163 to all 344 other rated wells or all 418 other wells on the map. Each list has its own search, null, and test inversion. The take rises to 0.84–0.86, and the total claim on the canal stays at 92 to 93 percent of need. The slope for fields without a well does not move, though that group loses 28 percent of its fields with all wells. A field's own well shows no protection on any list. At the same location such fields do not respond less: own well minus none is −9.72 (standard error 12.12) on the main list beyond 5 km of a pump or with none above, and about zero in 2022–2023. The lists may hold idle wells.

Table A19: Three Lists of Local Pumps, Maricopa–Stanfield

	Main	Rated wells	All wells
Fields with a local pump, of 2,383	431	851	952
Best score	32.56	33.89	32.68
Best: take, κ, κ_J	0.70, 1, 0.98	0.84, 1, 0.94	0.86, 0.82, 0.90
Bar	14.29	13.51	15.07
Slope, no well	−18.17 (5.22)	−18.83 (5.87)	−18.72 (5.84)
Water use in the cut	−8.5 (2.4)	−8.2 (2.7)	−8.5 (2.8)
Farmed share in the cut	−5.9 (1.4)	−5.5 (1.5)	−5.5 (1.6)
Fallowing on total stress	10.7 (3.8)	10.4 (3.7)	10.1 (3.6)
Field and year effects	16.7 (5.0)	16.1 (5.1)	15.6 (5.0)
Total stress of 1 or more	19.6 (6.1)	20.7 (6.2)	20.5 (5.8)
Canal claim, share of total need	0.923	0.925	0.927

Notes: Each column at its own best take, all rules. No draw of 200 reaches the best score on any list. At each list's best take, test inversion rejects equal sharing and upstream first only along a line ($p = 0.005$).

Other checks in Maricopa–Stanfield. The bar flips signs by owner, so neighbors on a branch get

independent signs. Flipping by canal branch raises the bar from 14.29 to 25.30, 21.10, 21.34, and 26.00, for branches of at most 100, 200, 300, and 500 fields; the best score, 32.56, stays above each, and at most 3 of 200 draws reach it. With S_t times kilometers from the head beside it, the slope is −14.53 (standard error 4.79), and the term itself is −21.16 (10.79). While CAP flows a field's place on the canal matters, and once CAP is gone the pumps above it do: the delivered share's correlation with kilometers from the head is −0.44 in 2016–2021 and −0.10 in 2022–2023.

The model does not explain everything. Next month's shortfall is not zero: next, this, and last month's together are −14.19 (5.27), −4.99 (6.21), and −31.54 (8.29). Monthly pumping is imputed, and farmers may know shortages ahead; I have tested neither, and the annual designs do not depend on it. In the cut, fields far from the head lose less, 9.6 per standard deviation of kilometers (standard error 2.4; before 3.0, 1.8). The model does not explain that, and it rests on the two end tenths. Water use responds mostly through fallowing: on fields farmed in all eight years the estimate is −0.9 (2.0). The farmed share has a contrast before the cut, 2.6 (1.2). Fallowing's response comes from the cut years, and by year it is 96.0 (26.7), 17.6 (14.0), 19.7 (16.9), −23.8 (8.2), 10.6 (7.4), and 12.9 (5.6) over 2018–2023. Next year's stress is not clean in 2017–2020 alone, at 24.3 (10.3).

The record of which wells ran. Table A20 gives Maricopa–Stanfield the kind of well record Central Arizona has. A flat volume costs little. Losing the record of which wells ran costs nearly everything. Maricopa–Stanfield's 2019 volumes beat capacity when both enter one regression, and the advantage sits in 2019–2020 (−77.60, standard error 25.01, against 62.38, 23.73), so 2019 pumping may partly reflect that year's shortage. I cannot separate the two.

Table A20: Maricopa–Stanfield on Central Arizona's Kind of Well Record

Well record	Monthly slope	Water use in the cut	Farmed share	Fallowing
Its own 2019 volumes, reported total	−18.33 (5.20)	−8.9 (2.4)	−5.6 (1.4)	8.1 (2.7)
2019 volumes, flat total	−15.71 (4.95)	−7.8 (2.4)	−4.6 (1.5)	7.6 (2.9)
Capacity, reported total	−7.10 (4.34)	−7.0 (2.5)	−3.8 (1.4)	6.6 (2.5)
Capacity and idle wells, reported total	−5.84 (4.48)	−1.8 (2.6)	−1.2 (1.5)	0.2 (2.7)
Capacity and idle wells, flat total	−1.17 (5.11)	0.1 (2.6)	−0.1 (1.5)	−1.4 (2.9)

Notes: Fields without a well, 121 rules at a take of 0.70, monthly-sample fields. Cut: per standard deviation, cut minus before.

A12 Further Tests

Randomizing the model's inputs. Table A21 gives the shuffles. In the Upper Gila, dates are the column to cite: position draws stay narrow even with owners moved, 0.54 and 0.65 of the standard error, because the variation is between canals and neither shuffle moves whole canals. In Hohokam at a take of 1.00 the wrong-signed slope is unusual, but at 0.70 it lies inside the draws. In Central Arizona the date shuffle is not a fair test, because its CAP follows demand less wells month by month, so moving it across years creates shortages the real water never had.

New Magma. The search clears the bar with 7,162 cells, at takes of 0.60 to 1.00 and κ of 0.46 to 1. Table A22 gives its slopes. Its fallowing gap opens as water falls: fields without and with a well fallow 8.3 and 8.9 percent in 2016, 35.7 and 11.9 in 2022, and 46.0 and 19.8 in 2023. The record is direct, and the district owns no wells by its audit. The cut shows no slope, while its water at the field falls to 0.68 and 0.40 of usual need. Fields with a well have the wrong sign in that design (water use 11.9, standard error 3.7; farmed 7.4, 2.2, with 6.1, 2.3, before the cut). Kilometers from the head go the wrong way both before and after the cut (5.5, 2.6; before 6.5, 3.0).

Table A21: Shuffled Dates, Fields, and Owners: Draws Reaching the Real Value, of 1,000

District, take	Statistic	Real	Dates	Fields	Owners
Maricopa–Stanfield, 0.70	Slope, no well	−18.17	1 (0.002)	0 (0.001)	0 (0.001)
	All rules, score	20.34	7 (0.008)	0 (0.001)	0 (0.001)
	Best of 121 rules, score	31.47	6 (0.007)	0 (0.001)	0 (0.001)
	Best of 121 over equal sharing	38.61	19 (0.020)	0 (0.001)	0 (0.001)
New Magma, 0.70	Slope, no well	−12.58	0 (0.001)	0 (0.001)	10 (0.011)
	All rules, score	22.17	6 (0.007)	0 (0.001)	1 (0.002)
	Best of 121 rules, score	38.90	13 (0.014)	0 (0.001)	0 (0.001)
	Upstream first minus equal sharing	43.42	3 (0.004)	0 (0.001)	0 (0.001)
Harquahala, 0.62	Slope, no well	−34.79	0 (0.001)	0 (0.001)	28 (0.029)
	No well minus well	−32.61	11 (0.012)	0 (0.001)	38 (0.039)
	All rules, score	6.01	58 (0.059)	95 (0.096)	219 (0.220)
Upper Gila, reported, 1.00	Slope, no well	−6.31	7 (0.008)	0 (0.001)	0 (0.001)
	Slope, with a well	−6.22	0 (0.001)	0 (0.001)	0 (0.001)
	Best of 121 rules, score	57.45	3 (0.004)	0 (0.001)	0 (0.001)
	Equal sharing minus upstream first	37.05	1 (0.002)	0 (0.001)	0 (0.001)
San Carlos, 1.00	Slope, no well, at or above	2.42	228	108	113
	Best of 121 rules, score	2.79	582 (0.582)	754 (0.754)	902 (0.902)
Central Arizona, 0.64	Slope, no well	7.94	911 (0.911)	1,000 (1.000)	956 (0.956)
Hohokam, 1.00	Slope, no well, at or above	18.08	1	7	45
Hohokam, 0.70	Slope, no well, at or above	11.52	92	103	220

Notes: In parentheses, the one-sided p-value, on the side the model predicts: a negative slope, a high score. Central Arizona's counts are on that low side. Scores are blind to sign. In Maricopa–Stanfield moved owners reach −0.09 on average (standard deviation 4.31), so the real slope lies 3.46 clustered standard errors from their mean. Maricopa–Stanfield's own best rule, 32.56, is reached by 2 date draws.

Table A22: New Magma: Slopes by Rule and Take, 2016–2023

Rule, take	S_t times no well	No well	With a well	No well minus well
Best cell, 0.70	0.11 (8.59)	−7.93 (3.24)	1.02 (3.78)	−8.95 (4.64)
Upstream first along a line, 0.70	−0.76 (8.50)	−7.53 (3.22)	0.23 (3.56)	−7.75 (4.52)
All rules, 0.70	4.30 (10.52)	−12.58 (5.41)	1.53 (8.29)	−14.10 (8.20)
All rules, 1.00	3.60 (12.57)	−10.94 (7.32)	1.03 (7.67)	−11.97 (9.13)
Upstream first along a line, 1.00	−6.53 (8.96)	−2.27 (2.87)	−0.43 (2.90)	−1.84 (3.88)

Notes: 100 times log monthly ET per unit of the shortfall; standard errors clustered by owner and by year-month. 24,880 field-months of 524 fields of 249 owners; 430 fields without a well.

Harquahala. Most of its fields have a well, so the search has little to find, and failing the bar says the data cannot pick a rule, not that the slope is absent. The score across all fields is a test of prediction in the district. The test of the slope is the slope itself and the draws of Table A21. The slope holds without its largest owners (−28.62, standard error 10.69, without one; −29.13, 12.00, without three) and without any one year (−19.18 to −43.43). S_t times no well has the wrong sign at 58.43 (28.38) beside the slopes, but it is 16.41 (22.21) alone. Across position draws the two correlate at −0.93, so the term is tied to the slope and not a separate finding. Table A23 gives the cut, with USGS field surveys for idle land. Access to a well has the right sign, and I cannot tell. Low well capacity goes with more idle land, though much of that gap opened in 2021, before CAP fell.

Table A23: Harquahala: Wells and Idle Land in the Cut

USGS idle land, points	2023 against 2020–2021	Before: 2021 minus 2020	2024 against 2020–2021
No well, all fields	9.88 (14.89)	−4.47 (8.40)	6.50 (17.71)
No well, owners of 40–640 acres	21.97 (14.75)	0.73 (7.35)	9.51 (19.38)
ln capacity, fields with a well	−13.09 (7.31)	−10.82 (6.81)	−13.75 (8.25)
Below 1.5 cfs per 100 acres	24.63 (5.73)	19.26 (7.31)	32.41 (8.94)

Notes: Controls: alfalfa share in 2016–2021, ln field acres, ln owner acres; owner clusters. Access: 752 fields, 63 owners, 73 fields without a well; size range: 191 fields, 35 owners, 52 without a well. Capacity: 679 fields with a well, 34 owners; below 1.5: 11 owners. Raw idle share in 2023, acre-weighted: 71.3 percent without a well, 45.2 with one.

Central Arizona. Table A24 gives five versions of its wells. Version A is the frozen recipe from Maricopa–Stanfield, and it is the version with a monthly search. Version C comes from its documents. Wells where they stand have the right sign in all five, on water use and on fallowing, and never a wrong sign I can tell from zero. All five are versions of one dataset, not separate evidence. Water use goes to about zero without the South Main canal, whose contrast is about −60 in every version and half of which is a rise before the cut. The pooled version moves with the crop calendar and does not survive baseline cotton as a control. At a take of 0.64, version A's slope is 7.94 (standard error 5.59); own need beside it leaves 8.05 (5.59), so demand does not make the sign. Equal sharing is not rejected here ($p = 0.527$). The monthly score is not a fair test either, because CAP carries no supply shock within a year. What would settle it is the district's annual withdrawal by well.

Table A24: Central Arizona: Five Versions of Its Wells

	A. Frozen recipe	B. Wells first	C. From documents	D. Connected wells	E. CAP first
Wells	217 canal, 217 local	as A	all 434 district-run	217 connected	all 434 district-run
Monthly volume	CAP first	wells first	wells first	wells first	CAP first
Fields	1,721 no pump	1,721 no pump	2,372	2,372	2,372
Wells where they stand					
Water use	−8.5 (3.1)	−7.4 (3.2)	−3.7 (2.4)	−2.7 (2.7)	−4.6 (2.3)
with baseline cotton	−6.4 (3.1)		−3.1 (2.5)	−2.5 (2.6)	−3.2 (2.3)
without South Main	−2.3 (3.1)		0.3 (2.6)	1.7 (2.7)	−0.6 (2.4)
Farmed share	−0.7 (1.6)	0.0 (1.7)	0.2 (1.5)	1.7 (1.9)	−0.4 (1.5)
Fallowing	2.3 (4.4)	4.0 (4.5)	6.5 (4.8)	0.1 (5.5)	4.4 (4.9)
The same well water pooled					
Water use	−12.1 (2.8)	−5.1 (2.5)	−5.7 (2.3)	−5.7 (2.3)	−14.2 (3.1)
with baseline cotton	−3.1 (2.6)		1.2 (2.3)	1.2 (2.3)	−2.9 (3.3)
Farmed share	−1.0 (1.6)	1.7 (1.5)	1.4 (1.4)	1.4 (1.4)	−2.4 (2.0)
Fallowing	3.0 (7.7)	20.7 (8.6)	20.9 (9.7)	21.1 (9.7)	−12.0 (11.7)

Notes: Per standard deviation of the predicted rise, cut minus before, owner clusters. Fallowing: per unit of total stress on own demand, field and year effects, fields farmed the year before. Take 0.70 where a take applies. No version has the record of which wells ran, and Central's network has not had my marks or the checks against decoys.

Hohokam. The search clears its bar only at takes of 0.94 to 1.00, the edge of the grid, and it rests on a wrong-signed slope. The slope is a contrast between parts of the tree (35.48, standard error 13.45, between; 1.73, 10.31, within), and Table A25 sets out the parts. The records miss water. In 2023 CAP was 2,482 acre-feet, all of it in June–August, and 1,521 of the 2,989 acres without a well were farmed. Their ET above fallow fields was 2,959 acre-feet, 1,419 of it in months with no CAP and almost no rain, against 1,861 acre-feet of CAP at the field. So the slope does not measure a shortage there. Mislinked fields, a few owners, one crop, field rain and heat, a satellite artifact, and registered capacity do not account for it. Why it is positive and not zero remains unknown. Fields without and with a well fallowed alike, 39.2 and 40.2 percent of acres in 2022 and 46.8 and 50.9 in 2023, and private pumping did not rise when CAP fell. So a registered well may not be usable water here.

Table A25: Hohokam: The Slope by Part of the Tree

Part	Fields, no well / well	Mean shortfall	Slope, no well, take 1.00	2023 fallow %, no well / well
West main	39 / 178	0.536	18.76 (8.12)	18 / 34
North branch	18 / 163	0.247	18.20 (11.49)	44 / 50
Mid north	40 / 156	0.177	−11.66 (20.65)	68 / 50
Kleck	12 / 75	0.274	−25.42 (19.32)	75 / 66
Head of main	8 / 78	0.154	−53.84 (29.83)	97 / 67

Notes: Parts of the tree, cut by a field-count rule after the results; descriptive. Fallow share by acres. The cut, take 1.00, per standard deviation: water use −1.0 (6.8), farmed share 2.4 (3.9). All-rules slope at 0.70: 11.52 (7.57).

San Carlos. The search finds nothing: the best score, 4.27, lies under a bar of 57.28, and 176 of 200 draws reach it. The series falls to zero in short months while wells and CAP keep farms irrigating, so the monthly water misses water that reaches fields. Fallowing on total stress with field and year effects is 45.3 (standard error 7.3) over all rules, 18.9 (5.2) for upstream first along a line, and 96.6 (15.4) for equal sharing. The rule with no position gives the largest slope, and fields with a well move alike (52.4, 15.2). Within a field and year, equal sharing's stress, which is only the crop calendar, explains 58.5 percent of all-rules stress, and kilometers times the year's stress 0.8 percent. I added that check after seeing the size of the slope. The 2022–2023 contrast is not clean, because 2018 and 2021 are short years in the baseline and 2023 is a full one. It gives water use of −6.1 (3.1), or −8.7 (3.3) net of the earlier contrast, and −0.6 (6.5) with a well.

Wellton-Mohawk and the Yuma districts. Wellton-Mohawk's water at the field runs 1.30 to 1.42 times usual demand over March–October. Only June and July 2022 are short, at a shortage of 0.10 and 0.12. Far fields gain water, in both periods: 7.0 per standard deviation of kilometers from the head (standard error 1.2), before 5.6 (1.4). In the Yuma districts every month is above demand at $e = 0.70$, with water at 1.43 to 2.51 times usual demand.

Shortages on paper. Table A26 lowers e in the four districts with senior rights and reruns everything. Only e changes. No case shows a negative slope everywhere that grows as e falls. Four of the 15 runs clear the bar, against fewer than one by chance, and two of those have the wrong sign, so a clear bar alone does not show a shortage. The sign and the slope must come with it. In Yuma Irrigation the best rule's slope hardly moves as e falls (−51.75, 16.73; −50.54, 18.29; −51.05, 18.84; −50.67, 19.79), because the best take falls with it (0.98, 0.74, 0.48, 0.22), and the data pick one pattern. At a take near 1 that is e near 0.6, against consumptive use over diversions of 0.53 to 0.56 in Reclamation's accounting. Its short months at 0.6 are August 2016, May and June 2017, July 2020, June 2021, July 2022, and May, June, and July 2023. I read e near 0.6 after the results, and its fallowing does not respond.

Table A26: Shortage on Paper: Lower Efficiency in the Senior-Rights Districts

District	e	Months short, of 64	Best / bar	All rules, take 0.70	All rules, take 1.00	Fallowing
Wellton-Mohawk	0.60	15	2.96 / 10.00	9.20 (6.97)	−0.54 (5.81)	−0.4 (4.5)
	0.50	40	4.48 / 8.74	−7.39 (5.14)	−8.38 (5.95)	6.4 (5.7)
	0.40	52	5.97 / 9.49	−10.00 (5.68)	−4.05 (5.84)	−2.3 (5.0)
	0.30	63	3.82 / 8.22	−9.75 (5.79)	−9.86 (7.12)	−11.4 (6.6)
Yuma Irrigation	0.60	9	40.46 / 34.48	no short month	−88.73 (31.90)	
	0.50	22	34.00 / 34.39	−92.62 (43.84)	−15.11 (10.06)	−13.2 (7.9)
	0.40	45	37.17 / 36.20	−12.01 (9.49)	12.78 (9.41)	−5.3 (5.4)
	0.30	56	35.51 / 49.35	15.00 (9.45)	11.78 (12.30)	−17.3 (6.5)
Yuma County Water Users	0.60	1	2.07 / 3.65	60.60 (35.11)	39.63 (25.07)	
	0.50	5	12.04 / 11.41	19.39 (8.44)	13.51 (7.65)	−36.7 (24.1)
	0.40	25	6.26 / 13.75	4.97 (3.28)	6.15 (3.11)	−10.4 (7.0)
	0.30	46	25.49 / 11.17	2.79 (3.56)	2.84 (3.65)	−2.5 (5.2)
Yuma Mesa	0.50	2	5.05 / 6.97	19.94 (23.93)	14.01 (10.17)	
	0.40	12	3.97 / 10.18	−6.70 (1.96)	−2.47 (2.85)	−19.7 (5.8)
	0.30	41	10.65 / 15.38	3.97 (5.59)	3.78 (6.52)	−12.3 (3.4)

Notes: Slopes for fields without a well, 100 times log monthly ET per unit of the shortfall. Fallowing: on total stress, all rules, take 1.00, field and year effects, fields without a well; blank where there is too little stress. Both of Yuma County Water Users' runs that clear the bar have the wrong sign. Yuma Mesa at 0.60 has no short month and was not run. Water in Reclamation's accounting, consumptive use over diversions: Yuma Mesa 0.52–0.70, Yuma Irrigation 0.53–0.56, about 0.7 in the other two.

A13 Other Districts

Section 8.3 runs the model on districts whose monthly record can show a shortage. Three more districts have enough water data to run it, and I report them here as a separate category: Cortaro–Marana, Maricopa Water District, and Salt River Valley (Table A27). Maricopa Water District holds 9,802 acres, just under the line of 10,000. Roosevelt and Buckeye have water figures for single years only, so I do not run them. The recipe is that of the districts added later (Appendix A2), on trees I did not review further.

Water. In Cortaro–Marana and Maricopa Water District the monthly CAP record is a minority of the water, 22 to 33 and 22 to 45 percent. I add the district's other water from the state's annual totals: what it withdrew and received, less CAP and, in Maricopa Water District, less deliveries to non-farm users. Its March–October part follows what the canals still need after CAP, and all of it enters at the head. That water exceeds the need in every year, by at least 17 percent in Cortaro–Marana and more than double in Maricopa Water District. Salt River Valley has no record of farm water by month. Its one monthly series is CAP water stored at the project, at most 9.3 percent of the project's deliveries, and I do not treat it as farm water. Its allocation stayed full, stated for most years and inferred from storage for the rest. At $e = 0.70$ the full allocation of 3.3 acre-feet an acre meets the model's demand in every year but 2016, when it falls 3 percent short. So I give Salt River Valley no short month.

What the model predicts. No month runs short in any of the three, under any rule. The shortfall is zero, so the search and the slopes have nothing to work on, as in the three Yuma districts. The water alone predicts no rise in fallowing and no loss that grows with distance from the head.

Results. Fallowing rose in all three, by 16 to 27 points, about as much as in the CAP districts, although the model finds no shortage. So a rise in fallowing does not by itself mean a water shortage. Loss mostly does not grow with distance. In Cortaro–Marana farther fields kept more, not less, and in Maricopa Water District I cannot tell the slopes from zero. The exception is water use in Salt River Valley in the cut, −5.0 (standard error 2.5), which the water does not predict.

My reading came after these numbers, and I do not test it: land left farming for reasons other than water, above all the growth of cities. Of Salt River Valley's fields 94 percent are near a

Table A27: Other Districts: No Shortage in the Model, and Fallowing Still Rose

	Cortaro–Marana	Maricopa Water District	Salt River Valley
Fields in the district	403	732	1,240
Acres in the district	10,271	9,802	16,647
Fields placed on the tree	369	593	783
Placed fields with a well	3	169	189
Short months, of 64	0	0	0
Percent of fields fallow			
2016–2021	12.0	28.0	23.4
2022–2023	27.9	50.4	50.5
Change	15.9	22.4	27.1
Percent of acres fallow			
2016–2021	7.1	26.4	17.8
2022–2023	16.3	50.2	42.2
Near a city			
Percent of fields near a city	17	62	94
Change in fallow, near a city	45.8	20.0	27.5
Change in fallow, elsewhere	9.8	26.3	20.2
Per standard deviation of km from the head, no well			
Water use, 2022–2023 against 2016–2021	12.5 (2.8)	4.8 (3.6)	−5.0 (2.5)
Water use, 2020–2021 against 2016–2019	9.4 (2.7)	3.0 (2.5)	1.4 (1.7)
Farmed, 2022–2023 against 2016–2021	17.2 (3.1)	0.2 (2.1)	−2.7 (1.9)
Farmed, 2020–2021 against 2016–2019	5.7 (1.9)	0.3 (2.5)	1.2 (1.7)

Notes: Short months: March–October 2016–2023, at the four corner rules and takes of 0.70 and 1.00; in Salt River Valley by the record of full allocations, not from a water series. Fallow: the main-sample flag, by year, all fields of the district. Near a city: at least 300 buildings within 1 km of the field's outline, from Microsoft's building footprints; changes are from 2016–2021 to 2022–2023, in points. Km: changes in the outcome, times 100, on km from the head per standard deviation, with the alfalfa share and log acres, clustered by owner. Standard deviations of km: 5.2, 4.2, and 18.8.

city, and 62 percent of Maricopa Water District's, against 0 to 7 percent in Maricopa–Stanfield, Central Arizona, Harquahala, and Hohokam. In both districts the rise began before the cut: from 13 percent in 2018 to 43 in 2021 in Salt River Valley, and from 25 percent in 2019 to 42 in 2021 in Maricopa Water District. Neither fell in 2023, the wettest year of the period on the Salt and the Agua Fria. In Cortaro–Marana the rise sits near the city: 46 points among the 17 percent of fields near one, and 10 elsewhere. The reading does not explain everything. Within Maricopa Water District the rise is no larger near a city, 20 points against 26 elsewhere. It may also bear on New Magma, the one short district that lies near cities: 54 percent of its fields are near one, and its fallowing rose 24 points there and 11 elsewhere.

A14 Water Loss and the Yield Proxy

The yield proxy is each crop's GCVI from Sentinel-2 in the weeks that form its yield, from five-day curves with clouds and spikes removed: for cotton the maximum over windows starting July 1 to August 31, for corn and sorghum the May–August maximum, for small grains and winter wheat the March–April maximum, and for alfalfa and other hay the sum of the peaks before each cutting in March–October. A field-year's seasonal water use relative to demand is the log of its thermal-model ET (SSEBop, eeMETRIC, geeSEBAL, and DisALEXI) over its crop's demand, summed over the months up to the end of its crop's window; this demand is the crop coefficient of the district's crop calendar times AZMET reference evapotranspiration, without the local calibration factor or rain. Within fields, the proxy rises by 1.03 percent per percent of that ratio for cotton, 1.80 for alfalfa and other hay, 1.28 for corn and sorghum, and 1.31 for small grains. The model's seasonal shortfall is its demand-weighted mean over the crop's water months. Table A28 gives

the estimates for Maricopa–Stanfield. I leave out owner by year effects, because an owner's fields share a location and those effects would remove the variation. In 2017–2021 the seasonal shortfall hardly moves within a field, and I cannot tell. With the cut years, one standard deviation lowers the proxy by 2.8 percent.

Table A28: The Seasonal Shortfall, Water Use, and the Yield Proxy, Maricopa–Stanfield

	ln yield proxy		ln water use relative to demand	
Years	No well	With a well	No well	With a well
2017–2021	34.43 (32.48)	−32.90 (73.94)	15.39 (11.52)	−54.40 (28.30)
2017–2023	−37.48 (14.99)	−74.93 (37.42)	−12.24 (8.33)	−42.66 (23.11)

Notes: Log points (times 100) for a seasonal shortfall from 0 to 1; standard errors clustered by owner. Field by crop group and crop group by year effects. All rules at a take of 0.70. 6,551 field-years of 1,967 fields in 2017–2021, and 8,534 of 1,973 in 2017–2023. Standard deviation of the shortfall within those effects, fields without a well: 0.026 and 0.074.

A15 The Policy Simulations

Each rule changes the model's shortfalls. A field-month's ET changes by

$$\Delta \mathrm{ET}_{it} \;=\; \mathrm{ET}_{it}\left[\exp(\beta\,\Delta\ell_{it}/100) - 1\right],$$

where β is the estimated slope of 100 times ln ET on the shortfall for the field's group and $\Delta\ell$ the change in its shortfall. The change in revenue is the crop's yield response to water (FAO-33 K_y: cotton 0.85, corn 1.25, durum 1.00, winter wheat 1.00, sorghum 0.90, alfalfa 1.10, barley 1.00, other hay 1.00) times the relative change in March–October ET, times the field's acres and the NASS Arizona revenue per acre for the crop and year. Other crops carry no dollar value. The figures are gross revenue, in partial equilibrium: slopes, pumping, and prices stay as estimated. The rules divide the district's water at the district level and are not routed along the canals. Maricopa–Stanfield's slopes are all rules at a take of 0.70, −18.17 (standard error 5.22) without a well and −47.57 (21.69) with one, and the ranges use each group's slope plus and minus 1.96 standard errors. For all fields together both groups' slopes move at once, so that range is a sensitivity range, not a confidence interval. Table A29 gives the results.

A16 Administrative Allocation Cuts: The CAP Curtailment Schedule

Central Arizona Project shortage tiers curtailed the Agricultural Settlement Pool. Realized pool deliveries were unchanged under Tier 0 in 2020, because 2019 deliveries already sat below the cap. Under Tier 1 in 2022 they fell 65 percent net of mitigation and 88 percent in realized deliveries. Under Tier 2a in 2023 they went to zero. I classify six high-use districts as treated from CAP delivery accounting. The 2004 settlement made the pool junior. The treated districts were not chosen at random and differ in groundwater access. The comparisons below therefore describe what happened in the cut districts. They are not a natural experiment.

Average outcomes. Table A30 shows a large break in fallowing. Relative to 2019 and the comparison districts, fallowing in 2022 rises 13.3 percentage points under the nearest-district assignment. It rises 18.9 points when treatment is restricted to fields inside the district polygons. The satellite indices of all fields decline as well. The boundary restriction strengthens these estimates. It also worsens the pre-trends, and the estimates are sensitive to the comparison pool. The nearest

Table A29: What Shortfalls Cost and What a Remedy Would Need: Changes in Gross Crop Revenue

Rule	Fields	ET change, AF/yr	Revenue change, $/yr	$/acre/yr
A. Maricopa–Stanfield, 2016–2021 (31,434 acres without a well, 11,049 with)				
Full coverage (additional water)	no well	434	212,680 (91,300 to 338,533)	6.8
	with a well	285	139,503 (14,117 to 277,864)	12.6
	all	719	352,183 (105,417 to 616,398)	8.3
Equal sharing of the same water	all	−39	3,177 (−497 to 16,787)	0.1
No-well fields first (2023 rule)	no well	434	212,680 (91,300 to 338,533)	6.8
	with a well	−1,091	−489,707 (−856,271 to −56,508)	−44.3
	all	−657	−277,027 (−517,737 to 34,792)	−6.5
if fields with a well lose at the no-well slope	all	−7	14,604 (2,958 to 35,103)	0.3
if their wells replace the water	all	434	212,680 (91,300 to 338,533)	5.0
Canal pumps pooled	no well	−17	−6,306 (−8,041 to −3,261)	−0.2
	with a well	7	1,679 (−58 to 7,417)	0.2
B. The best rule alone, Maricopa–Stanfield				
Full coverage	no well	270	139,616 (54,238 to 229,625)	4.4
No-well fields first	with a well	−571	−251,489 (−446,788 to −35,904)	−22.8
C. Upper Gila, reported diversions, 2016–2023 (20,965 acres)				
Water reaching crops, 0.7 to 0.8	all	122	57,846 (28,376 to 87,431)	2.8
Water reaching crops, 0.7 to 0.9	all	231	109,448 (53,608 to 165,673)	5.2
Pooled river	all	−16	−6,904 (−7,690 to −4,532)	−0.3
Full coverage	all	1,883	821,843 (400,924 to 1,254,187)	39.2

D. A new well for fields without a well, Maricopa–Stanfield

Fields	Benefit, $/acre/yr	AF/acre/yr to pump	Benefit, $/AF	Benefit / capital
All	6.9	0.09	80.3	4%
Least short third	1.3	0.02	83.5	1%
Middle third	3.9	0.05	81.5	2%
Most short third	17.2	0.22	79.7	11%
Canal pump within 2 km above	4.0	0.05	78.4	2%
None within 2 km above	12.0	0.15	81.5	7%

Notes: Ranges: each group's slope plus and minus 1.96 standard errors. The first rule, full coverage, takes more water than the district had; the others divide the same water. Full coverage for fields without a well is 0.5 percent of their March–October water use and of their $40,785,835 of gross revenue a year. No-well fields first moves 3,364 acre-feet a year, or $63.2 of their gain per acre-foot; on the best rule, 3,283 and $42.5. Well: Maricopa–Stanfield's 2024 cost of $1,115,297 for one 2,000-gallon-per-minute well with pipeline, about 2,000 acre-feet a year, serving 446 acres: $2,501 per acre, $163 per acre a year at 5 percent over 30 years, and $36 per acre-foot of capacity, before energy and operation (Maricopa–Stanfield Irrigation and Drainage District 2024). Lining: the model has no loss that grows with distance, so lining appears as the share of water reaching crops, and the size of that rise is an assumption. 308.5 km of tree line on the ten canals; Reclamation costs of $82,021 to $328,084 per km for a 10-foot wetted perimeter, or $5,336 to $21,342 per km a year; so a rise to 0.8 is worth 1 to 4 percent of that cost, and to 0.9, 2 to 7 percent. The model counts no return of seepage to the aquifer.

assignment labels 31 percent of treated fields outside the district polygon. The delivery collapse coincides with a large break in fallowing and production in the treated districts. The comparison does not show that the CAP cut alone caused the full magnitudes. Conditioning on fields that stay planted removes the fallowing response and changes the crop mix, so it cannot recover a clean effect on planted fields.

Table A30: Tier 1 and Average Field Outcomes

Treatment geography	Outcome	Estimate	District SE	Webb p
2022 treated–control change relative to 2019				
Nearest-district assignment	Fallowing	+0.133	0.032	0.015
	$\ln(1 + \text{GCVI})$	−0.164	0.071	0.052
	$\ln(1 + \text{NDMI})$	−0.035	0.018	0.120
	$\ln(1 + \text{ET})$	−0.140	0.081	0.249
Distance-zero boundary proxy	Fallowing	+0.189	0.031	0.001
	$\ln(1 + \text{GCVI})$	−0.205	0.071	0.003
	$\ln(1 + \text{NDMI})$	−0.048	0.021	0.036
	$\ln(1 + \text{ET})$	−0.210	0.075	0.024

Notes: Field-weighted district-year event-study comparisons with district and year effects; 2019 is the reference year. The nearest-district assignment uses the paper's nearest-district label; the boundary restriction retains only fields geographically inside the named district polygon. Fallowing is in probability units; satellite outcomes enter as $\ln(1 + x)$. Treatment varies by district and year, so standard errors cluster by irrigation district. Webb wild-cluster p-values use 9,999 null-imposed draws. The table is descriptive: joint pre-trend tests fail for several outcomes, particularly under the boundary restriction, and estimates vary with the comparison pool.

Groundwater. The ADWR Groundwater Site Inventory gives a balanced panel of 65 monitoring sites in five treated districts and 61 sites in eight comparison districts, each observed every year of 2018–2023. Every district has equal weight, and sites have equal weight within a district. Relative to 2021, depth to groundwater in the treated districts rose 8.33 feet more than in the comparison districts by 2023 ($p = 0.008$; Webb $p = 0.011$). The 2022 difference is 2.23 feet ($p = 0.546$). Net of each site's 2018–2021 trend, the 2023 difference is 12.47 feet ($p = 0.065$; Webb $p = 0.042$). The p-values cluster by district, and the Webb p-values use 9,999 null-imposed draws. Greater depth means a lower water table. The records show the aquifer falling after the cut. They do not measure pumping volumes, who pumped, or how much of the fall the cut caused.

Supplementary Material for Online Publication Only

The supplement documents the data (Appendices S1–S6) and the water deliveries behind the 2008–2015 comparison of Section 7.2.4 (Appendix S7). It then holds material the paper leans on less: the model within the season, the timing of planting, the farmed-status check, and which fields stopped farming (Appendices S8–S11).

S1 Geospatial Sources and Field Panel

- **Irrigation canal network.** The USGS National Hydrography Dataset Plus (NHDPlus) provides canal geometries, flow attributes, and unique identifiers (NHDPlusID). After filtering out ephemeral streams (NHD feature code 46007), 147,892 canal segments are retained from the 547,842 raw flowlines across Arizona.
- **Field boundaries.** The USDA Crop Sequence Boundaries (CSB) dataset provides satellite-derived field boundaries spanning 2016–2023, each field with a unique identifier. I keep the

fields in Arizona (FIPS state code 04), across all 15 counties.

- **Digital elevation model (DEM).** USGS 1/3 arc-second elevation tiles provide the high-resolution elevation rasters used to determine water-flow directions within the canal network by calculating elevation gradients along canal segments.
- **Crop-type classification (CDL).** The USDA CropScape Cropland Data Layer provides annual satellite-based crop-type classification at 30 m resolution for 2016–2023. The crop label sets the crop-by-year effects and each field's predicted demand. A field-year without a crop code is fallow, and the monthly water-use rule of Appendix S2 marks further field-years bare (Appendix S6). The crop sequences also train the fallow-type classifier.

The panel contains the 54,946 fields of Section 4.1, observed in every crop year of 2016–2023. The irrigation network of Section 5 links 28,427 of them to canals. Sample counts differ across variables because satellite outcomes, drought measures, and network variables are observed on different supports.

S2 Satellite Outcomes

Water use. Monthly evapotranspiration comes from the OpenET ensemble, version 2.0 (the gridMET-based monthly collection, band `et_ensemble_mad`), in millimetres per month. I take its mean over each field polygon at 30 m. A crop year runs from November of the year before to October, and all twelve months enter the water model. Crop year 2016 has no November or December. The water model uses the log of water use in active field-months and drops the few months without positive use.

Yield proxy. GCVI $= B8/B3-1$ comes from Sentinel-2 harmonized surface reflectance, through Google Earth Engine. I keep scenes with less than 20 percent cloud cover and mask cloud shadow, medium- and high-probability cloud, and cirrus with the scene-classification layer (classes 3, 8, 9, and 10). In each window I take the maximum at each pixel and then the mean over the field at 20 m. Crop year y has two windows, November 1 of $y-1$ to May 1 of y for winter crops, and May 1 to October 1 of y; the end dates are exclusive. The yield proxy is the log of the larger of the two maxima, on planted field-years of crop years 2019–2023. GCVI responds to the chlorophyll in the canopy, and satellite vegetation indices track harvested yield in ground data (Lobell et al. 2020).

Fallow. A field-year is fallow when the Cropland Data Layer gives it no crop code, or when its monthly water use shows no crop. The rule scores each month m of the calendar year by $s_m = (\mathrm{ET}_m - F)/(R-F)$. F is the median water use in that calendar month of the county's fields coded fallow/idle (CDL code 61) in 2016–2018 and 2020. R is the 90th percentile of water use of the county's annual-crop fields in that month and year. A score of 0 is the fallow level, and 1 is what the top tenth of cropped fields reach. Months with $R-F$ below 10 mm carry no information and are skipped. A field-year's activity is its second-highest monthly score, so one wet month does not make a crop. The field-year is bare when its activity falls below the county cutoff, the midpoint between the median activity of the county's fields coded 61 and that of its annual-crop fields in the same years. Counties with too few fields in either group use statewide values. Tree and vine crops keep their CDL status, since their canopies never reach the water use of annual crops, and so do field-years with fewer than six informative months. The rule runs on the calendar year, and

its flag applies to crop year y. A rule on May–October water use alone would call a winter crop bare, such as the vegetables grown in Yuma from November to March.

S3 Drought Measures

I take SPEI from the SPEIbase global gridded data (Vicente-Serrano, Beguería, and López-Moreno 2010), at about 0.5 degrees, sampled at each field's centroid. I use the 24-month and the 3-month index, each averaged over May–August. The window ends before the crop season ends. It covers the months when irrigation demand peaks and allocations bind. SPEI-24 follows reservoir storage and the reliability of surface water over several years. SPEI-3 marks short dry spells.

I set drought for each irrigation district and crop year. I average the May–August SPEI-24 over the district's fields on the irrigation network. Each field counts once, whatever its size. The district is in drought when the mean is below -0.6. Every field of the district then has the same drought value. An irrigation district here is whoever delivers the water, so a reservation counts as its own district. The yield and fallowing models also control for each field's own continuous SPEI-3 and SPEI-24.

Drought hits irrigation districts in different years. In 2021, 97 percent of the fields in junior-rights districts were in drought. Fields in senior-rights districts had their drought years in 2019, when 46 percent of them were in drought, and in 2022. Drought covers 59 percent of field-months in 2008–2015 and 13 percent in 2016–2023 (Figure 7). SPEI is an atmospheric index. It marks years when surface water runs short, not deliveries or cuts. Appendix A16 reports the CAP cuts.

S4 Water Rights, Groundwater, and Irrigation Districts

- **Surface-water rights.** Surface-water rights data from the Arizona Department of Water Resources (ADWR) "Statement of Claimant Locator" database provide ownership, priority dates, and locations. These records represent Place of Use (POU) designations, the locations where water may legally be applied for beneficial use. Each field takes the priority year of the nearest recorded right.[17] A significant limitation is that water rights are recorded as point locations rather than polygons: a single point cannot precisely define which fields fall under a given right, and the service area may extend beyond the recorded coordinates. The water-rights controls therefore rely on distance-based matching from field centroids to the nearest recorded right location, introducing measurement error that likely biases toward zero the estimates of how legal priority affects crop outcomes. The nearest-neighbor match imposes no maximum distance; matches beyond 0.2 km are flagged as too distant for the water-rights controls.

- **Irrigation districts.** I obtain irrigation-district boundary shapefiles from ADWR and spatially assign each field to the nearest district, generating a district identifier, a within-district indicator, and distance to the nearest district. The nearest identifier is a geographic assignment for every field, not evidence of membership or service. The institutional profiles therefore distinguish fields inside the named boundary from outside fields and describe "inside" as a service proxy rather than a legal entitlement.

- **Grandfathered groundwater rights.** The Arizona Grandfathered Groundwater Rights database identifies historical parcel boundaries with established groundwater-extraction rights (Irrigation Grandfathered Rights, IGFR). These rights, which predate Arizona's 1980 Groundwater Management Act, provide legal access to groundwater, an important alternative source during surface-water shortages.

- **Well registry.** The ADWR Wells55 registry provides well locations, use codes, application status, and reported installation dates. I calculate the number of wells within the grandfathered-rights boundaries associated with each field as the primary access measure.

- **Groundwater conditions.** The ADWR Groundwater Site Inventory provides dated depth-to-water readings and measurement-source metadata. The monitoring panel of Appendix A16 retains direct readings with blank measurement remarks and credible agency source codes, collapses duplicate site-date records by their median, and requires the same site in every year from 2018 through 2023. Depth is a groundwater-condition outcome: greater depth means a lower water table, not a metered withdrawal volume.

- **Physical supply.** The source audit verifies ordinary agricultural water used within mapped service areas from district, Reclamation, CAP, ADWR, and related primary records. Its fixed 30-district set covers 95.6% of service-supported member fields. The strict classification retains unresolved cases; the approved broader-system sensitivity classifies six Harquahala/Gila Valley encompassing-system cases as mixed without imputing a groundwater share. The resulting division is groundwater only (3 districts), mixed (22), and surface water only (5).

[17] Fields with no matched water right are assigned a priority year of 2025, treating them as having the lowest possible seniority. This affects approximately 72% of observations, reflecting the coarse point-based matching described below.

- **Pricing and governance.** The audit records volumetric, per-acre, flat, tiered, time-based, and energy-pass-through charges, plus rotation schedules and shortage protocols. A descriptive code establishes that a mechanism is documented at some point in 2016–2023; a separate static code requires evidence suitable for a time-invariant interaction variable throughout the analysis period. Unknown, conflicted, and period-ineligible entries remain missing. The strict static view has too little yes–no support for pricing or governance interaction regressions.
- **Seniority.** Priority is classified within the district's operational allocation system rather than translated to a universal rank. The priority categories are senior (6 districts), junior (10), mixed (5), unknown (6), and not applicable (3). The junior-rights and senior-rights groups of Section 7.2.1 use only the affirmatively junior and senior districts, and only the fields inside them; mixed and unknown districts are in neither group.

The integrated audit workbook joins the 81 live panel identifiers to pricing/governance evidence and preserves the independently reviewed 30-district priority/supply scope used for analysis. Table S1 reports the complete category distribution and the yes/known denominators; it does not code undocumented mechanisms as zero.

Table S1: Irrigation-District Characteristics and Documentation Coverage

Category or mechanism	Districts	Service-supported plots	Plot share	Documented yes/known	Static yes/known
Panel A: Physical supply (broader-system sensitivity)					
Groundwater only	3	1,642	5.9%		
Mixed surface/groundwater	22	23,714	84.6%		
Surface water only	5	2,663	9.5%		
Panel B: Operational priority					
Senior	6	10,879	38.8%		
Mixed priority	5	3,588	12.8%		
Junior	10	7,870	28.1%		
Unknown	6	4,040	14.4%		
Not applicable	3	1,642	5.9%		
Panel C: Pricing and governance audit coverage					
Volumetric pricing	–	–	–	5/5	3/3
Per-acre assessment	–	–	–	26/26	18/18
Flat recurring fee	–	–	–	2/2	1/1
Tiered/progressive pricing	–	–	–	2/2	2/2
Time-based pricing	–	–	–	1/1	—
Energy pass-through	–	–	–	0/1	0/1
Rotation schedule	–	–	–	3/4	1/2
Shortage protocol	–	–	–	12/12	8/8

Notes: The fixed 30-district set accounts for 28,019 service-supported member fields (95.6% of all member fields). Physical supply uses the approved broader-system sensitivity and does not impute a numeric groundwater share. Priority categories are operational within system; mixed, unknown, and not-applicable values are not forced into the senior/junior binary. "Documented" means the mechanism is established at some point in 2016–2023; "static" requires full-period eligibility for a time-invariant regression interaction variable. Entries are yes/known district counts, so 3/3 means three documented yes districts and no documented no districts, not a three-district balanced comparison.

S5 Soil, Market Access, and Secondary Controls

- **Soil quality (SSURGO).** The USDA NRCS Soil Survey Geographic Database provides each field's soil properties: salinity (EC), clay and sand content, organic carbon, available water storage, bulk density, pH, erosion factors, and a productivity index.
- **Market access (OpenStreetMap).** I calculate distances from each field centroid to the nearest city, large city, highway, and rural place, and to the nearest irrigation district.
- **Land status.** Distances to reservations, state trust land, Reclamation land, and protected areas (GAP status 1–2).

- **Community Water System service areas.** An indicator for whether the field centroid lies inside an ADEQ CWS service-area polygon.

The yield and fallowing models hold these variables in their control set, with field acres, slope, coordinates, distance to the canal, network degree and head/tail indicators, and the water-right and well variables. The soil, distance, and land variables do not change within a field, so the field effects absorb them. The water-right and well variables enter interacted with drought (Section 6.4), and SPEI-3 and SPEI-24 enter as they are (Appendix S3). The water model has field by crop-year effects, which absorb every control that varies only by field and year. The complete control list is in the replication package.

S6 Fallow Classification

A field-year has a crop code when its CDL code lies outside a 24-code non-crop set (fallow/idle, water, developed, barren, forest, shrubland, pasture, wetlands, and similar categories, plus missing codes). It is **active** when it has a crop code and the monthly water-use rule of Appendix S2 does not mark it bare, and **fallow** otherwise. Activity sets the fallowing outcome and predicted demand, which is zero for a fallow field.

Not all fallow is equally informative. Fallow type comes from a machine-learning classifier trained on 8-year crop-sequence patterns, which assigns each fallow field-year a predicted rotation probability: **non-rotation (stress) fallow** requires probability < 0.3, likely fallowed due to water constraints or economic stress, and **rotation fallow** requires probability > 0.7, representing planned agronomic rotation. Fallow field-years with predicted probabilities in the middle band $[0.3, 0.7]$ are excluded from both classifier-based shares. These two shares, within 3 km, are the nearby-fallowing controls of the yield model (equation (5)).

The shares are computed in geographic space: for each field-year, the shares of fields within 3 km that are stress-fallow and rotation-fallow, regardless of canal direction. They absorb local shocks that spread across nearby fields. The fallowing model (equation (6)) leaves them out, since they measure fallowing too.

S7 Water Deliveries, 2008–2023

Table S2 records the deliveries behind the 2008–2015 comparison of Section 7.2.4. In 2008–2014, CAP agricultural deliveries stayed within 1 percent of schedule and no shortage was declared, although SPEI drought was more severe than in 2020–2022. The 2015 reduction was voluntary and compensated. Realized deliveries first fell under Tier 1 in 2022. The facts come from official records compiled in the replication package.

S8 The Model Within the Season

This appendix extends the identity of Section 3.2 to the months of a season, for the in-season tests of Section 8. Let S_{mt} be the water that enters the canal in month m of season t, w^*_{jm} user j's scheduled delivery in that month, with $\sum_m w^*_{jm} = w^*_j$, and ε_{jmt} the amount by which j's diversion departs from that schedule, with $\sum_m \varepsilon_{jmt} = \varepsilon_{jt}$. Equation (1) for one month is

$$A_{imt} \;=\; S_{mt} \;-\; \sum_{j<i}\big(w^*_{jm} + \varepsilon_{jmt}\big) \;+\; \rho \sum_{j<i} \lambda_{ij}\big(w^*_{jm} + \varepsilon_{jmt}\big), \tag{14}$$

Table S2: Institutional Water-Delivery Timeline, 2008–2023

Years	Deliveries and storage	Source
2008–2014	CAP agricultural-pool deliveries within ≈0.9% of the 400 kAF schedule; no shortage declared	CAP delivery reports
2010–2014	Meteorological drought more severe than 2020–22 by SPEI; Lake Mead year-end 10.301–14.644 MAF (1,086.30–1,132.83 ft), above the 1,075 ft shortage trigger	Reclamation RISE
2015	CAP agricultural-pool deliveries fell to 318,279 AF; Forbearance I conservation of 80,922 AF, voluntary and compensated (not a shortage cut)	CAP delivery and conservation ledgers
2008–2023	SRP Salt–Verde allocation 3.0 AF/acre (2008–14), raised to 3.3 in 2015–23; available schedules show no allocation cut	Phoenix Water Resource Plan
2020	Tier 0 declared; realized CAP ag-pool deliveries flat against 2019 (255,415 versus 256,143 AF); the reduction was absorbed by credited forbearance (CAWCD DCP contribution 180,608 AF) and unused pool headroom, so orders sat near the reduced cap	Reclamation accounting; CAP delivery ledgers
2022	Tier 1 shortage (projected Mead 1,065.85 ft); CAP ag pool realized 30,825 AF (≈88% year-over-year fall)	Reclamation Tier 1 determination; CAP delivery report
2011–2022	San Carlos Reservoir below 1% of usable capacity (2011–13, 2015, 2018–19, 2021 zero in July, 2022)	USGS station 09469000
2023	Tier 2a; wet-year refill (San Carlos to 226 kAF); no mandatory Yuma senior (Priority 1–3) cut documented in 2021–23	USGS 09469000; Reclamation

Notes: Compiled from official records in the replication package's delivery-fact audit; CAP and Reclamation series are high-confidence, post-2019 SRP schedules and Yuma senior-priority status medium-confidence. Sources: CAP delivery and conservation ledgers (https://library.cap-az.com); Bureau of Reclamation RISE reservoir series and shortage determinations (https://data.usbr.gov); USGS daily storage, station 09469000 (https://waterservices.usgs.gov); City of Phoenix 2021 Water Resource Plan.

and the wedge, cushion, shortfall, and outcome take the month index: $\Delta_{imt} = \sum_{j<i}(1-\rho\lambda_{ij})\,\varepsilon_{jmt}$, $\ell_{im}(S_{mt}) = A^0_{imt} - w^*_{im}$ with A^0_{imt} the value of equation (14) at $\varepsilon_{jmt} = 0$, $q_{imt} = [\Delta_{imt} - \ell_{im}(S_{mt})]_+$, and $Y_{imt} = Y^0_{im} - \beta\, q_{imt}/A^0_{imt} + u_{imt}$. A monthly departure can be negative. In the delivery model of Section 8.1, the crop demand of the land above i moves with the part of ε_{jmt} that follows what that land planted that year. Planting is set before the season (Appendix S9) and is taken as given. The four cases below can hold together.

(a) A canal capacity that binds at the peak. The canal carries at most $\bar{S}_t$ in a month, and deliveries are scheduled within it. Below capacity, the district releases what the farms divert, so $q_{imt} = 0$. At capacity, $S_{mt} = \bar{S}_t$ cannot rise. A departure above i then takes its water during delivery, before the canal reaches i, which is the residual claiming of Section 3.2 within one month. Once the cushion is used up,

$$\frac{\partial q_{imt}}{\partial \varepsilon_{jmt}} \;=\; 1 - \rho\lambda_{ij} \;\geq\; 0 \quad (j < i), \qquad \frac{\partial q_{imt}}{\partial \varepsilon_{kmt}} \;=\; 0 \quad (k > i). \tag{15}$$

Schedules follow crop need, so the canal reaches capacity in the peak months. The cushion is thinner where the field's own crop needs more ($\partial \ell_{im}/\partial w^*_{im} = -1$) and when the capacity is cut ($\partial \ell_{im}/\partial \bar{S}_t = 1$). So the upstream slope is negative only in months at capacity and strongest at the field's own peak demand, and the downstream slope is zero. The gap opens when the canal water is cut, which is the in-season form of drought amplification. Without an account (case d), the seasonal leak $\varepsilon_{jt} \geq 0$ still makes $\Delta_{it} = \sum_m \Delta_{imt}$ rise toward the tail, and case (a) alone keeps the level gradient.

(b) Branch releases routed by orders. Let flow F_{mt} reach the junction above i, short of the orders O_{mt} of the branches below it, and let the district split it in proportion to orders. With $O_{bmt} = \sum_{k\in b}(w^*_{km} + \varepsilon_{kmt})$ the orders on i's branch b, which include the farms below i,

$$A_{imt} \;=\; F_{mt}\,\frac{O_{bmt}}{O_{mt}} \;-\; \sum_{j<i,\; j\in b} \big(1-\rho\lambda_{ij}\big)\big(w^*_{jm} + \varepsilon_{jmt}\big),$$
$$\frac{\partial A_{imt}}{\partial \varepsilon_{kmt}} \;=\; F_{mt}\,\frac{O_{mt} - O_{bmt}}{O^2_{mt}} \;>\; 0 \quad (k > i,\; k \in b).$$

More orders below i send more water down its branch, and a field that is short takes the added water as it passes. Downstream demand then raises a field's water. The same term enters for the farms above i on its branch, so the contrast in the field's water between the two directions stays $-(1-\rho\lambda_{ij})$, as in case (a). With splits fixed in advance, the term and the downstream slope are zero.

(c) District wells along the canal. Let district pump p add $g_{pmt} \leq \bar{g}_p$ to the canal in month m, where $\bar{g}_p$ is its registered capacity, so that $\sum_{p<i} g_{pmt}$ joins the right side of equation (14). A pump runs on demand. If it sits between upstream land j and field i and runs below capacity, it replaces what j diverts, and $\partial A_{imt}/\partial \varepsilon_{jmt} = 0$. Land with no pump in between keeps $\partial A_{imt}/\partial \varepsilon_{jmt} = -(1-\rho\lambda_{ij})$, so the penalty should sit in that land. Capacity can only add water ($\partial A_{imt}/\partial \bar{g}_p \geq 0$), so a field served by more capacity should lose less. A pump at capacity is a source like the canal head in case (a). The fields it serves draw on it in order, and those nearer the pump take first. When the canal carries only pumped water, as in Central Arizona and Maricopa–Stanfield in 2023,

each pump is the source for the fields it serves, and the competition runs among them. A farm with its own well replaces a shortfall by pumping at cost c_g (Section 3.4), so its ET should fall little.

(d) Seasonal allotments per acre. Central Arizona and Maricopa–Stanfield ration water by a seasonal allotment per acre, and Central Arizona cut each farmer's allotment in proportion (CAIDD 2022). Deliveries are charged against the allotment, so a farm that diverts more in one month diverts less later. When every farm uses its whole allotment, $\sum_m \varepsilon_{jmt} = 0$ for each j, and

$$\Delta_{it} \;=\; \sum_{m} \Delta_{imt} \;=\; \sum_{j<i} \bigl(1 - \rho\lambda_{ij}\bigr) \sum_{m} \varepsilon_{jmt} \;=\; 0. \tag{16}$$

The pro-rata case of Appendix A1 keeps a leak beyond the scaled entitlement, which cumulates toward the tail. The account leaves none. The seasonal shortfall then does not rise toward the tail, as long as the canal falls below capacity later in the season so that each farm can take the rest of its allotment. If schedules also match what farms expect to divert in each month, $E[\varepsilon_{jmt}] = 0$, and then $E[\Delta_{imt}] = 0$ at every position. The fixed head-to-tail gradient of the first prediction in Section 3.2, which comes from the mean of the wedge, disappears. Only the spread of Δ_{imt} still grows toward the tail, which can leave a second-order gradient through the kink in q_{imt}. The monthly wedge itself remains. A farm above i that diverts more than its schedule in a month at capacity leaves i short then, by equation (15), and i gets the water back later, when its crop needs less. Central Arizona wrote that shortages "usually occur during periods of above normal temperatures in the growing season when there is high demand, or at the end of the year when the mandated reduced allotment has been used" (CAIDD 2022). Diversions that escape the meter are a seasonal leak again, and the gradient returns in proportion to them. The allotments themselves need not be equal, as long as they do not fall toward the tail. In Maricopa–Stanfield they ranged from 1.7 to 2.8 acre-feet per entitled acre, by where the grower's fields were (Section 8.5). In the CAP districts the field effects of equation (12) absorb any fixed gradient, and the estimates of Section 8 use only the variation within the season. Statewide, the usual position does not lower the yield proxy, but fields that usually sit low on junior canals are left fallow more often in drought (Section 7.2.2).

What the data show. Section 8 takes cases (a), (b), and (d) together. In months when the water reaching a district falls short of its crop demand, fields with more crop demand upstream receive less, and fields without a well use less water; shuffled supply dates and shuffled positions do not reproduce the slope (Section 8.4). The search over rules speaks to case (b): sharing a junction's water by the claims along each branch is one rule, and the data keep rules in which side lines take some priority and reject equal sharing (Section 8.4). Case (c) appears in two forms. District wells on the canals mark which fields keep water, and competition on the canal matters little for a field with a pump above it (Section 8.5). A field's own well shows no protection in Maricopa–Stanfield, while in Harquahala and New Magma the loss falls on fields without a well (Section 8.4). In Central Arizona, where district wells come first and the records do not say which wells ran, I cannot tell (Section 8.6).

S9 Planting Decided Before the Season

Predicted demand (Section 6.3) and the crop demand of the delivery model (Section 8.1) count what each field planted that year, from the Cropland Data Layer, with the calendar of its crop. The Cropland Data Layer gives one status per field and year, assigned after the season. If farmers had decided during the season whether to farm a field, in response to the water they received, upstream planting and a field's water in the same months would be set together. The Sentinel-2 vegetation index shows when the status was set. I use the five-day GCVI composites of all 6,818 fields inside the boundaries of the four CAP districts in 2017–2023 and call a field green in a window when its GCVI is at least 3.0 (Table S3). Thresholds of 2.5 and 4.0 give the same pattern.

Land fallowed in the shortage years was not planted first. Of the fields fallow in 2022 and farmed in 2021, 11.7 percent were green at some time from March through June. Among newly fallowed fields, the share was 12.9 to 19.0 percent in 2017–2019 and 10.3 percent in 2021. The median ET of the newly fallowed fields from January through June 2022 was 177 mm, close to that of newly fallowed fields in other years and far below the 477 mm of farmed fields. Fields that were planted stayed planted. Of the cotton fields green by July 14, 2.2 percent had no green window from July 15 through October 31 in 2022 and 1.5 percent in 2021, against 0.5 to 2.1 percent in 2017–2019. The share of all fields that are green fell more after mid-July than in spring in 2021 and 2022, by 5.6 and 7.3 points relative to 2017–2019. It fell even more in 2020, by 10.5 points, when deliveries were near normal. This drop does not follow the shortage. Corn is a large part of it: in every year from 2017 to 2022, half or more of the corn fields green by July 14 had no green window after it.

In 2021 and 2022, what fields planted was set before the canal water fell short. The response within the season is in water use, the outcome of the design. The year 2023 differs: 24.1 percent of its newly fallowed fields were green in spring and idle afterwards, and 4.1 percent of its green cotton fields had no green window after mid-July.

Table S3: Planting and Fallowing Were Set Before the Season

	2017	2018	2019	2020	2021	2022	2023
Panel A. Fields fallow in the year							
Newly fallowed fields	416	498	636	601	551	1,407	1,545
percent green at any time March–June	19.0	12.9	15.4	16.6	10.3	11.7	24.1
median ET, January–June (mm)	150	161	168	180	143	177	227
Fallow two years in a row: median ET, January–June (mm)	114	98	188	190	129	121	212
Farmed fields: median ET, January–June (mm)	459	444	461	494	493	477	487
Panel B. Fields planted in the year							
Cotton fields green by July 14	1,314	1,141	698	674	931	626	314
percent with no green window July 15–October 31	0.5	2.1	1.4	0.4	1.5	2.2	4.1
Corn fields green by July 14	338	546	623	643	740	408	446
percent with no green window July 15–October 31	50.6	51.8	66.1	80.7	83.0	61.8	42.4
All fields: change in the share green, late season minus spring (points)	−0.6	3.4	−2.9	−10.5	−5.6	−7.3	−14.5

Notes: All 6,818 fields inside the boundaries of Central Arizona, Maricopa–Stanfield, Hohokam, and Harquahala, in every year 2017–2023. Status in the year (fallow or farmed, and the crop) from the Cropland Data Layer; newly fallowed: fallow in the year and farmed the year before. Green: Sentinel-2 GCVI of at least 3.0 in a five-day composite with at least half of the field observed. ET: OpenET ensemble, sum over January–June, median over fields. Panel B, last row: the share of fields green in each five-day window minus its 2017–2019 mean, averaged over July 15–October 31, minus the same average over March–June, in percentage points. CAP deliveries were near normal in 2020, fell to 40 to 60 percent of normal from July 2021, and were 15 to 40 percent of normal through 2022.

S10 The Farmed Status Against Satellite Records

The Cropland Data Layer gives one label per field and year, farmed or fallow. The fallow rule of Appendix S2 starts from it, and the fallowing of Appendix S11 counts fields by it. Two errors are possible. A crop label on land that grew nothing hides fallowing. A fallow label on land that grew a crop adds fallowing that did not happen. Satellite records can find both. I check the label of all 6,818 fields inside the four CAP districts in 2017–2023 against two of them: the Sentinel-2 green windows of Appendix S9 and annual OpenET (Table S4).

Crop labels on land that grew nothing are rare. Over the seven years, 3.4 percent of farmed acres have no green window, between 2.2 and 4.5 percent in each year (Panel A). Grain labels fail most often, on 6 to 7 percent of grain acres in 2019, 2020 and 2022. Cotton labels fail on at most 1.8 percent of cotton acres. Farmed fields with no green window used a median of 512 mm of water a year. That is about half the 959 mm of green farmed fields and more than the 328 mm of fallow fields with no green window. Some are fallow land with a crop label, and some are thin crops. At a GCVI threshold of 2.0, 1.7 percent of farmed acres have no green window, and their median of 376 mm is close to the 311 mm of fallow land.

Fallow labels on green land are more common. Over the seven years, 28 percent of fallow field-years have a green window. Most turn green in winter. Of these fields, 68 percent peak between December and March, and the median one stays green for 35 days. Fallow fields green for less than 30 days used a median of 425 mm of water a year, about 100 mm more than fallow fields that never turn green. They are weeds after winter rain. One fallow field-year in six stays green for 30 days or more, with a median of 605 mm, nearly two thirds of a farmed field (Panel B). In these records a short winter crop and a dense stand of weeds look alike.

Counted from satellite records, farmed land fell as much in 2022 as the Cropland Data Layer says, and less in 2023 (Panel C). In 2022, farmed acres were 75 percent of their 2017–2019 level by the Cropland Data Layer, 77 percent counting fields green for 30 days or more, and 77 percent counting fields with at least 600 mm of annual ET. In 2023 the three counts are 57, 67 and 64 percent. The gap is widest in Maricopa–Stanfield, 57 percent against 67 percent by water use. Water use per farmed acre stays between 3.2 and 3.4 acre-feet under every count and in every year. The Cropland Data Layer and the water-use count agree on 90 percent of field-years. The 2022 fallowing and the flat water use per farmed acre hold when farmed land is counted from satellite records. The Cropland Data Layer may overstate the 2023 fallowing by 7 to 10 points of the 2017–2019 level.

Table S4: The 2022 Fallowing Holds Under Satellite Counts

	2017	2018	2019	2020	2021	2022	2023
Panel A. Fields the crop map calls farmed							
Farmed fields	5,860	5,858	5,597	5,471	5,409	4,304	3,328
percent of farmed acres with no green window	2.2	3.5	4.2	3.4	3.0	4.5	3.5
percent of grain acres with no green window	0.8	2.1	6.0	6.4	3.6	7.2	2.1
percent of cotton acres with no green window	0.3	0.9	0.9	1.3	1.0	1.8	1.2
median annual ET, green fields (mm)	964	975	963	947	952	935	969
median annual ET, fields with no green window (mm)	525	503	493	469	485	501	641
Panel B. Fields the crop map calls fallow							
Fallow fields	958	960	1,221	1,347	1,409	2,514	3,490
percent with a green window	20.1	22.9	27.4	35.2	27.7	23.3	32.5
percent green for 30 days or more	12.1	13.0	12.0	18.3	13.1	15.0	22.2
median annual ET, fields with no green window (mm)	241	292	312	293	328	345	362
median annual ET, fields green for 30 days or more (mm)	870	651	633	546	663	608	587
Panel C. Farmed acres (2017–2019 = 1) and ET per farmed acre (acre-feet)							
Farmed acres, crop map	1.01	1.02	0.97	0.95	0.95	0.75	0.57
Farmed acres, green for 30 days or more	1.03	1.02	0.96	0.97	0.97	0.77	0.67
Farmed acres, annual ET of at least 600 mm	1.02	1.02	0.96	0.96	0.97	0.77	0.64
ET per farmed acre, crop map	3.29	3.29	3.26	3.25	3.28	3.22	3.32
ET per farmed acre, green for 30 days or more	3.34	3.36	3.35	3.29	3.33	3.25	3.18
ET per farmed acre, annual ET of at least 600 mm	3.35	3.37	3.35	3.33	3.34	3.28	3.31

Notes: All 6,818 fields inside the boundaries of Central Arizona, Maricopa–Stanfield, Hohokam, and Harquahala, in every year 2017–2023. Farmed or fallow from the Cropland Data Layer. Green window: Sentinel-2 GCVI of at least 3.0 in a five-day composite with at least half of the field observed, over the calendar year, as in Table S3. ET: OpenET ensemble, January–December sum. Panel C counts a field as farmed by the Cropland Data Layer, by at least six green windows, or by annual ET of at least 600 mm (about 2 acre-feet per acre). Farmed acres are relative to their 2017–2019 mean, and ET per farmed acre is the ET of the counted fields divided by their acres. GCVI thresholds of 2.5 and 4.0 leave 2.4 and 6.0 percent of farmed acres without a green window over the seven years.

S11 Who Left Fields Fallow

This appendix asks which CAP fields stopped farming when CAP water was cut. A field enters year t when it was farmed in $t-1$, and the outcome is whether it is fallow in t. The sample is the 4,641 CAP fields inside the boundaries of the four districts in 2018–2023. Each field trait enters with its own slope in every year, with district by year effects and the crop of the year before. Table S5 reports how each slope changed in 2022 and in 2023 against its 2018–2021 mean. Standard errors are clustered by canal network component, of which there are 89, and 20 in the three Pinal districts.

The crop of the year before carries most of the change. Against alfalfa, a cotton field was 24 points more likely to stop in 2022 than in 2018–2021, and a corn or sorghum field 18 points (Panel A). Measured against 2017–2019 rates, about 27,000 acres were added to fallow land in 2022. Cotton fields account for 40 percent of them, corn and sorghum fields for 23 percent, fallow fields that did not return for 20 percent, and alfalfa for 11 percent. Smaller fields stop more in every year, and the gap does not widen in the shortage years.

In the three Pinal districts, more old wells in the field's groundwater right lowered stopping in 2023, when CAP deliveries were near zero, by 4.9 points per standard deviation. In 2022 the change is about zero. Deeper groundwater went with more stopping in 2022 (4.0 points per standard deviation, standard error 2.0). Service by a district pump changes sign between 2022 and 2023.

Fields near the head of the canal network stopped more in the shortage years. One standard deviation toward the end lowered stopping in 2023 by 4.0 points. The network head is the top of the mapped district canals, not the point where CAP water enters. I therefore place the six CAP turnouts named in the districts' delivery records at the points where district canal lines

leave the aqueduct, and check each against public CAP, Reclamation, district and map sources. From each turnout I follow the field flow graph down from the most upstream fields near it that lie no higher than the turnout. For the 1,469 fields downstream of a turnout on the graph, one standard deviation farther downstream lowered stopping by 3.4 points in 2022 and by 3.3 points in 2023 against 2018–2021 (Panel B). In 2022, 35 percent of the farmed fields in the fifth nearest the turnout stopped, against 21 percent in the farthest fifth. The result holds with other rules for the entry fields and with the network position in the model. Distance along the canals that ignores the direction of flow does not predict stopping, and neither does straight-line distance to the aqueduct. Fields lower below the aqueduct stopped less in 2022, and height and position overlap.

None of the measured traits explains why fields near the turnout stopped more. Normal-year fallowing, the yield proxy, and July–August water use in 2016–2019 barely change with position, and adding them leaves the turnout slope unchanged. District wells upstream at any distance leave it unchanged as well, although fields near the turnout have fewer of them. Depth to groundwater, soil carbon, distance to the nearest town and field size leave the slope on network position unchanged. Maricopa–Stanfield varied its 2022 allotments by where a grower's fields were (Section 8.5). A proxy for local well supply does not explain the slope there, and the allotments by grower are not public.

Neighbors' planting does not move fallowing through the canal. For every field, farmed or not, I measure the crop water demand of the land linked to it through its district's canals: the farmed share of that land weighted by crop coefficients, upstream of the field minus downstream, leaving out linked land within a distance set from district size. I build it from two expectations of what the competing land will plant. One is its planting the year before. The other is the chance that each neighboring field is farmed, given its own history and what fields with the same history in its district did that year. For upstream minus downstream land with no district pump in between, neither changes fallowing in any period (Panel C). With allotments per acre the season's water is fixed, and competition moves water between months among the farmed fields. Among fields served by the same district pump, fields whose nearer sharers were expected to plant more were left fallow more often in 2022 and 2023, by 3.6 and 4.3 points per ten points (standard errors 1.7 and 2.0), against 2.3 in 2020–21. A swap test by pump, which trades the near and far sharers of a random half of the pumps, is a strict test. It gives $p = 0.21$ in 2022 and in 2023, so I treat the pattern as suggestive. Crop choice moves with the expected planting of the land upstream, already in 2020–21, which points to crop patterns shared along a canal, not to competition.

In the five Upper Gila canals (717 fields), fallowing follows the river. In the short springs of 2018 and 2021, 23 and 24 percent of the fields farmed the year before were left fallow, against 5 to 10 percent in the other years. Across canal-years the share that stopped rises with the share of March to June months with canal water below normal (correlation 0.53). In 2022, also short in spring, 8 percent stopped. The same crops left first. In 2021, 55 percent of the corn and sorghum fields and 57 percent of the grain fields were left fallow, against 15 percent of the alfalfa and other hay fields.

Table S5: Who Left Fields Fallow

	All fields		Pinal, with groundwater	
	2022	2023	2022	2023
Panel A. Field traits; change of the slope against 2018–2021, percentage points				
Cotton the year before (against alfalfa)	+23.6* (3.8)	+20.7* (3.3)	+23.9* (3.5)	+19.8* (3.3)
Corn or sorghum the year before	+17.6* (2.1)	+18.5* (4.7)	+19.4* (1.4)	+11.2* (3.8)
Small grains the year before	+1.3 (3.8)	+15.2* (3.9)	−2.4 (4.3)	+3.6 (3.8)
Toward the end of the canal network, per SD	−1.7 (1.4)	−4.0* (1.2)	−1.4 (2.1)	−3.6 (1.8)
Lower below the CAP aqueduct, per SD	−3.2* (1.4)	−3.0 (2.1)	−1.0 (2.0)	−1.2 (1.5)
Log acres, per SD	+0.5 (0.7)	+1.4 (0.9)	−0.7 (0.8)	+2.5* (0.6)
Served by a district pump	−5.5 (6.1)	+4.6 (6.7)	−16.4* (4.6)	+12.2* (4.4)
Old wells in the groundwater right, per SD	–	–	−0.7 (1.1)	−4.9* (1.5)
Depth to groundwater, per SD	–	–	+4.0 (2.0)	−1.3 (2.2)
Panel B. Farther downstream from the CAP turnout, per SD; change against 2018–2021				
Entries within 5 km of the nearest	−3.4* (0.5)	−3.3* (0.4)		
Entries within 2 km	−3.5* (0.6)	−2.3* (0.5)		
Entries within 10 km	−2.6* (0.5)	−1.9* (0.7)		
Within 5 km, with the network position	−2.5* (0.8)	−1.7* (0.8)		
Panel C. Neighbors' planting and fallowing, per ten points; change against 2017–2019				
Canal, upstream minus downstream, expected	+0.2 (1.0)	−0.6 (1.0)		
Canal, upstream minus downstream, previous year	+0.2 (0.8)	0.0 (0.7)		
Pump sharers, near minus far, expected	+3.6* (1.7)	+4.3* (2.0)		

Notes: Panels A and B: fields of the 4,641 farmed in year $t-1$; linear probability of being fallow in t, in percentage points. Each trait enters with its own slope in every year from 2018 to 2023, with district by year effects and the crop of $t-1$; a cell is the 2022 or 2023 slope minus the 2018–2021 mean slope. Continuous traits per standard deviation; crop against alfalfa or other hay. Panel A also controls for distance to the canal, a field above its canal, distance to the nearest town, soil organic carbon, normal upstream farming, and the pump capacity serving the field. The Pinal columns add depth to groundwater (2008–2014), old wells in the groundwater right and an own registered well, on the fields of Central Arizona, Maricopa–Stanfield and Hohokam with groundwater data. Panel B: position downstream of a CAP turnout on the field flow graph for the 1,469 fields that lie downstream of one; entry fields are the most upstream fields within the stated distance along the canals of the field nearest the turnout and no higher than the turnout; the other fields enter with indicators. Standard errors in Panels A and B clustered by canal network component. Panel C: fields with linked land farmed in 2016–2019 (6,335; 2,023 served by a district pump with sharers on both sides in Central Arizona and Maricopa–Stanfield); exposure of June to August; field and district by year effects; standard errors two-way by field and canal block by year. * $p < 0.05$.